Differential Geometry

Adam Kelly (ak2316@cam.ac.uk)

July 27, 2026

Differential geometry is what happens when you take the calculus you learned in first year and try to do it on something that isn't flat. The objects of study are *smooth manifolds* – spaces which, close up, look like a piece of Euclidean space, but which globally may be curved, knotted or twisted in interesting ways – and the tools of study are the derivative and the integral, suitably reinterpreted. The central discovery of the subject, and the one which gives it its character, is that curvature is not merely a feature of how a space sits inside a bigger space: a two-dimensional creature confined to a surface can measure the curvature of its world without ever leaving it. This is Gauss's *Theorema Egregium*, and everything in the second half of this course flows from it, culminating in the Gauss–Bonnet theorem, which ties the total curvature of a compact surface to a purely topological invariant.

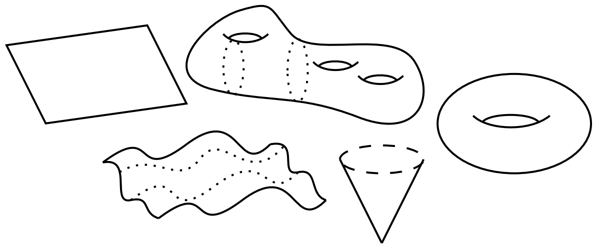
This article constitutes my notes for the 'Differential Geometry' course, held in Michaelmas 2022 at Cambridge. These notes are *not a transcription of the lectures*, and differ significantly in quite a few areas. Still, all lectured material should be covered.

Contents

§1 Differential Topology

§1.1 Manifolds

The idea of a manifold is to generalize our typical boring and flat 'Euclidean space' with an object where everything isn't necessarily flat. For example, we can take some surfaces as drawn below.



We really care about surfaces where, in a sufficiently small region around each point, things look like we are in normal flat Euclidean space. This, along with a desire for things to be nice and smooth (as we shall define) will manifest into our basic notion of a *manifold*.

Definition 1.1 (Smooth)

For an open subset $U \subseteq \mathbb{R}^n$, we say a function $f : U \to \mathbb{R}^m$ is **smooth**[a] if it has continuous partial derivatives of all orders.

[a] For a function $f : X \to \mathbb{R}^m$ with X not open, we say that f is smooth if the above definition holds for some open neighborhood V_x containing x for each $x \in X$.

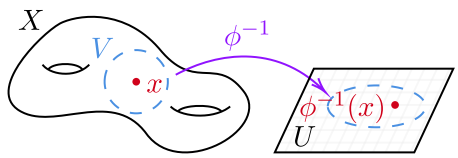
Definition 1.2 (Diffeomorphism)

A smooth function $f : X \to Y$ is a **diffeomorphism** if it's a bijection with smooth inverse. If a diffeomorphism between X and Y exists we say they are **diffeomorphic**.

Definition 1.3 (Manifold)

We say that $X \subseteq \mathbb{R}^n$ is a **k-dimensional manifold** if each $x \in X$ has a neighborhood $V \subseteq X$ diffeomorphic to an open set of $\mathbb{R}^k$.

With neighborhoods around each point looking like $\mathbb{R}^k$, we can sensibly imagine that one could locally put a coordinate system on such neighborhoods, one coming from the diffeomorphism to $\mathbb{R}^k$. This motivates the definition of a *chart* on a manifold.



Definition 1.4 (Chart)

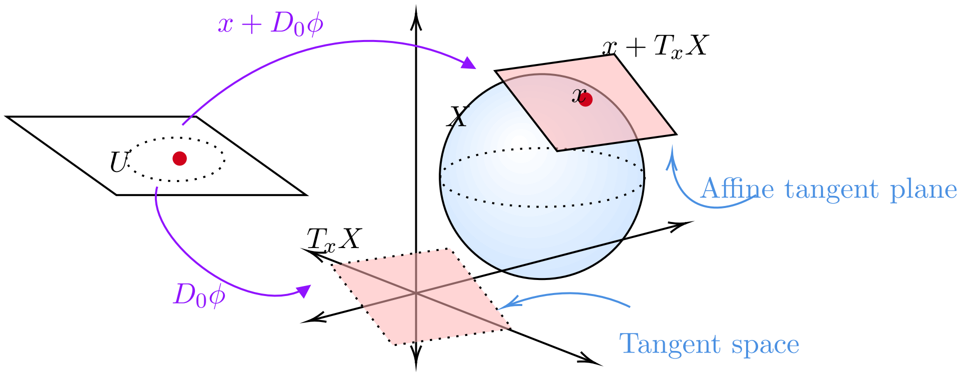
A diffeomorphism $\phi : U \to V$, where U is an open set of $\mathbb{R}^k$ is a **parameterisation** of the neighborhood V. The inverse diffeomorphism $\phi^{-1} : V \to U$ is called a **chart** on V.

Of course, if we have manifolds X and Z with $Z \subseteq X$ we say that Z is a **submanifold** of X. In this case, the **codimension** of Z in X is $\dim X - \dim Z$.

§1.2 Tangent Spaces and Derivatives

This is a course in differential geometry, and we've pinned down the 'geometry' part, now it's time to pin down the 'differential part'. We want to generalise the notion of a derivative for smooth functions on manifolds.

The idea you should have in your head of derivatives are that they are the best linear approximation to a given function. From this perspective, defining the derivative on manifolds requires us to answer two questions: what space should be our 'linear approximation' to a manifold that our derivative should act on, and then what is our linear approximation? Answering the first of these questions will lead us to the notion of **tangent spaces**.



Definition 1.5 (Tangent Space of a Manifold)

Let $X \subseteq \mathbb{R}^n$ be an arbitrary smooth manifold, and let $x \in X$. Choose a parameterisation $\phi : U \to X$ with $\phi(0) = x$ where U is an open set of $\mathbb{R}^k$. Then thinking of ϕ as a map from U to $\mathbb{R}^N$, we define the **tangent space** T_xX to be the image of the derivative of ϕ at 0, $D_0\phi(\mathbb{R}^k)$.

In this definition we had to pick some parameterisation ϕ, but one would hope that our resulting space is independent of this choice. Indeed, this is the case.

Lemma 1.6 (Tangent Spaces are Paramaterisation Independent)

T_xX does not depend on the parameterisation ϕ chosen.

Proof. Suppose $\psi : V \to X$ is another choice of parameteristaion, and suppose without loss of generality that $\phi(0) = \psi(0) = x$. By shrinking both U and V, we may assume that $\phi(U) = \psi(V)$. Then the map $h = \psi^{-1} \circ \phi : U \to V$ is a diffeomorphism. Now we write $\phi = \psi \circ h$ and we differentiate. Using the chain rule we have: $D_0\phi = D_0\psi \circ D_0 h$. Since $D_0 h$ is an invertible linear map, it follows at once that $D_0\phi$ and $D_0\psi$ have the same image. □

We are now ready to define the derivative of a smooth map $f : X \to Y$ between arbitrary manifolds, and we will see that it comes rather naturally. We would hope that for X and Y open subsets of euclidean space, our new definition should coincide with our previous definition of the derivative. Also, we would expect that the chain rule should still hold. This is going to force our definition.

Let $\phi : U \to X$ and $\psi : V \to Y$ be parameterisations, with $\phi(0) = x$, $\psi(0) = y$ and U, V being open subsets of $\mathbb{R}^k$ and $\mathbb{R}^l$ respectively. Then for small enough U, the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "f"] & Y \\ U \arrow[u, "\phi"] \arrow[r, "h=\psi^{-1}\circ f\circ\phi"] & V \arrow[u, "\psi"'] \end{tikzcd}$$

Then $D_0\psi$, $D_0\phi$, $D_0 h$ are all well defined, and as we said previously we would hope the chain rule still holds, which implies that the following square of linear maps commutes:

$$\begin{tikzcd} T_xX \arrow[r, "df_x"] & T_yY \\ \mathbb{R}^k \arrow[u, "d\phi_0"] \arrow[r, "dh_0"] & \mathbb{R}^l \arrow[u, "d\psi_0"'] \end{tikzcd}$$

Then since $D_0\phi$ is an isomorphism, only one definition of $D_x f$ will meet our requirements, namely

$$D_x f = D_0\psi \circ D_0 h \circ D_0\phi^{-1}.$$

Again, one would hope that this definition of $D_x f$ is independent of ϕ and ψ, and this is the case (and the same proof as before works). Also, we have forced the chain rule still to hold.

Theorem 1.7 (Chain Rule)
If $f : X \to Y$ and $g : Y \to Z$ are smooth maps between manifolds, then

$$D_x(g \circ f) = D_{f(x)} g \circ D_x f.$$

Proof. This is immediate from the ordinary chain rule. Choose smooth extensions F of f and G of g to open neighbourhoods of x and $f(x)$ in the ambient Euclidean spaces, arranged so that $G \circ F$ is defined; then $G \circ F$ is a smooth extension of $g \circ f$, and

$$D_x(g \circ f) = D(G \circ F)|_x \,|_{T_xX} = \left(DG|_{f(x)} \circ DF|_x \right) |_{T_xX} = D_{f(x)} g \circ D_x f,$$

where the middle equality is the chain rule in Euclidean space and the last uses that $D_x f$ maps $T_x X$ into $T_{f(x)} Y$. □

§1.2.1 The Inverse Function Theorem

Let $f : X \to Y$ be a smooth map between manifolds. We say that f is a **local diffeomorphism** at x if f maps a neighbourhood of x diffeomorphically onto a neighbourhood of $f(x)$.

Theorem 1.8 (Inverse Function Theorem)

Suppose that $f : X \to Y$ is a smooth map whose derivative df_x at the point x is an isomorphism. Then f is a local diffeomorphism at x.

Proof Sketch. This can be proven using the inverse function theorem for smooth functions between open sets of Euclidean space, as was shown in Analysis II. □

§1.3 Regular Values and Sard's Theorem

Let $f : X \to Y$ be a smooth map between manifolds. Let C be the set of all points $x \in X$ such that $df_x : T_x X \to T_{f(x)} Y$ is not surjective.

Definition 1.9

A point in C will be called a **critical point**. A point in $f(C)$ will be called a **critical value**. A point in the complement of $f(C)$ will be called a **regular value**.

Remark. Note that if $\dim X < \dim Y$, then $C = X$ and the preimage of a regular value is the empty set.

Theorem 1.10 (Preimage Theorem)

Let y be a regular value of $f : X \to Y$ with $\dim X \geq \dim Y$. Then the set $f^{-1}(y)$ is a submanifold of X with $\dim f^{-1}(y) = \dim X - \dim Y$.

Proof. Let $x \in f^{-1}(y)$. Since y is a regular value, the derivative df_x maps $T_x X$ onto $T_y Y$. The kernel of df_x is a subspace K of $T_x X$ of dimension $p := \dim X - \dim Y$. Suppose $X \subset \mathbb{R}^N$ and let $T : \mathbb{R}^N \to \mathbb{R}^p$ be any linear map such that $\operatorname{Ker}(T) \cap K = \{0\}$. Consider the map $F : X \to Y \times \mathbb{R}^p$ given by

$$F(z) = (f(z), T(z)).$$

The derivative of F is given by

$$dF_x(v) = (df_x(v), T(v))$$

which is clearly nonsingular by our choice of T. By the inverse function theorem, F is a local diffeomorphism at x, i.e. F maps some neighbourhood U of x diffeomorphically onto a neighbourhood V of $(y, T(x))$. Hence F maps $f^{-1}(y) \cap U$

diffeomorphically onto $(\{y\} \times \mathbb{R}^p) \cap V$ which proves that $f^{-1}(y)$ is a manifold with $\dim f^{-1}(y) = p$. $\square$

The case $\dim X = \dim Y$ is particularly important. The theorem says that if y is a regular value of f, then $f^{-1}(y)$ is a 0 -dimensional manifold, i.e. collection of points. If X is compact, this collection must be finite (why?) so we obtain:

Corollary 1.11

Let $f : X \to Y$ be a smooth map between manifolds of the same dimension. If X is compact and y is a regular value, $f^{-1}(y)$ consists of a finite set of points.

We can actually say a bit more:

Theorem 1.12 (Stack of Records Theorem)

Let $f : X \to Y$ be a smooth map between manifolds of the same dimension with X compact. Let y be a regular value of f and write $f^{-1}(y) = \{x_1, \ldots, x_k\}$. Then there exists a neighbourhood U of y in Y such that $f^{-1}(U)$ is a disjoint union $V_1 \cup \cdots \cup V_k$, where V_i is an open neighbourhood of x_i and f maps each V_i diffeomorphically onto U.

Proof. By the inverse function theorem we can pick disjoint neighbourhoods W_i of x_i such that f maps W_i diffeomorphically onto a neighbourhood of y. Observe that $f(X - \cup_i W_i)$ is a compact set which does not contain y. Now take

$$U := \cap_i f(W_i) - f(X - \cup_i W_i).$$

$\square$

If we let $\# f^{-1}(y)$ be the cardinality of $f^{-1}(y)$, the theorem implies that the function $y \mapsto \# f^{-1}(y)$ is locally constant as y ranges over regular values of f.

The Preimage theorem gives a particularly nice way of producing manifolds. For example, if we consider the map $f : \mathbb{R}^{n+1} \to \mathbb{R}$ given by $f(x) = |x|^2$ we can check easily that 1 is a regular value of f (do it!). Hence $S^n = f^{-1}(1)$ is a smooth manifold of dimension n. By switching our mental wavelength a little we can produce quite interesting examples as follows:

Example 1.13 (Orthogonal Group)

Let $O(n)$ be the group of orthogonal matrices of size $n \times n$, i.e., a matrix $A \in O(n)$ if and only if $AA^t = I$, where I is the identity matrix. The space of all $n \times n$ matrices $M(n)$ is just $\mathbb{R}^{n^2}$, so we can think of $O(n)$ as living inside $\mathbb{R}^{n^2}$. We will show that $O(n)$ is a manifold of dimension $\frac{n(n-1)}{2}$.

Let $S(n) \subset M(n)$ be the space of all symmetric matrices. Since it is a vector space, is clearly a submanifold of $M(n)$ and it has dimension $\frac{n(n+1)}{2}$.

Let $f : M(n) \to S(n)$ be the smooth map $f(A) = AA^t$. Since $O(n) = f^{-1}(I)$ it suffices to show that I is a regular value of f. We compute:

$$\begin{aligned} df_A(H) &= \lim_{s \to 0} \frac{f(A+sH) - f(A)}{s} \\ &= \lim_{s \to 0} \frac{(A+sH)(A+sH)^t - AA^t}{s} \\ &= HA^t + AH^t. \end{aligned}$$

Fix A with $AA^t = I$. We must prove that given any $C \in S(n) = T_I S(n)$ there is $H \in M(n) = T_A M(n)$ such that

$$df_A(H) = HA^t + AH^t = C.$$

Since C is symmetric, we can write $C = \frac{1}{2}C + \frac{1}{2}C^t$ so if we can solve $HA^t = \frac{1}{2}C$ we are done. Multiplying by A on the right and using $AA^t = I$ we obtain $H = \frac{1}{2}CA$ which solves $df_A(H) = C$. Thus I is a regular value of f.

Note that $O(n)$ is both a group and a manifold. In fact, the group operations are smooth, that is, the maps $(A, B) \mapsto AB$ and $A \mapsto A^{-1} = A^t$ are smooth. (Why?) A group that is manifold, and whose group operations are smooth, is called a Lie group.

§1.3.1 Sard's Theorem

The Preimage theorem raises the question: is it easy to find regular values? How are abundant are they?

Recall that an arbitrary set A in $\mathbb{R}^n$ has measure zero if it can be covered by a countable number of rectangular solids with arbitrary small total volume. In other words given $\varepsilon > 0$, there exists a countable collection $\{R_1, R_2, \dots\}$ of rectangular solids in $\mathbb{R}^n$, such that A is contained in $\cup_i R_i$ and

$$\sum_i \operatorname{Vol}(R_i) < \varepsilon.$$

Let X be manifold. An arbitrary subset $A \subset X$ has **measure zero** if, for every local parametrization ϕ of $X, \phi^{-1}(A)$ has measure zero in Euclidean space.

The following deep theorem, tells us that there are plenty of regular values.

Theorem 1.14 (Sard's Theorem, 1942)

The set of critical values of a smooth map $f : X \to Y$ has measure zero.

Since a set of measure zero cannot contain a non-empty open set we obtain:

Corollary 1.15

The regular values of any smooth map $f : X \to Y$ are dense in Y.

§1.3.2 Morse Functions

A particularly useful class of maps to a one-dimensional target are those whose critical points are as simple as possible.

Definition 1.16

Let X be a smooth manifold. A smooth function $f : X \to \mathbb{R}$ is a **Morse function** if at every critical point x of f the Hessian of f, computed in any chart, is non-singular. Equivalently, the critical points of f are non-degenerate, and in particular isolated.

Sard's theorem is exactly what one needs to show that Morse functions are abundant: for $X \subseteq \mathbb{R}^N$ and almost every $a \in \mathbb{R}^N$, the function $f_a(x) = f(x) + \langle a, x \rangle$ is Morse. This, and the Morse lemma (near a non-degenerate critical point one can choose coordinates in which f is a quadratic form), are developed on the first example sheet; they are the starting point of Morse theory, which reconstructs the topology of X from the critical points of a single function on it.

§1.4 Transversality

We know that the solutions of the equation $f(x) = y$ form a smooth manifold, provided that y is a regular value of $f : X \to Y$. Suppose now that we replace y by a submanifold $Z \subset Y$ and we ask, when is $f^{-1}(Z)$ a submanifold of X ?

Definition 1.17 (Transversal)

A smooth map $f : X \to Y$ is said to be **transversal** to a submanifold $Z \subset Y$ if for every $x \in f^{-1}(Z)$ we have

$$\operatorname{Image}(df_x) + T_{f(x)}Z = T_{f(x)}Y.$$

We write $f \pitchfork Z$.

Note that if $Z = \{y\}$, the notion of transversality reduces to the notion of regular value.

With this definition, we can now state a more general version of the Preimage theorem.

Theorem 1.18

If the smooth map $f : X \to Y$ is transversal to a submanifold $Z \subset Y$, then $f^{-1}(Z)$ is submanifold of X. Moreover, $f^{-1}(Z)$ and Z have the same codimension.

Proof (Non-examinable, sketch only). It is not hard to see that Z can be written in a neighbourhood of a point $y = f(x)$ as the zero set of a collection of functions $h_1, \dots, h_r$, where r is the codimension of Z in Y. Let $H := (h_1, \dots, h_r)$. Then near $x, f^{-1}(Z)$ is the zero set of the function $H \circ f$. Thus if $0 \in \mathbb{R}^r$ is a regular value of $H \circ f$ we are done. But $dH_y \circ df_x$ is surjective if and only if

$$\operatorname{Image}(df_x) + T_{f(x)}Z = T_{f(x)}Y$$

since $dH_y : T_yY \to \mathbb{R}^r$ is onto with kernel T_yZ. □

Example 1.19

Consider $f : \mathbb{R} \to \mathbb{R}^2$ given by $f(t) = (0, t)$ and let Z be the x axis in $\mathbb{R}^2$. Then f is transversal to Z, but for example the map $h(t) = \left(t, t^2\right)$ is not.

An important special case occurs when f is the inclusion of a submanifold X of Y and Z is another submanifold of Y. In this case the condition of transversality reduces to

$$T_x X + T_x Z = T_x Y$$

for every $x \in X \cap Z$. This condition is quite easy to visualize and when it holds we say that X and Z are transversal (we write $X \pitchfork Z$). We now have:

Theorem 1.20

The intersection of two transversal submanifolds of Y is a submanifold of codimension given by

$$\operatorname{codim}(X \cap Z) = \operatorname{codim} X + \operatorname{codim} Z.$$

One of the main virtues of the notion of transversality is its stability i.e. it survives after small perturbations of the map f. You can convince yourself of this property by looking at pictures of transversal submanifolds of Euclidean space. Another very important virtue is genericity: smooth maps may be deformed by arbitrary small amounts into a map that is transversal to Z. (This is non-obvious and we refer to Chapter 2 in the book by Guillemin and Pollack for details.)

§1.5 Manifolds with Boundary

Consider the closed half-space

$$\mathbb{H}^k := \left\{(x_1, \ldots, x_k) \in \mathbb{R}^k : x_k \geq 0\right\}.$$

The boundary $\partial \mathbb{H}^n$ is defined to be the hyperplane $x_k = 0$ in $\mathbb{R}^k$.

Definition 1.21

A subset $X \subset \mathbb{R}^N$ is called a smooth k-manifold with boundary if each $x \in X$ has a neighbourhood diffeomorphic to an open set in $\mathbb{H}^k$. As before, such a diffeomorphism is called a chart on X. The boundary of X, denoted ∂X, is given by the set of points that belong to the image of $\partial \mathbb{H}^k$ under some local parametrization. Its complement is called the interior of X, $\operatorname{Int}(X) = X - \partial X$.

Remark (Warning)**.** Do not confuse the boundary or interior of X as defined above with the topological notions of interior and boundary as a subset of $\mathbb{R}^N$.

The tangent space is defined as before, so that $T_x X$ is a k-dimensional vector space even for points $x \in \partial X$. The interior of X is a k-manifold without boundary and ∂X is a manifold without boundary of dimension $k-1$ (this requires a proof!).

Here is an easy way of generating examples.

Lemma 1.22

Let X be a manifold without boundary and let $f : X \to \mathbb{R}$ be a smooth function with 0 as a regular value. Then the subset $\{x \in X : f(x) \geq 0\}$ is a smooth manifold with boundary equal to $f^{-1}(0)$.

Proof. The set where $f > 0$ is open in X and is therefore a submanifold of the same dimension as X. For a point $x \in X$ with $f(x) = 0$, the same proof as in Theorem 1.10 shows that x has a neighbourhood diffeomorphic to a neighbourhood of a point in $\mathbb{H}^k$. □

As an easy application of the lemma, consider the unit ball B^k given by all $x \in \mathbb{R}^k$ such that $|x| \leq 1$. By considering the function $f(x) = 1 - |x|^2$ it follows that B^k is a smooth manifold with boundary S^{k-1}.

Theorem 1.23

Let $f : X \to Y$ be a smooth map from an m-manifold with boundary to an n-manifold, where $m > n$. If y is a regular value, both for f and for the restriction of f to ∂X, then $f^{-1}(y)$ is a smooth $(m-n)$-manifold with boundary equal to $f^{-1}(y) \cap \partial X$

Proof. Recall that being a submanifold is a local property so without loss of generality we can suppose that $f : \mathbb{H}^m \to \mathbb{R}^n$ with $y \in \mathbb{R}^n$ a regular value. Consider $z \in f^{-1}(y)$. If z belongs to the interior of $\mathbb{H}^m$, then as in Theorem 1.10, $f^{-1}(y)$ is a smooth manifold near z.

Let z be now in $\partial\mathbb{H}^m$. Since f is smooth, there is a neighbourhood U of z in $\mathbb{R}^m$ and a smooth map $F : U \to \mathbb{R}^n$ such that F restricted to $U \cap \mathbb{H}^m$ is f. By shrinking U if necessary we can assume that F has no critical points in U (why?). Hence $F^{-1}(y)$ is a smooth manifold of dimension $m - n$.

Let $\pi : F^{-1}(y) \to \mathbb{R}$ be the projection $(x_1, \ldots, x_m) \mapsto x_m$. Observe that the tangent space of $F^{-1}(y)$ at a point $x \in \pi^{-1}(0)$ is equal to the kernel of

$$dF_x = df_x : \mathbb{R}^m \to \mathbb{R}^n.$$

Hence 0 must be a regular value of π since we are assuming that y is a regular value of f restricted to $\partial\mathbb{H}^m$.

But $F^{-1}(y) \cap \mathbb{H}^m = f^{-1}(y) \cap U$ is the set of all $x \in F^{-1}(y)$ with $\pi(x) \geq 0$ and by Lemma 1.22, is a smooth manifold with boundary equal to $\pi^{-1}(0)$. □

It is not hard to guess what is the appropriate version of this theorem for the more general case of a map $f : X \to Y$ and a submanifold Z of Y. Suppose X has boundary, but Y and Z are boundaryless. The next theorem is stated without proof.

Theorem 1.24

Suppose that both $f : X \to Y$ and $f \mid \partial X : \partial X \to Y$ are transversal to Z. Then $f^{-1}(Z)$ is a manifold with boundary given by $f^{-1}(Z) \cap \partial X$ and codimension equal to the codimension of Z.

§1.6 Degree Modulo 2

Let X be a smooth boundaryless manifold. Then $X \times [0,1]$ is a manifold with boundary $\partial X = X \times \{0\} \cup X \times \{1\}$.

Definition 1.25

Two maps $f, g : X \to Y$ are called **smoothly homotopic** if there exists a smooth map $F : X \times [0,1] \to Y$ with $F(x,0) = f(x)$ and $F(x,1) = g(x)$ for all $x \in X$. The map F is called a **smooth homotopy** between f and g. The relation of smooth homotopy is an equivalence relation. (Why?) The equivalence class to which a map belongs is its **homotopy** class. Let $f_t : X \to Y$ be the 1-parameter family of maps given by $f_t(x) = F(x,t)$.

Definition 1.26

The diffeomorphism f is **smoothly isotopic** to g if there exists a smooth homotopy $F : X \times [0,1] \to Y$ from f to g such that for each $t \in [0,1]$, f_t maps X diffeomorphically onto Y.

Lemma 1.27 (Homotopy Lemma)

Let $f, g : X \to Y$ be smooth maps which are smoothly homotopic. Suppose X is compact, has the same dimension as Y and $\partial X = \emptyset$. If y is a regular value for both f and g, then

$$\# f^{-1}(y) = \# g^{-1}(y) \pmod 2.$$

Proof. The proof relies on the following important fact, the 'Classification of 1-manifolds': Every compact connected 1-manifold is diffeomorphic to $[0,1]$ or S^1.

As every compact manifold is the disjoint union of finitely many compact connected manifolds we have that the boundary of any compact 1-manifold consists of an even number of points.

Let $F : X \times [0,1] \to Y$ be a smooth homotopy between f and g. Assume for the time being that y is also a regular value for F. Then $F^{-1}(y)$ is a compact 1-manifold with boundary equal to

$$F^{-1}(y) \cap (X \times \{0\} \cup X \times \{1\}) = f^{-1}(y) \times \{0\} \cup g^{-1}(y) \times \{1\}.$$

Thus the cardinality of the boundary of $F^{-1}(y)$ is just $\# f^{-1}(y) + \# g^{-1}(y)$. By the corollary above

$$\# f^{-1}(y) = \# g^{-1}(y) (\mathrm{mod} 2).$$

If y is not a regular value of F we proceed as follows. From Theorem 1.12 we know that $w \mapsto \# f^{-1}(w), w \mapsto \# g^{-1}(w)$ are locally constant as w ranges over regular values. Thus there are neighbourhoods V and W of y, consisting of regular values of f and g respectively for which

$$\# f^{-1}(w) = \# f^{-1}(y)$$

for all $w \in V$, and

$$\#g^{-1}(w) = \#g^{-1}(y)$$

for all $w \in W$. By Sard's theorem we can choose a regular value z of F in $V \cap W$. Then

$$\#f^{-1}(y) = \#f^{-1}(z) = \#g^{-1}(z) = \#g^{-1}(y)(\text{mod}2)$$

as desired. □

Lemma 1.28 (Homogeneity Lemma)

Let X be a smooth connected manifold, possibly with boundary. Let y and z be points in $\text{Int}(X)$. Then there exists a diffeomorphism $h : X \to X$ smoothly isotopic to the identity such that $h(y) = z$.

Proof. Let us call two points y and z "isotopic" if there exists a diffeomorphism h isotopic to the identity that maps y to z. It is evident that this is an equivalence relation. Since X is connected, it suffices to show that each equivalence class is a open set.

To prove that equivalence classes are open we will construct an isotopy h_t of $\mathbb{R}^k$ such that h_0 is the identity, each h_t is the identity outside the open unit ball around the origin, and $h_1(0)$ is any specified point in the open unit ball. This will suffice, since we can now parametrize a small neighbourhood of y in $\text{Int}(X)$ and use h_t to construct an isotopy that will move y to any point nearby.

Let $\varphi : \mathbb{R}^k \to \mathbb{R}$ be a smooth function such that $\varphi(x) > 0$ for $|x| < 1$ and $\varphi(x) = 0$ for $|x| \geq 1$ (why do they exist?). Given a unit vector $u \in \mathbb{R}^k$ consider the ordinary differential equation in $\mathbb{R}^k$ given by

$$\frac{dx}{dt} = u\varphi(x).$$

Let $F_t : \mathbb{R}^k \to \mathbb{R}^k$ be the flow of this differential equation, i.e., for each $x \in \mathbb{R}^k$, the curve $t \mapsto F_t(x)$ is the unique solution passing through x. Standard theorems on differential equations tell us that:

1. F_t is defined for all $x \in \mathbb{R}^k$ and $t \in \mathbb{R}$ and smooth;
2. F_0 is the identity;
3. $F_{t+s} = F_t \circ F_s$.

Clearly for each t, F_t leaves all points outside the unit ball fixed. For appropriate choices of u and t, F_t will map the origin to any point in the open unit ball. (I may replace this proof by the one given by Guillemin and Pollack on page 142 of their book which does not use flows.) □

In what follows suppose that X is compact and without boundary and Y is connected and with the same dimension as X. Let $f : X \to Y$ be a smooth map.

Theorem 1.29 (Degree mod 2)

If y and z are regular values of f then

$$\#f^{-1}(y) = \#f^{-1}(z)(\text{mod}2).$$

This common residue class is called the **degree mod 2** of f (denoted $\deg_2(f)$) and only depends on the homotopy class of f.

Proof. Given y and z, let h be the diffeomorphism smoothly isotopic to the identity such that $h(y) = z$ given by the Homogeneity Lemma. Observe that z is also a regular value of $h \circ f$. Since $h \circ f$ is homotopic to f, the Homotopy Lemma tells us that

$$\#(h \circ f)^{-1}(z) = \#f^{-1}(z) \pmod 2.$$

But $(h \circ f)^{-1}(z) = f^{-1}\left(h^{-1}(z)\right) = f^{-1}(y)$ and therefore

$$\#f^{-1}(y) = \#f^{-1}(z) \pmod 2$$

as desired. Let g be smoothly homotopic to f. By Sard's theorem, there is a point $y \in Y$ which is a regular value for both f and g (why?). By the Homotopy Lemma

$$\deg_2(f) = \#f^{-1}(y)(\text{mod}2) = \#g^{-1}(y)(\text{mod}2) = \deg_2(g)$$

which completes the proof. □

Example 1.30

The identity map of a compact boundaryless manifold X has $\deg_2 = 1$, since any point has exactly one preimage, while a constant map has $\deg_2 = 0$, since a point other than the image value has none. Since $\deg_2$ is a homotopy invariant, the identity is never homotopic to a constant map.

This innocuous observation has real force.

Corollary 1.31

There is no smooth map $f : B^{k+1} \to S^k$ restricting to the identity on $S^k = \partial B^{k+1}$; that is, the sphere is not a smooth retract of the ball.

Proof. If such an f existed, then $F : S^k \times [0, 1] \to S^k$ given by $F(x, t) = f(tx)$ would be a smooth homotopy between the constant map $x \mapsto f(0)$ and the identity, contradicting the example above. □

The previous example yields the smooth version of a famous fixed point theorem.

Theorem 1.32 (Smooth Brouwer Fixed Point Theorem)

Any smooth map $f : B^k \to B^k$ has a fixed point.

Proof. Suppose f has no fixed point. We will construct a map $g : B^k \to S^{k-1}$ which restricts to the identity on S^{k-1} which contradicts Corollary 1.31. For $x \in B^k$,

> let $g(x) \in S^{k-1}$ be the point where the line segment starting at $f(x)$ and passing through x hits the boundary. One can write a formula for g to show smoothness. $\square$

In fact, any continuous map $f : B^k \to B^k$ has a fixed point (Brouwer fixed point theorem). One can prove this using the smooth version approximating f by polynomials. Indeed, by the Weierstrass approximation theorem, given $\varepsilon > 0$, there is a polynomial P such that $|f(x) - P(x)| < \varepsilon$ for all $x \in B^k$. Set $Q(x) = P(x)/(1+\varepsilon)$. Now Q maps B^k into B^k and $|f(x) - Q(x)| < 2\varepsilon$ for all $x \in B^k$. Suppose $f(x) \neq x$ for all x. Then the continuous function $|f(x) - x|$ must take a positive minimum τ on B^k. If now let $\varepsilon = \tau/2$ the Q from above does not have a fixed point, which contradicts the smooth Brouwer fixed point theorem.

§1.6.1 Intersection Numbers Modulo 2

What we said above about degree modulo 2 can be generalized to the case of a smooth map $f : X \to Y$ and Z a submanifold of Y. Suppose that:

1. X is compact without boundary;
2. Z is closed and without boundary;
3. $f \pitchfork Z$
4. $\dim X + \dim Z = \dim Y$.

Under these conditions $f^{-1}(Z)$ is a closed 0-dimensional submanifold of X and hence it consists of finitely many points. Define the **mod 2 intersection number** of the map f with $Z, I_2(f, Z)$, to be the cardinality of $f^{-1}(Z)$ modulo 2. An analogue of the Homotopy Lemma (for intersection numbers) also holds here: if f_0 and f_1 are maps transversal to Z and homotopic, then $I_2(f_0, Z) = I_2(f_1, Z)$. In fact, we can define $I_2(f, Z)$ for a map f which is not necessarily transversal to Z. Using that transversality is generic, we can find a map g homotopic to f such that $g \pitchfork Z$ and we now set $I_2(f, Z) := I_2(g, Z)$. It does not matter which g we choose as long as g is homotopic to f, thanks to the Homotopy Lemma for intersection numbers.

Let us have a closer look at the important special case in which X itself is a compact submanifold of Y and Z a closed submanifold of complementary dimension. In this case f is the inclusion map $X \hookrightarrow Y$, and if $X \pitchfork Z, I_2(X, Z) := I_2(f, Z)$ is just $\#(X \cap Z) \bmod 2$. If $I_2(X, Z) \neq 0$, it means that no matter how X is deformed, we cannot move it away completely from Z.

Example 1.33

Let Y be the 2-torus $S^1 \times S^1$. Let $X = S^1 \times \{1\}$ and $Z = \{1\} \times S^1$. Then $I_2(X, Z) = 1$ and we cannot smoothly pull the circles apart. When $\dim X = \frac{1}{2} \dim Y$, we can consider $I_2(X, X)$ the self-intersection number modulo 2. If X is the central curve in a Möbius band, then $I_2(X, X) = 1$.

§1.7 Abstract Manifolds and Whitney's Embedding Theorem

One can actually define manifolds without making any reference to the ambient space $\mathbb{R}^N$. You will study manifolds in this more abstract setting in Part III and in the Riemann Surfaces course. In any case, here is the definition:

Definition 1.34

An n-dimensional smooth manifold is a second countable Hausdorff space X together with a collection of maps called **charts** such that:

1. a chart is a homeomorphism $\phi : U \to \phi(U)$, where U is open in X and $\phi(U)$ is open in $\mathbb{R}^n$
2. each point $x \in X$ belongs to the domain of some chart;
3. for charts $\phi : U \to \phi(U) \subset \mathbb{R}^n$ and $\psi : V \to \phi(V) \subset \mathbb{R}^n$, the map $\phi \circ \psi^{-1} : \psi(U \cap V) \to \phi(U \cap V)$ is smooth;
4. the collection of charts is maximal with respect to the properties above.

A set of charts satisfying the first three properties is called an **atlas**. An atlas can always be enlarged uniquely to give a maximal atlas as in the definition. If in the definition we require the maps $\phi \circ \psi^{-1}$ just to be of class $C^k (k \geq 1)$ then we say that we have a manifold of class C^k. (Recall that a map is of class C^k if it has continuous partial derivatives up to order k.)

The definition is set up so that it is plain how to define a smooth map between manifolds. An **immersion** is a smooth map $f : X \to Y$ such that df_x is injective for all $x \in X$. A **submersion** is a smooth map $f : X \to Y$ such that df_x is surjective for all $x \in X$. An **embedding** is a smooth map $f : X \to Y$ which is an immersion and a homeomorphism onto its image.

The next theorem tells us that we did not lose much by restricting out attention to manifolds as subsets of Euclidean space.

Theorem 1.35 (Whitney's embedding theorem)

A smooth n-manifold X can be embedded into $\mathbb{R}^{2n+1}$.

The proof of this theorem can be found in most books about manifolds. In fact, Whitney proved the much harder result that X can be embedded in $\mathbb{R}^{2n}$.

§2 Curves in $\mathbb{R}^3$

Before we look at surfaces it is worth spending some time on the simplest interesting manifolds of all: one-dimensional ones, that is, curves. Curves are simple enough that we can say everything about them, and the story we will tell – attach a moving frame to the object, differentiate it, and read off the geometry from the coefficients that appear – is exactly the story we will tell for surfaces later, but with all the technicalities stripped away.

The punchline is that a curve in $\mathbb{R}^3$ has precisely two local invariants, its *curvature* and its *torsion*, and that these determine the curve completely up to a rigid motion of space. Curvature says how fast the curve is turning, and torsion says how fast it is failing to be planar.

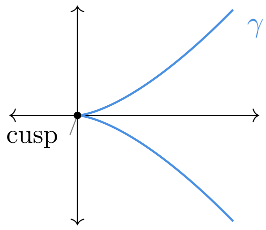
§2.1 Regular Curves and Arc Length

We should first be careful about what we mean by a curve. There is a difference between a curve as a *map* and a curve as a *subset* of $\mathbb{R}^3$, and it is the map we want.

Definition 2.1 (Smooth Curve)

A **smooth curve** in $\mathbb{R}^n$ is a smooth map $\gamma : (a, b) \to \mathbb{R}^n$, where $-\infty \leq a < b \leq \infty$. Its image $\gamma((a, b))$ is called the **trace** of the curve. We say γ is **regular** if $\gamma'(t) \neq 0$ for all $t \in (a, b)$.

Regularity is exactly the condition that γ is an immersion, and it is what stops the trace from having corners. It really is needed: the map $\gamma(t) = (t^2, t^3)$ is a perfectly smooth map $\mathbb{R} \to \mathbb{R}^2$, but $\gamma'(0) = 0$ and the trace has a cusp at the origin.



For a regular curve there is a preferred way to parametrise: by the distance travelled along it.

Definition 2.2 (Arc Length)

Let $\gamma : (a, b) \to \mathbb{R}^n$ be a smooth curve and fix $t_0 \in (a, b)$. The **arc length** of γ starting at t_0 is

$$s(t) = \int_{t_0}^{t} \|\gamma'(u)\| \, \mathrm{d}u.$$

We say γ is parametrised by arc length, or has **unit speed**, if $\|\gamma'(t)\| = 1$ for all t.

Lemma 2.3 (Unit Speed Reparametrisation)

Every regular curve admits a unit speed reparametrisation. That is, if $\gamma : (a, b) \to \mathbb{R}^n$ is regular then there is a diffeomorphism φ from an interval (c, d) onto (a, b) such that $\tilde{\gamma} = \gamma \circ \varphi$ has unit speed.

Proof. With s as above we have $s'(t) = \|\gamma'(t)\| > 0$ by regularity, so s is strictly increasing and smooth, hence a diffeomorphism from (a, b) onto an interval (c, d); its inverse $\varphi = s^{-1}$ is smooth by the inverse function theorem. Setting $\tilde{\gamma} = \gamma \circ \varphi$, the chain rule gives

$$\tilde{\gamma}'(s) = \gamma'(\varphi(s))\varphi'(s) = \frac{\gamma'(\varphi(s))}{\|\gamma'(\varphi(s))\|},$$

which has unit norm. $\square$

From now on, when a curve is unit speed we will write s for the parameter and use a dot for $\mathrm{d}/\mathrm{d}s$; for a general parameter we write t and use a prime. Everything we define will be checked to be independent of the parametrisation, so we may always compute in whichever one is convenient.

§2.2 Curvature and the Frenet–Serret Formulae

Let γ be a unit speed curve. Its velocity $T = \dot{\gamma}$ is a unit vector, the **unit tangent**. Differentiating $\langle T, T\rangle = 1$ gives

$$\langle \dot{T}, T\rangle = 0,$$

so the acceleration of a unit speed curve is always perpendicular to its velocity. All the acceleration goes into turning, none into speeding up, and the size of that turning is our first invariant.

Definition 2.4 (Curvature)

The **curvature** of a unit speed curve γ is

$$\kappa(s) = \|\ddot{\gamma}(s)\| = \left\|\dot{T}(s)\right\| \geq 0.$$

For a curve given in an arbitrary regular parametrisation it would be painful to have to find the arc length reparametrisation before computing anything, so it is worth recording a direct formula.

Proposition 2.5

If γ is a regular curve in $\mathbb{R}^3$ with arbitrary parameter t, then

$$\kappa = \frac{\|\gamma' \times \gamma''\|}{\|\gamma'\|^3}.$$

Proof. Write $v = \|\gamma'\|$ and let s be arc length, so $\mathrm{d}s/\mathrm{d}t = v$ and $\gamma' = vT$. Differentiating,

$$\gamma'' = v'T + v\frac{\mathrm{d}T}{\mathrm{d}t} = v'T + v^2\dot{T}.$$

Since $T \times T = 0$ and $\dot{T} \perp T$ with $\left\|\dot{T}\right\| = \kappa$, we get

$$\gamma' \times \gamma'' = vT \times (v'T + v^2\dot{T}) = v^3(T \times \dot{T}),$$

and $\left\|T \times \dot{T}\right\| = \|T\| \left\|\dot{T}\right\| = \kappa$ as the two are orthogonal. Rearranging gives the result. □

Example 2.6 (Lines and Circles)

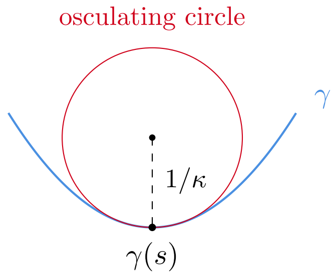
A straight line $\gamma(s) = p + sq$ with $\|q\| = 1$ has $\ddot{\gamma} = 0$, so $\kappa \equiv 0$. Conversely if $\kappa \equiv 0$ then $\ddot{\gamma} \equiv 0$, so $\gamma(s) = p + sq$ for constants p, q: a unit speed curve is a straight line if and only if its curvature vanishes identically.

The circle of radius a traversed at unit speed is $\gamma(s) = (a\cos(s/a), a\sin(s/a))$, and

$$\ddot{\gamma}(s) = -\tfrac{1}{a}\left(\cos(s/a), \sin(s/a)\right),$$

so $\kappa \equiv 1/a$. Small circles are very curved, large circles are nearly straight, which is as it should be.

The circle example gives the standard geometric interpretation of κ: at a point where $\kappa(s) > 0$ there is a unique circle which agrees with γ to second order at $\gamma(s)$, namely the circle of radius $1/a = 1/\kappa(s)$ lying in the plane spanned by $\dot{\gamma}$ and $\ddot{\gamma}$ and centred on the concave side. This is the **osculating circle**, and $1/\kappa(s)$ is the **radius of curvature**.



Now suppose $\kappa(s) > 0$ everywhere. Then $\dot{T}/\kappa$ is a unit vector orthogonal to T, and we can complete to an orthonormal basis of $\mathbb{R}^3$.

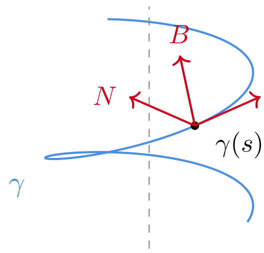
Definition 2.7 (Frenet Frame)

Let γ be a unit speed curve in $\mathbb{R}^3$ with $\kappa > 0$ everywhere. The **principal normal** and **binormal** are

$$N = \frac{\dot{T}}{\kappa}, \qquad B = T \times N.$$

The triple (T, N, B) is a right-handed orthonormal basis of $\mathbb{R}^3$ at each point, called the **Frenet frame** of γ.

The plane spanned by T and N is the **osculating plane**; it is the plane the curve is doing its best to lie in. The whole of the local theory of curves comes from differentiating the frame and expressing the answer back in terms of the frame.



Theorem 2.8 (Frenet–Serret Formulae)

Let γ be a unit speed curve in $\mathbb{R}^3$ with $\kappa > 0$. Then there is a smooth function τ, the **torsion** of γ, such that

$$\begin{aligned} \dot{T} &= \kappa N, \\ \dot{N} &= -\kappa T + \tau B, \\ \dot{B} &= -\tau N. \end{aligned}$$

Equivalently, writing the frame as the rows of a matrix,

$$\frac{\mathrm{d}}{\mathrm{d}s}\begin{pmatrix} T \\ N \\ B \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix}\begin{pmatrix} T \\ N \\ B \end{pmatrix}.$$

Proof. Since (T, N, B) is an orthonormal basis at each point we may expand each derivative in it, so we must simply identify the nine coefficients $\langle \dot{T}, T\rangle$, $\langle \dot{T}, N\rangle$ and so on. Differentiating the six relations $\langle X, Y\rangle = \delta_{XY}$ for $X, Y \in \{T, N, B\}$ gives

$$\langle \dot{X}, Y\rangle + \langle X, \dot{Y}\rangle = 0,$$

so the matrix of coefficients is antisymmetric and only three entries need to be found. By definition $\dot{T} = \kappa N$, which gives the first row and, by antisymmetry, the $\langle \dot{N}, T\rangle = -\kappa$ entry and $\langle \dot{B}, T\rangle = 0$. Defining $\tau := \langle \dot{N}, B\rangle$ fills in the rest, and τ is smooth since N and B are. $\square$

Remark. Note the sign convention: some books define τ with the opposite sign. Ours is the one for which the right-handed helix (below) has positive torsion.

Torsion measures the rate at which the osculating plane rotates, or equivalently the rate at which the curve pulls out of that plane. So it should vanish exactly for planar curves, and it does.

Proposition 2.9

A unit speed curve γ in $\mathbb{R}^3$ with $\kappa > 0$ has $\tau \equiv 0$ if and only if its trace is contained in a plane.

Proof. If $\tau \equiv 0$ then $\dot{B} = -\tau N = 0$, so B is a constant vector b. Fix s_0 and consider $f(s) = \langle \gamma(s) - \gamma(s_0), b\rangle$. Then $f(s_0) = 0$ and $\dot{f} = \langle T, b\rangle = 0$, so $f \equiv 0$ and γ lies in the plane through $\gamma(s_0)$ with normal b.

Conversely, suppose γ lies in a plane with unit normal b. Differentiating $\langle \gamma - \gamma(s_0), b\rangle = 0$ twice gives $\langle T, b\rangle = 0$ and $\langle \dot{T}, b\rangle = 0$, so b is orthogonal to both T and N and hence $B = \pm b$ is constant. Therefore $\dot{B} = 0$ and, since $N \neq 0$, $\tau \equiv 0$. $\square$

As with curvature, it is convenient to have a formula valid in an arbitrary parametrisation.

Proposition 2.10

If γ is a regular curve in $\mathbb{R}^3$ with $\gamma' \times \gamma'' \neq 0$, then

$$\tau = \frac{\langle \gamma' \times \gamma'', \gamma'''\rangle}{\|\gamma' \times \gamma''\|^2}.$$

Proof. Again put $v = \|\gamma'\|$. We computed $\gamma' = vT$ and $\gamma'' = v'T + v^2\kappa N$, so differentiating once more and collecting only the terms we need,

$$\gamma''' = (\text{multiple of } T) + (\text{multiple of } N) + v^3\kappa\tau B,$$

using $\dot{N} = -\kappa T + \tau B$. From the proof of the curvature formula, $\gamma' \times \gamma'' = v^3 \kappa B$. Hence

$$\langle \gamma' \times \gamma'', \gamma''' \rangle = v^3\kappa \cdot v^3 \kappa\tau = v^6\kappa^2\tau = \left\|\gamma' \times \gamma''\right\|^2 \tau,$$

as claimed. □

Example 2.11 (The Helix)

Let $a > 0$, $b \neq 0$ and consider the helix $\gamma(t) = (a\cos t, a\sin t, bt)$. We have

$$\gamma' = (-a\sin t, a\cos t, b), \quad \gamma'' = (-a\cos t, -a\sin t, 0), \quad \gamma''' = (a\sin t, -a\cos t, 0),$$

so $\|\gamma'\| = \sqrt{a^2+b^2}$ and

$$\gamma' \times \gamma'' = (ab\sin t, -ab\cos t, a^2), \qquad \left\|\gamma' \times \gamma''\right\| = a\sqrt{a^2+b^2}.$$

Therefore

$$\kappa = \frac{a\sqrt{a^2+b^2}}{(a^2+b^2)^{3/2}} = \frac{a}{a^2+b^2}, \qquad \tau = \frac{\langle \gamma' \times \gamma'', \gamma''' \rangle}{a^2(a^2+b^2)} = \frac{a^2 b}{a^2(a^2+b^2)} = \frac{b}{a^2+b^2}.$$

Both are constant. Letting $b \to 0$ we recover the circle of radius a, with $\kappa = 1/a$ and $\tau = 0$.

The helix is in fact the *only* curve with constant curvature and torsion, which is a consequence of the following, the fundamental theorem of the local theory of curves. It says that (κ, τ) is a complete set of local invariants.

Theorem 2.12 (Fundamental Theorem of Curve Theory)

Let $\kappa : (a,b) \to (0,\infty)$ and $\tau : (a,b) \to \mathbb{R}$ be smooth. Then there is a unit speed curve $\gamma : (a,b) \to \mathbb{R}^3$ with curvature κ and torsion τ, and any two such curves differ by a rigid motion of $\mathbb{R}^3$ (that is, by a map $x \mapsto Ax + c$ with $A \in \mathrm{SO}(3)$).

Proof. Existence. The Frenet–Serret formulae are a linear system of ordinary differential equations for the nine components of (T, N, B), with smooth coefficients. By Picard's theorem there is a unique solution on all of (a,b) with any prescribed initial value at s_0; take the initial value to be the standard basis (e_1, e_2, e_3).

We must check the solution stays orthonormal. Consider the six functions $\langle X, Y \rangle$ for $X, Y \in \{T, N, B\}$. Differentiating and substituting the equations shows these six functions themselves satisfy a linear system of ODEs, one solution of which is the constant δ_{XY}; by uniqueness, and since they agree at s_0, they are equal to it for all s. So (T, N, B) remains orthonormal, and by continuity it stays right-handed.

Now set $\gamma(s) = \int_{s_0}^{s} T(u)\,\mathrm{d}u$. Then $\dot{\gamma} = T$ has unit length, $\ddot{\gamma} = \dot{T} = \kappa N$ has norm κ since $\kappa > 0$, so the curvature of γ is κ and its principal normal is N; the binormal is then $T \times N = B$ (right-handedness) and $\langle \dot{N}, B \rangle = \tau$, as required.

Uniqueness. Suppose γ_1, γ_2 both have curvature κ and torsion τ, with frames (T_i, N_i, B_i). The two frames at s_0 are right-handed orthonormal bases, so there is a unique $A \in \mathrm{SO}(3)$ carrying the first to the second; replacing γ_1 by $A\gamma_1 + (\gamma_2(s_0) - A\gamma_1(s_0))$ – a rigid motion, which changes neither κ nor τ – we may assume the two curves and their frames agree at s_0. Consider

$$f(s) = \|T_1 - T_2\|^2 + \|N_1 - N_2\|^2 + \|B_1 - B_2\|^2 .$$

Differentiating and using Frenet–Serret for both curves, every term cancels: for instance

$$\tfrac{1}{2}\frac{\mathrm{d}}{\mathrm{d}s}\|T_1 - T_2\|^2 = \langle \kappa N_1 - \kappa N_2, T_1 - T_2\rangle,$$

and adding the analogous expressions for N and B the κ and τ terms cancel in pairs, giving $\dot{f} = 0$. Since $f(s_0) = 0$ and $f \geq 0$ we get $f \equiv 0$, so $T_1 = T_2$ everywhere and hence $\gamma_1 - \gamma_2$ is constant; as they agree at s_0 they are equal. □

Corollary 2.13

A unit speed curve in $\mathbb{R}^3$ with constant curvature $\kappa_0 > 0$ and constant torsion τ_0 is a circular helix (a circle if $\tau_0 = 0$).

Proof. Solve $a/(a^2 + b^2) = \kappa_0$, $b/(a^2 + b^2) = \tau_0$ for $a > 0$ and b: taking $a = \kappa_0/(\kappa_0^2 + \tau_0^2)$ and $b = \tau_0/(\kappa_0^2 + \tau_0^2)$ works. The corresponding helix, reparametrised at unit speed, has the given curvature and torsion, so by uniqueness our curve is a rigid motion of it. □

§2.3 Plane Curves

For curves in the plane the theory simplifies, but in one respect it improves: a plane curve has a preferred normal direction available whether or not κ vanishes, and so we can attach a *sign* to the curvature.

Definition 2.14 (Signed Curvature)

Let $\gamma : (a, b) \to \mathbb{R}^2$ be a unit speed plane curve with tangent $T = \dot{\gamma}$. Let N_s be T rotated anticlockwise by $\pi/2$, so that (T, N_s) is a positively oriented orthonormal basis of $\mathbb{R}^2$. The **signed curvature** κ_s is defined by

$$\dot{T} = \kappa_s N_s.$$

This makes sense because $\dot{T} \perp T$, so $\dot{T}$ is automatically a multiple of N_s. Differentiating $\langle N_s, N_s\rangle = 1$ and $\langle T, N_s\rangle = 0$ gives $\dot{N}_s = -\kappa_s T$, so we have a two-dimensional Frenet system

$$\dot{T} = \kappa_s N_s, \qquad \dot{N}_s = -\kappa_s T,$$

and clearly $\kappa = |\kappa_s|$. The sign records which way the curve is turning: anticlockwise where $\kappa_s > 0$, clockwise where $\kappa_s < 0$. Reversing the orientation of the curve reverses the sign.

Since T is a unit vector in the plane, we can write it as an angle, and it is worth knowing that the angle can be chosen smoothly.

Lemma 2.15 (Turning Angle)

Let $\gamma : (a, b) \to \mathbb{R}^2$ be a unit speed curve. Then there is a smooth function θ :

$(a, b) \to \mathbb{R}$ with

$$T(s) = (\cos\theta(s), \sin\theta(s)),$$

unique up to adding a constant in $2\pi\mathbb{Z}$, and $\dot\theta = \kappa_s$.

Proof. Write $T = (T_1, T_2)$. Pick θ_0 with $T(s_0) = (\cos\theta_0, \sin\theta_0)$ and define

$$\theta(s) = \theta_0 + \int_{s_0}^{s} \left(T_1\dot T_2 - T_2\dot T_1\right)\,\mathrm{d}u,$$

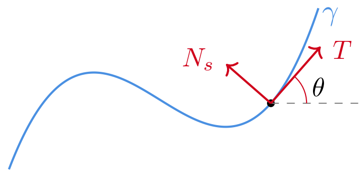
which is smooth. Using $T_1^2 + T_2^2 = 1$ one checks that both

$$\frac{\mathrm{d}}{\mathrm{d}s}(T_1\cos\theta + T_2\sin\theta) \quad\text{and}\quad \frac{\mathrm{d}}{\mathrm{d}s}(T_2\cos\theta - T_1\sin\theta)$$

vanish identically; evaluating at s_0 then gives $T_1\cos\theta + T_2\sin\theta \equiv 1$ and $T_2\cos\theta - T_1\sin\theta \equiv 0$, which forces $T = (\cos\theta, \sin\theta)$. Finally

$$\dot T = \dot\theta(-\sin\theta, \cos\theta) = \dot\theta N_s,$$

so $\dot\theta = \kappa_s$. Uniqueness is clear since two such angle functions differ by a continuous $2\pi\mathbb{Z}$-valued function. $\square$



Now let $\gamma : [0, L] \to \mathbb{R}^2$ be a **closed** unit speed curve, meaning that γ and all its derivatives agree at 0 and L (so that γ extends to an L-periodic smooth curve on $\mathbb{R}$). Then $T(0) = T(L)$, so $\theta(L) - \theta(0) \in 2\pi\mathbb{Z}$.

Definition 2.16 (Turning Number)
The **turning number** of a closed plane curve γ of length L is the integer

$$n(\gamma) = \frac{1}{2\pi}(\theta(L) - \theta(0)) = \frac{1}{2\pi}\int_0^L \kappa_s\,\mathrm{d}s.$$

The circle traversed anticlockwise has turning number 1, and traversed k times has turning number k; a figure eight has turning number 0. What is intuitively obvious but genuinely delicate to prove is that a curve which does not cross itself cannot do anything exotic.

Theorem 2.17 (Hopf's Umlaufsatz)
A simple closed plane curve has turning number ± 1, with the sign $+$ if it is traversed anticlockwise.

Proof Sketch. Translate so that γ lies in the upper half plane with $\gamma(0)$ on the x-axis, so the tangent at $s = 0$ is horizontal. On the triangle $\Delta = \{(s_1, s_2) : 0 \le s_1 \le$

$s_2 \leq L\}$ define the *secant map*

$$\Psi(s_1, s_2) = \frac{\gamma(s_2) - \gamma(s_1)}{\|\gamma(s_2) - \gamma(s_1)\|}$$

off the diagonal and the two corners, extended by $\Psi(s,s) = T(s)$ and $\Psi(0,L) = -T(0)$. Simplicity of γ is exactly what makes Ψ well defined and continuous on Δ. The turning number is the degree of Ψ restricted to the diagonal edge, and since Δ is convex we may deform that edge to the other two edges without changing the degree. Along those two edges the secant vector sweeps monotonically through a half turn each (the curve lies on one side of the x-axis), giving a total of one full turn. □

§2.4 The Isoperimetric Inequality

We now turn to two global results about plane curves: statements which involve the curve as a whole rather than a neighbourhood of a point. The first is the oldest theorem in the subject, going back to the legend of Dido: among all closed curves of a given length, the circle encloses the most area.

The proof is a beautiful piece of Fourier analysis, resting on the following inequality, which says that a mean-zero periodic function cannot oscillate slower than the fundamental frequency of its period.

Lemma 2.18 (Wirtinger's Inequality)
Let $f : \mathbb{R} \to \mathbb{R}$ be a C^1 function of period L with $\int_0^L f \, \mathrm{d}t = 0$. Then

$$\int_0^L |f'|^2 \, \mathrm{d}t \geq \frac{4\pi^2}{L^2} \int_0^L |f|^2 \, \mathrm{d}t,$$

with equality if and only if $f(t) = a\cos(2\pi t/L) + b\sin(2\pi t/L)$ for some constants a, b.

Proof. Write $f(t) = \sum_{n\in\mathbb{Z}} a_n e^{2\pi int/L}$ and $f'(t) = \sum_{n\in\mathbb{Z}} b_n e^{2\pi int/L}$ for the Fourier series of f and f'; these converge absolutely and uniformly since f is C^1. The hypothesis $\int_0^L f = 0$ says $a_0 = 0$, and periodicity of f gives $b_0 = \frac{1}{L}\int_0^L f' = 0$. Integrating by parts,

$$b_n = \frac{1}{L}\int_0^L f'(t) e^{-2\pi int/L} \, \mathrm{d}t = \frac{2\pi in}{L} \cdot \frac{1}{L}\int_0^L f(t) e^{-2\pi int/L} \, \mathrm{d}t = \frac{2\pi in}{L} a_n.$$

Now Parseval's identity gives

$$\int_0^L |f'|^2 \, \mathrm{d}t = L \sum_{n\neq 0} |b_n|^2 = \frac{4\pi^2}{L} \sum_{n\neq 0} n^2 |a_n|^2 \geq \frac{4\pi^2}{L} \sum_{n\neq 0} |a_n|^2 = \frac{4\pi^2}{L^2} \int_0^L |f|^2 \, \mathrm{d}t,$$

using $n^2 \geq 1$. Equality holds if and only if $a_n = 0$ for every $|n| \geq 2$, which is the stated form. □

Theorem 2.19 (Isoperimetric Inequality)
Let $\Omega \subseteq \mathbb{R}^2$ be a bounded domain whose boundary $\partial\Omega$ is a simple closed C^1 curve, of length L and enclosing area A. Then

$$L^2 \geq 4\pi A,$$

with equality if and only if Ω is a disc.

Proof. Parametrise $\partial\Omega$ by arc length $s \in [0, L]$, writing the position vector as $X(s) = (x(s), y(s))$, and let ν be the outward unit normal along $\partial\Omega$. Translating Ω changes neither L nor A, so we may assume $\int_0^L X \, \mathrm{d}s = 0$.

Applied to the position vector field $X(x, y) = (x, y)$, whose divergence is 2, the divergence theorem gives

$$2A = \int_\Omega \nabla \cdot X \, \mathrm{d}A = \int_{\partial\Omega} \langle X, \nu \rangle \, \mathrm{d}s \leq \int_0^L \|X\| \, \mathrm{d}s \leq \left(\int_0^L \|X\|^2 \, \mathrm{d}s \right)^{1/2} L^{1/2},$$

the last step by Cauchy–Schwarz. Now x and y are C^1 and L-periodic with vanishing mean, so Wirtinger's inequality applies to each, and since $(x')^2 + (y')^2 = 1$,

$$\int_0^L \|X\|^2 \, \mathrm{d}s \leq \frac{L^2}{4\pi^2} \int_0^L \left((x')^2 + (y')^2 \right) \, \mathrm{d}s = \frac{L^3}{4\pi^2}.$$

Substituting, $2A \leq (L^{3/2}/2\pi) \cdot L^{1/2} = L^2/2\pi$, which is the inequality.

If equality holds then in particular equality holds in Cauchy–Schwarz, so $\|X\|$ is constant along $\partial\Omega$: every boundary point is the same distance from the origin, and $\partial\Omega$ is a circle. $\square$

Remark. The same argument in fact shows a little more. If we had used Green's theorem in the form $A = \int_0^L xy' \, \mathrm{d}s$ instead, adding and subtracting x^2 gives the identity

$$\frac{L^2}{2\pi} - 2A = \int_0^L \left((x')^2 - \tfrac{4\pi^2}{L^2} x^2 \right) \, \mathrm{d}s + \int_0^L \left(\tfrac{2\pi}{L} x - y' \right)^2 \, \mathrm{d}s$$

after rescaling the parameter to have period 2π, which exhibits the deficit $L^2 - 4\pi A$ as a sum of two explicitly non-negative quantities.

§2.5 The Four Vertex Theorem

Our second global result concerns the curvature function of a closed plane curve. On a circle κ_s is constant; on any other simple closed curve it must vary, and the theorem says it must do so in a reasonably complicated way.

Definition 2.20 (Vertex)
A **vertex** of a plane curve γ is a point where $\kappa_s' = 0$, that is, a stationary point of the signed curvature.

An ellipse which is not a circle has exactly four vertices, the ends of the two axes, so the following is sharp.

Theorem 2.21 (Four Vertex Theorem)

Every simple closed plane curve has at least four vertices.

We will prove this for *convex* curves, meaning that the curve bounds a convex region; equivalently (after choosing the anticlockwise orientation) $\kappa_s \geq 0$ everywhere. The general case is true but substantially harder, and was only settled in full by Osserman in 1985.

Proof for convex curves. Parametrise γ by arc length on $[0, L]$. Since κ_s is a smooth function on a circle it attains a maximum and a minimum, so there are at least two vertices. Vertices where κ_s changes from increasing to decreasing alternate with those where it changes from decreasing to increasing, so the number of vertices, if finite, is even; hence it suffices to rule out the case of exactly two.

So suppose the only vertices are $p = \gamma(s_1)$ and $q = \gamma(s_2)$, at which κ_s attains its maximum and minimum. Then κ_s is strictly increasing on one of the two arcs of γ between p and q and strictly decreasing on the other. Let ℓ be the line through p and q. As γ is convex, ℓ meets the curve only at p and q, and the two arcs lie strictly on opposite sides of ℓ. Choose coordinates so that ℓ is the x-axis, and write $\gamma = (x, y)$; then $\kappa_s' \cdot y$ has a constant sign along the curve, and is not identically zero, so

$$\int_0^L \kappa_s'(s) y(s) \, \mathrm{d}s \neq 0.$$

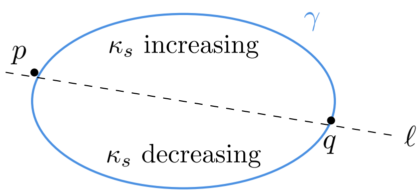
On the other hand, the Frenet equations in the plane read $\dot{T} = \kappa_s N_s$, that is, with $T = (\dot{x}, \dot{y})$ and $N_s = (-\dot{y}, \dot{x})$,

$$\ddot{x} = -\kappa_s \dot{y}, \qquad \ddot{y} = \kappa_s \dot{x}.$$

Integrating by parts around the closed curve (all boundary terms vanish by periodicity),

$$\int_0^L \kappa_s' y \, \mathrm{d}s = -\int_0^L \kappa_s \dot{y} \, \mathrm{d}s = \int_0^L \ddot{x} \, \mathrm{d}s = [\dot{x}]_0^L = 0,$$

a contradiction. $\square$



Remark. A converse also holds (Gluck): any strictly positive smooth periodic function with at least two maxima and two minima is the curvature function of a simple closed convex curve.

§3 Surfaces, Length and Area

We now specialise the general machinery of Section 1 to the case that interests us most: two-dimensional manifolds sitting inside $\mathbb{R}^3$. This is where differential geometry began, with Gauss's work on surveying, and it remains the setting in which every important idea in the subject can be seen in its simplest non-trivial form.

Everything in Section 1 applies verbatim. But we are now going to do *geometry* rather than topology: we will measure lengths, angles and areas on our surfaces, and then measure how they curve. The extra structure needed to do this comes from the ambient Euclidean inner product, and it will be packaged into two quadratic forms, the first and second fundamental forms.

§3.1 Smooth Surfaces

Definition 3.1 (Smooth Surface)
A **smooth surface** is a 2-dimensional smooth manifold $\Sigma \subseteq \mathbb{R}^3$. As in Section 1, a diffeomorphism $\sigma : V \to U \subseteq \Sigma$ from an open set $V \subseteq \mathbb{R}^2$ onto an open subset U of Σ is called an **allowable parametrisation** (or *allowable parameterisation*) of U.

We will consistently write $\sigma : V \to U \subseteq \Sigma$ for an allowable parametrisation, with coordinates (u, v) on V, and abbreviate the partial derivatives of σ by

$$\sigma_u = \frac{\partial \sigma}{\partial u}, \quad \sigma_v = \frac{\partial \sigma}{\partial v}, \quad \sigma_{uu} = \frac{\partial^2 \sigma}{\partial u^2}, \quad \text{and so on.}$$

Regularity of σ – which is part of it being a parametrisation – says exactly that σ_u and σ_v are linearly independent at every point, so $\{\sigma_u, \sigma_v\}$ is a basis for the tangent plane.

It is worth recording the standard ways in which surfaces turn up, all of which we already have the tools to handle.

Example 3.2 (Ways to Build Surfaces)
(i) *Graphs.* If $f : V \to \mathbb{R}$ is smooth on an open $V \subseteq \mathbb{R}^2$ then $\sigma(u, v) = (u, v, f(u, v))$ parametrises the graph of f, which is therefore a smooth surface. Every surface looks like this locally.

(ii) *Level sets.* If $f : W \to \mathbb{R}$ is smooth on an open $W \subseteq \mathbb{R}^3$ and c is a regular value, then $f^{-1}(c)$ is a smooth surface by the preimage theorem. For instance $f(x, y, z) = x^2 + y^2 + z^2$ gives the spheres, and $f(x, y, z) = x^2 + y^2 - z^2$ gives the hyperboloids.

(iii) *Surfaces of revolution.* Let $\eta(u) = (f(u), 0, g(u))$ be a regular injective curve in the xz-plane with $f > 0$. Rotating about the z-axis gives

$$\sigma(u, v) = (f(u) \cos v, f(u) \sin v, g(u)) .$$

(iv) *Ruled surfaces.* Given a curve η and a nowhere-zero vector field w along it, $\sigma(u, v) = \eta(u) + v\, w(u)$ sweeps out a surface (where regular): a union of straight lines. Cylinders, cones and the helicoid all arise this way.

§3.2 Orientability and Tangent Planes

Let's now use the framework we have developed to discuss a geometric property of surfaces: *orientability*.

Definition 3.3 (Orientation-Preserving Maps)

Let V, V' be open sets in $\mathbb{R}^2$, and let $f : V \to V'$ be a diffeomorphism. We say that f is **orientation-preserving** if $Df|_x$ has positive determinant for all $x \in V$.

Definition 3.4 (Orientability)

An abstract smooth surface Σ is **orientable** if it admits an atlas $\{(U_i, \phi_i)\}$ where the transition maps are all orientation-preserving. Such an atlas is a **orientation** of Σ.

We will now show that orientability is a 'topological' property of surfaces, in that it is preserved by diffeomorphisms.

Theorem 3.5

If Σ_1 and Σ_2 are diffeomorphic abstract smooth surfaces, then Σ_1 is orientable if and only if Σ_2 is orientable.

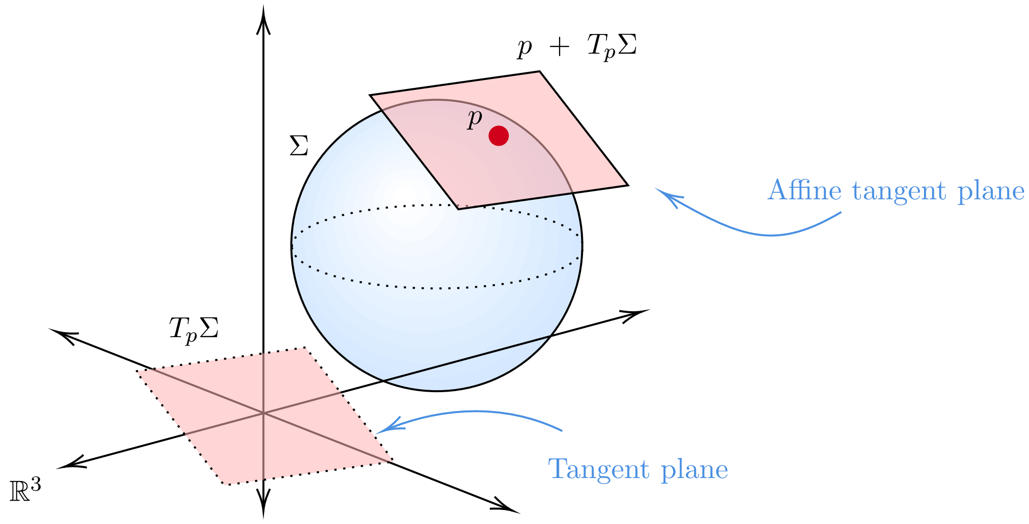
Proof. Let $f : \Sigma_1 \to \Sigma_2$ be diffeomorphism. Suppose Σ_2 is orientable and equipped with an oriented smooth atlas. Consider the atlas on Σ_1 of charts of the form $(f^{-1}(U), \phi \circ f|_{f^{-1}(U)})$, where (U, ψ) is a chart at $f(p)$ in the oriented atlas for Σ_2. Then the transition map between two such charts is exactly a transition map between charts in the Σ_2 atlas. $\square$

So what does orientability *feel like*, in a geometric sense? To discuss this, we are going to need the idea of *tangent planes*.

Definition 3.6 (Tangent Plane)

Let Σ be a smooth surface in $\mathbb{R}^3$, and $p \in \Sigma$. Let $\sigma : V \to U \subseteq \Sigma$ be an allowable parameterisation of Σ near o, so V is an open subset of $\mathbb{R}^2$ and U is open in Σ, such that $\sigma(0) = p$. The **tangent plane** $T_p\Sigma$ to p at Σ is the image of $D\sigma|_0 \subseteq \mathbb{R}^3$, which is a two-dimensional vector subspace of $\mathbb{R}^3$. The **affine tangent plane** is $p + T_p\Sigma$, which is an affine[a] subspace of $\mathbb{R}^3$.

[a]An *affine* subspace of a vector space is a translate of a linear subspace.



We should check that this notion makes geometric sense, in that it's independent of the choice of parametrisation used.

Lemma 3.7

$T_p\Sigma$ is independent of the choice of allowable paramterisation.

Proof. Suppose $\sigma : V \to U$ and $\tilde{\sigma} : \tilde{V} \to \tilde{U}$ are allowable parameterisations with $\sigma(0) = \tilde{\sigma}(0) = p$. Then there exists a transition map $\sigma^{-1} \circ \tilde{\sigma}$ which is a diffeomorphism of open sets in $\mathbb{R}^2$. So $D\left(\sigma^{-1} \circ \tilde{\sigma}\right)\big|_0$ is an isomorphism, and since $\tilde{\sigma} = \sigma \circ (\sigma^{-1} \circ \tilde{\sigma})$ the images of $D\tilde{\sigma}|_0$ and $D\sigma|_0$ agree. □

We can then sensibly define the *normal* to a smooth surface.

Definition 3.8

If Σ is a smooth surface in $\mathbb{R}^3$ and $p \in \Sigma$, the **normal direction** to Σ at p is $(T_p\Sigma)^\perp$, the Euclidean orthogonal complement to the tangent plane at p.

We can see that for every $p \in \Sigma$, there exist exactly two normalized normal vectors, and from this, we finally get our interpretation of orientability.

Definition 3.9

A smooth surface in $\mathbb{R}^3$ is **two-sided** if it admits a continuous global choice of unit normal vector.

Lemma 3.10 (Interpretation of Orientability)

A smooth surface in $\mathbb{R}^3$ is orientable if and only if it is two-sided.

Proof. Suppose Σ is orientable, with an atlas $\{(U_i, \sigma_i)\}$ whose transition maps all have positive Jacobian determinant. On each U_i define

$$n_i = \frac{(\sigma_i)_u \times (\sigma_i)_v}{\|(\sigma_i)_u \times (\sigma_i)_v\|},$$

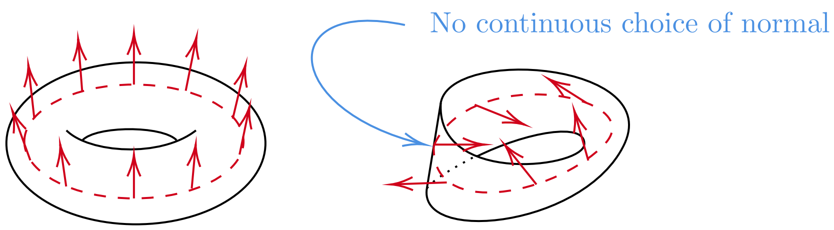
which is smooth and unit. If $p \in U_i \cap U_j$ and φ is the transition map, then as in the proof that area is well defined we have $(\sigma_j)_u \times (\sigma_j)_v = \det(D\varphi)\,(\sigma_i)_u \times (\sigma_i)_v$, and $\det(D\varphi) > 0$, so $n_i = n_j$ on the overlap. Hence the n_i glue to a global continuous (indeed smooth) unit normal, and Σ is two-sided.

Conversely suppose n is a global continuous unit normal. Take any atlas, and for each chart discard it in favour of the chart with u and v swapped if necessary, so that

$$\frac{\sigma_u \times \sigma_v}{\|\sigma_u \times \sigma_v\|} = n$$

on that patch – possible since the left hand side is a continuous unit normal, hence equal to $\pm n$ on each connected component, and swapping the coordinates changes its sign. For two such charts the same identity as above gives $\det(D\varphi) > 0$, so this atlas is an orientation. □

Some illustrations of an orientable and non-orientable surface are shown below.



Torus (Orientable)   Möbius Strip (Non-Orientable)

§3.3 The First Fundamental Form

Having pinned down what our surfaces are, we can start doing geometry on them: measuring lengths, angles and areas.

Recall that for a curve $\gamma : (a, b) \to \mathbb{R}^3$, the length of γ is given by

$$L(\gamma) = \int_a^b \|\gamma'(t)\| \, \mathrm{d}t,$$

and that this is independent of the parametrisation.[1] Now suppose the curve lies on a surface. We would like to compute its length without ever leaving the surface, that is, in terms of the coordinates (u, v) of a parametrisation.

So let Σ be a smooth surface in $\mathbb{R}^3$, let $\sigma : V \to U \subseteq \Sigma$ be an allowable parametrisation, and let $\gamma : (a, b) \to \mathbb{R}^3$ be a smooth curve whose image lies in U. Writing $\gamma(t) = \sigma(u(t), v(t))$, the chain rule gives $\gamma'(t) = \sigma_u u'(t) + \sigma_v v'(t)$, and so

$$\|\gamma'(t)\|^2 = Eu'(t)^2 + 2Fu'(t)v'(t) + Gv'(t)^2,$$

where

$$E = \langle \sigma_u, \sigma_u \rangle, \quad F = \langle \sigma_u, \sigma_v \rangle = \langle \sigma_v, \sigma_u \rangle, \quad G = \langle \sigma_v, \sigma_v \rangle,$$

[1] Feel free to check this – it is just the substitution rule.

and $\langle\cdot,\cdot\rangle$ is the Euclidean inner product. So three functions on V are all we need.

Definition 3.11 (First Fundamental Form)
The **first fundamental form** of Σ in the parametrisation σ is the expression

$$\mathrm{I} = E\,\mathrm{d}u^2 + 2F\,\mathrm{d}u\,\mathrm{d}v + G\,\mathrm{d}v^2,$$

equivalently the symmetric matrix $\begin{pmatrix} E & F \\ F & G \end{pmatrix} = (D\sigma)^\top (D\sigma)$ representing the restriction of the Euclidean inner product to $T_p\Sigma$ in the basis $\{\sigma_u, \sigma_v\}$.

Since the inner product is positive definite and $\{\sigma_u, \sigma_v\}$ is a basis, we always have $E > 0$, $G > 0$ and $EG - F^2 > 0$. In terms of the first fundamental form, if γ has image in the image of σ then

$$L(\gamma) = \int_a^b \sqrt{E(u')^2 + 2Fu'v' + G(v')^2}\,\mathrm{d}t,$$

and the angle θ between two tangent vectors $a\sigma_u + b\sigma_v$ and $c\sigma_u + d\sigma_v$ is given by

$$\cos\theta = \frac{Eac + F(ad+bc) + Gbd}{\sqrt{Ea^2 + 2Fab + Gb^2}\sqrt{Ec^2 + 2Fcd + Gd^2}}.$$

So *all* of the metric geometry of the surface, as far as a curve-measuring inhabitant of it is concerned, is contained in E, F and G.

Example 3.12 (Some First Fundamental Forms) (i) The plane $\sigma(u,v) = (u,v,0)$ has $\sigma_u = (1,0,0)$ and $\sigma_v = (0,1,0)$, so $\mathrm{I} = \mathrm{d}u^2 + \mathrm{d}v^2$, as it must.

(ii) The cylinder $\sigma(u,v) = (a\cos u, a\sin u, v)$ has $\sigma_u = a(-\sin u, \cos u, 0)$ and $\sigma_v = (0,0,1)$, so $\mathrm{I} = a^2\,\mathrm{d}u^2 + \mathrm{d}v^2$.

(iii) A surface of revolution $\sigma(u,v) = (f(u)\cos v, f(u)\sin v, g(u))$ has

$$\sigma_u = (f'\cos v, f'\sin v, g'), \quad \sigma_v = (-f\sin v, f\cos v, 0),$$

so $E = (f')^2 + (g')^2$, $F = 0$ and $G = f^2$. If the profile curve $\eta = (f, 0, g)$ is unit speed then this is simply

$$\mathrm{I} = \mathrm{d}u^2 + f(u)^2\,\mathrm{d}v^2.$$

(iv) The unit sphere, parametrised by latitude and longitude as

$$\sigma(u,v) = (\cos u\cos v,\ \cos u\sin v,\ \sin u),$$

is the surface of revolution with $f = \cos u$ and $g = \sin u$, so

$$\mathrm{I} = \mathrm{d}u^2 + \cos^2 u\,\mathrm{d}v^2.$$

(v) The torus $\sigma(u,v) = ((a + r\cos u)\cos v, (a + r\cos u)\sin v, r\sin u)$ (with $a > r > 0$) is the surface of revolution with $f = a + r\cos u$, $g = r\sin u$, so $E = r^2$, $F = 0$, $G = (a + r\cos u)^2$.

§3.4 Isometries and Conformal Maps

With length encapsulated by the first fundamental form, we get a nice way to capture when two surfaces have the same geometry in a *length* sense.

Definition 3.13 (Isometric)

Let Σ, Σ' be smooth surfaces in $\mathbb{R}^3$. A diffeomorphism $f : \Sigma \to \Sigma'$ is an **isometry** if $Df|_p : T_p\Sigma \to T_{f(p)}\Sigma'$ is a linear isometry for every p; equivalently, if f preserves the length of all curves. If such an f exists we say Σ and Σ' are **isometric**.

Definition 3.14 (Locally Isometric)

We say that Σ, Σ' are **locally isometric** near points $p \in \Sigma$ and $q \in \Sigma'$ if there exist open neighbourhoods U of p and U' of q which are isometric.

The whole point of the first fundamental form is the following, which turns a geometric question into a computation.

Lemma 3.15

Smooth surfaces Σ, Σ' in $\mathbb{R}^3$ are locally isometric near $p \in \Sigma$ and $q \in \Sigma'$ if and only if there exist allowable parametrisations $\sigma : V \to U \subseteq \Sigma$ and $\sigma' : V \to U' \subseteq \Sigma'$, *with the same domain* V, whose first fundamental forms are equal.

Proof. Suppose such parametrisations exist and set $f = \sigma' \circ \sigma^{-1} : U \to U'$, a diffeomorphism. In the bases $\{\sigma_u, \sigma_v\}$ and $\{\sigma'_u, \sigma'_v\}$, $Df|_p$ is the identity matrix, and the two first fundamental forms are equal, so $Df|_p$ carries an inner product to the same inner product, and is therefore a linear isometry.

Conversely, suppose $f : U \to U'$ is an isometry and let $\sigma : V \to U$ be any allowable parametrisation. Then $\sigma' = f \circ \sigma$ is an allowable parametrisation of U' on the same domain, and $\sigma'_u = Df|(\sigma_u)$, $\sigma'_v = Df|(\sigma_v)$. Since $Df|$ preserves inner products, $E' = E$, $F' = F$ and $G' = G$. □

Example 3.16 (The Plane and the Cylinder)

Parametrise the plane by $\sigma(u, v) = (au, v, 0)$, so that $\mathrm{I} = a^2\, \mathrm{d}u^2 + \mathrm{d}v^2$, and compare with the cylinder $\tilde{\sigma}(u, v) = (a \cos u, a \sin u, v)$ computed above. On $V = (0, 2\pi) \times \mathbb{R}$ the two first fundamental forms agree, so a cylinder is locally isometric to a plane: you really can roll a sheet of paper into a tube without stretching it. They are not globally isometric, since one is simply connected and the other is not.

Example 3.17 (The Sphere Cannot be Flattened)

No open subset of S^2 is isometric to an open subset of the plane. This is not obvious from the definition – it follows immediately from the Theorema Egregium of Section 5 – but it is worth flagging now as the reason that every world map distorts distances.

A weaker but very useful notion asks only that angles, not lengths, be preserved.

Definition 3.18 (Conformal)
A diffeomorphism $f : \Sigma \to \Sigma'$ is **conformal** if for every p there is $\lambda(p) > 0$ with

$$\langle Df|_p(w_1), Df|_p(w_2)\rangle = \lambda(p)\langle w_1, w_2\rangle$$

for all $w_1, w_2 \in T_p\Sigma$. Equivalently, in corresponding parametrisations $\mathrm{I}' = \lambda\,\mathrm{I}$.

Since the angle formula above is unchanged when E, F, G are all multiplied by the same positive function, a conformal map is precisely one that preserves angles (though not orientation of angles unless we also ask $\det Df > 0$). A parametrisation is called **isothermal** if in it

$$\mathrm{I} = \lambda(u,v)^2 \left(\mathrm{d}u^2 + \mathrm{d}v^2\right),$$

that is, if σ is a conformal map from a piece of the plane to the surface. Isothermal parametrisations always exist locally, though this is a genuinely hard theorem; they will be crucial when we come to minimal surfaces.

Example 3.19 (Stereographic Projection)
Stereographic projection $\pi : S^2 \setminus \{N\} \to \mathbb{R}^2$ from the north pole is conformal, with the inverse parametrisation

$$\sigma(u,v) = \frac{1}{1+u^2+v^2}\left(2u, 2v, u^2+v^2-1\right)$$

having first fundamental form

$$\mathrm{I} = \frac{4}{(1+u^2+v^2)^2}\left(\mathrm{d}u^2 + \mathrm{d}v^2\right),$$

so it is isothermal. This is why nautical charts made by stereographic projection get the angles right, and it is the reason for the appearance of $\pi \circ \mathcal{N}$ later in the Weierstrass representation.

§3.5 Area

With length pinned down by the first fundamental form, it is not too hard to pin down area.

Recall that the parallelogram spanned by two vectors w_1, w_2 has area $\|w_1 \times w_2\|$, and that

$$\|w_1 \times w_2\|^2 = \langle w_1, w_1\rangle\langle w_2, w_2\rangle - \langle w_1, w_2\rangle^2.$$

So if $\sigma : V \to U \subseteq \Sigma$ is an allowable parametrisation, the infinitesimal parallelogram spanned by $\sigma_u\,\mathrm{d}u$ and $\sigma_v\,\mathrm{d}v$ has area

$$\|\sigma_u \times \sigma_v\|\ \mathrm{d}u\,\mathrm{d}v = \sqrt{EG - F^2}\,\mathrm{d}u\,\mathrm{d}v.$$

This gives us a sensible way to define area, but we had better check it does not depend on the parametrisation.

Lemma 3.20
Let $\Omega \subseteq \Sigma$ be a bounded domain contained in the images of two allowable parametri-

sations $\sigma : V \to U$ and $\tilde{\sigma} : \tilde{V} \to \tilde{U}$. Then

$$\int_{\sigma^{-1}(\Omega)} \sqrt{EG - F^2}\, \mathrm{d}u\, \mathrm{d}v = \int_{\tilde{\sigma}^{-1}(\Omega)} \sqrt{\tilde{E}\tilde{G} - \tilde{F}^2}\, \mathrm{d}\tilde{u}\, \mathrm{d}\tilde{v}.$$

Proof. Let $\varphi = \sigma^{-1} \circ \tilde{\sigma}$ be the transition map, a diffeomorphism between open subsets of $\mathbb{R}^2$, and let $J = \det D\varphi$ be its Jacobian determinant. Since $\tilde{\sigma} = \sigma \circ \varphi$, the chain rule gives $D\tilde{\sigma} = D\sigma \circ D\varphi$, and hence

$$\tilde{\sigma}_{\tilde{u}} \times \tilde{\sigma}_{\tilde{v}} = J\,(\sigma_u \times \sigma_v),$$

since taking the cross product of the two columns of a 3×2 matrix and then post-composing with a 2×2 matrix scales by its determinant. Taking norms, $\sqrt{\tilde{E}\tilde{G} - \tilde{F}^2} = |J|\sqrt{EG - F^2} \circ \varphi$, and the claim is exactly the change of variables formula. $\square$

Definition 3.21 (Area)

Let Σ be a smooth surface in $\mathbb{R}^3$, let $\sigma : V \to U \subseteq \Sigma$ be an allowable parametrisation and let $\Omega \subseteq U$ be a bounded domain. Then

$$\operatorname{area}(\Omega) = \int_{\sigma^{-1}(\Omega)} \sqrt{EG - F^2}\, \mathrm{d}u\, \mathrm{d}v.$$

More generally, for continuous $h : \Sigma \to \mathbb{R}$ we write

$$\int_{\Omega} h\, \mathrm{d}A = \int_{\sigma^{-1}(\Omega)} h(\sigma(u,v))\sqrt{EG - F^2}\, \mathrm{d}u\, \mathrm{d}v,$$

and call $\mathrm{d}A = \sqrt{EG - F^2}\, \mathrm{d}u\, \mathrm{d}v$ the **area element** of Σ.

Remark. For regions not contained in a single parametrised patch, one patches these integrals together using a partition of unity. In practice we can nearly always find a single parametrisation covering everything except a finite union of curves, which contribute no area, and so we will not fuss about this.

Example 3.22 (Area of the Torus)

For the torus with $E = r^2$, $F = 0$, $G = (a + r\cos u)^2$ we have $\sqrt{EG - F^2} = r(a + r\cos u)$, so

$$\text{area} = \int_0^{2\pi} \int_0^{2\pi} r(a + r\cos u)\, \mathrm{d}u\, \mathrm{d}v = 4\pi^2 ra.$$

This is Pappus's theorem: the area is the circumference $2\pi r$ of the generating circle times the distance $2\pi a$ travelled by its centre.

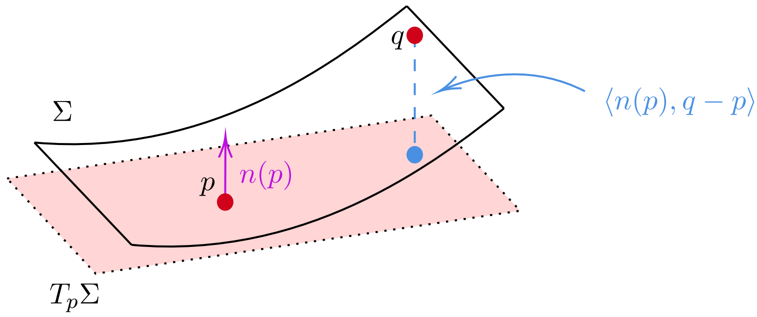
Remark. Isometries preserve area, since they preserve E, F, G. The converse is false: an area-preserving map need not be an isometry. Archimedes' theorem gives a beautiful example – the projection of the unit sphere onto the circumscribed cylinder, radially away from the axis, preserves area but certainly not lengths.

§4 The Gauss Map and Curvature

§4.1 The Second Fundamental Form

Intuitively, the first fundamental form pins down how length works along our surface. We now, through the second fundamental form, will try to pin down how *curved* our surface is (as an embedding of $\mathbb{R}^3$).

We are (informally) going to look at curvature at a point as measuring how much we deviate from the tangent plane by moving a small amount away from our point.



Let Σ be a smooth surface in $\mathbb{R}^3$, and let $\sigma : V \to U \subseteq \Sigma$ be an allowable parameterisation. Consider a point $p = \sigma(u, v)$, and some nearby point $q = \sigma(u + h, v + \ell)$. Then by Taylor's theorem we have

$$\begin{aligned}\sigma(u+h, v+\ell) = \sigma(u, v) &+ h\sigma_u(u, v) + \ell\sigma_v(u, v)\\ &+ \frac{1}{2}\left(h^2\sigma_{uu}(u, v) + 2h\ell\sigma_{uv}(u, v) + \ell^2\sigma_{vv}(u, v)\right)\\ &+ O(h^3, \ell^3).\end{aligned}$$

Recalling that the tangent plane is given by $T_p\Sigma = \operatorname{span}\{\sigma_u, \sigma_v\}$, we get the perpendicular distance from $q = \sigma(u + h, v + \ell)$ to the affine tangent plane $T_p\Sigma + p$ is given by

$$\begin{aligned}\langle n, \sigma(u+h, v+\ell) - \sigma(u, v)\rangle = \frac{1}{2}&\left(\langle n, \sigma_{uu}\rangle h^2 + 2\langle n, \sigma_{uv}\rangle h\ell + \langle n, \sigma_{vv}\rangle \ell^2\right)\\ &+ O(h^3, l^3).\end{aligned}$$

and so we can see that these inner products give us this difference. This inspires our definition of the second fundamental form.

Definition 4.1 (Second Fundamental Form)
The **second fundamental form** of Σ in the allowable parameterisation σ is

$$L\,\mathrm{d}u^2 + 2M\,\mathrm{d}u\,\mathrm{d}v + N\,\mathrm{d}v^2$$

where $L = \langle n, \sigma_{uu}\rangle$, $M = \langle n, \sigma_{uv}\rangle$, and $N = \langle n, \sigma_{vv}\rangle$, and n is the unit normal given by $n = \frac{\sigma_u \times \sigma_v}{\|\sigma_u \times \sigma_v\|}$.

We will say more about how this relates to curvature in the coming sections. It's good to know (but not proven in this course) that the two fundamental forms completely determine a smooth oriented connected surface in $\mathbb{R}^3$, up to rigid motion. This is the *fundamental theorem of surfaces in* $\mathbb{R}^3$.

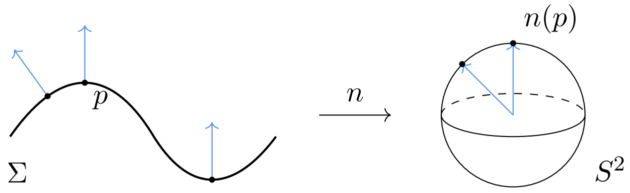
§4.2 The Gauss Map and the Shape Operator

We are now going to define what our notion of curvature actually is. This is done using the *Gauss map*, which packages the unit normal into a map to the sphere.

Definition 4.2 (Gauss Map)
Let Σ be a smooth oriented surface in $\mathbb{R}^3$. The **Gauss map** is

$$n : \Sigma \to S^2, \qquad p \mapsto n(p),$$

sending each point to its positively oriented unit normal, viewed as a point of the unit sphere.



Lemma 4.3
The Gauss map is smooth.

Proof. Smoothness is a local property, so we may check it in an allowable parametrisation σ compatible with the orientation, where

$$n \circ \sigma = \frac{\sigma_u \times \sigma_v}{\|\sigma_u \times \sigma_v\|}.$$

The numerator is smooth and the denominator is smooth and non-vanishing (since σ_u, σ_v are linearly independent), so $n \circ \sigma$ is smooth. □

The reason the Gauss map is so useful is a small miracle of dimension counting. Its derivative at p is a linear map

$$Dn|_p : T_p\Sigma \to T_{n(p)}S^2,$$

but $T_{n(p)}S^2$ is the plane orthogonal to $n(p)$ in $\mathbb{R}^3$, which is precisely $T_p\Sigma$. So $Dn|_p$ is an *endomorphism of the tangent plane*, and we may take its determinant, its trace, and its eigenvalues.

Definition 4.4 (Shape Operator)
The **shape operator** (or Weingarten map) of Σ at p is the linear map

$$S_p = -\,Dn|_p : T_p\Sigma \to T_p\Sigma.$$

The sign is a convention chosen so that the sphere with its outward normal has positive curvature. The key algebraic fact is the following.

Lemma 4.5

The shape operator is self-adjoint with respect to the first fundamental form, and

$$\mathrm{II}_p(w_1, w_2) = \langle S_p(w_1), w_2 \rangle.$$

In particular the second fundamental form is a symmetric bilinear form.

Proof. Work in an allowable parametrisation and abbreviate n for $n \circ \sigma$. Applying the chain rule to a curve $\sigma(u(t), v(t))$ shows that $Dn|_p(\sigma_u) = n_u$ and $Dn|_p(\sigma_v) = n_v$, so it is enough to check the claim on the basis $\{\sigma_u, \sigma_v\}$.

Differentiating the identities $\langle n, \sigma_u \rangle = 0$ and $\langle n, \sigma_v \rangle = 0$ gives

$$\begin{aligned} \langle n_u, \sigma_u \rangle &= -\langle n, \sigma_{uu} \rangle = -L, & \langle n_u, \sigma_v \rangle &= -\langle n, \sigma_{uv} \rangle = -M, \\ \langle n_v, \sigma_u \rangle &= -\langle n, \sigma_{vu} \rangle = -M, & \langle n_v, \sigma_v \rangle &= -\langle n, \sigma_{vv} \rangle = -N. \end{aligned}$$

The two middle expressions agree because mixed partials commute, which gives $\langle Dn|_p \sigma_u, \sigma_v \rangle = \langle \sigma_u, Dn|_p \sigma_v \rangle$: self-adjointness. Reading the same four identities the other way round says exactly that the matrix of II in the basis $\{\sigma_u, \sigma_v\}$ is the matrix of $\langle S_p(\cdot), \cdot \rangle$. □

The identities in that proof deserve to be displayed on their own, since they are the computational heart of the subject:

$$\begin{pmatrix} L & M \\ M & N \end{pmatrix} = -\begin{pmatrix} \langle n_u, \sigma_u \rangle & \langle n_u, \sigma_v \rangle \\ \langle n_v, \sigma_u \rangle & \langle n_v, \sigma_v \rangle \end{pmatrix} = -(Dn)^\top (D\sigma),$$

in perfect analogy with $\mathrm{I} = (D\sigma)^\top (D\sigma)$.

§4.3 Normal and Principal Curvatures

The second fundamental form has a direct geometric meaning in terms of curves on the surface.

Definition 4.6 (Normal Curvature)

Let γ be a unit speed curve on Σ with $\gamma(0) = p$. The **normal curvature** of γ at p is

$$\kappa_n = \langle \ddot{\gamma}(0), n(p) \rangle,$$

the component of the acceleration in the normal direction.

Proposition 4.7

The normal curvature of a unit speed curve γ on Σ at p depends only on the tangent vector $\dot{\gamma}(0)$, and

$$\kappa_n = \mathrm{II}_p(\dot{\gamma}(0), \dot{\gamma}(0)).$$

Proof. Along γ we have $\langle n(\gamma(t)), \dot{\gamma}(t) \rangle = 0$. Differentiating at $t = 0$,

$$\langle Dn|_p(\dot{\gamma}(0)), \dot{\gamma}(0) \rangle + \langle n(p), \ddot{\gamma}(0) \rangle = 0,$$

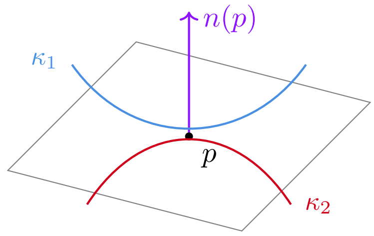
so $\kappa_n = \langle S_p(\dot\gamma(0)), \dot\gamma(0)\rangle = \mathrm{II}_p(\dot\gamma(0), \dot\gamma(0))$, which depends only on $\dot\gamma(0)$. □

This is sometimes called Meusnier's theorem. It says that all curves on Σ through p with the same tangent direction bend away from the tangent plane at the same rate; in particular we may as well compute with the most symmetric such curve.

Definition 4.8 (Normal Section)

Given a unit vector $w \in T_p\Sigma$, let Π be the plane through p spanned by w and $n(p)$. Then $\Pi \cap \Sigma$ is, near p, a smooth curve (the two are transversal), called the **normal section** of Σ at p in the direction w.

A normal section is a plane curve lying in Π, and its principal normal at p is $\pm n(p)$; so its signed curvature there is exactly $\mathrm{II}_p(w, w)$. Normal curvature is genuinely the curvature of the slice you get by cutting the surface with a plane containing the normal.



Since S_p is self-adjoint, the spectral theorem applies.

Definition 4.9 (Principal Curvatures)

The eigenvalues $\kappa_1 \geq \kappa_2$ of the shape operator S_p are the **principal curvatures** of Σ at p, and the corresponding unit eigenvectors are the **principal directions**. A point where $\kappa_1 = \kappa_2$ is called **umbilic**.

Corollary 4.10

There is an orthonormal basis $\{e_1, e_2\}$ of $T_p\Sigma$ with $S_p(e_i) = \kappa_i e_i$, and κ_1, κ_2 are the maximum and minimum values of the normal curvature over all unit tangent directions.

Proof. The first statement is the spectral theorem for a self-adjoint operator on a two-dimensional inner product space. For the second, write a unit vector as $w = \cos\theta\, e_1 + \sin\theta\, e_2$; then

$$\mathrm{II}_p(w, w) = \kappa_1 \cos^2\theta + \kappa_2 \sin^2\theta,$$

which is a convex combination of κ_1 and κ_2, attaining each at $\theta = 0, \pi/2$. □

The formula in that proof is *Euler's theorem*, and it is worth remembering: the normal curvature in the direction making angle θ with the first principal direction is $\kappa_1 \cos^2\theta + \kappa_2 \sin^2\theta$.

Definition 4.11 (Gaussian and Mean Curvature)

Let Σ be a smooth surface in $\mathbb{R}^3$. The **Gaussian curvature** and **mean curvature** of Σ at p are

$$K(p) = \det S_p = \det\left(Dn|_p\right) = \kappa_1\kappa_2, \qquad H(p) = \tfrac{1}{2}\operatorname{tr} S_p = \tfrac{1}{2}(\kappa_1+\kappa_2).$$

Note that reversing the orientation replaces n by $-n$, hence S_p by $-S_p$; so κ_1, κ_2 and H change sign, but K does not. In particular K is well defined even on a non-orientable surface, since every surface is locally orientable. Both are smooth functions of p, being polynomial expressions in the derivatives of a smooth parametrisation.

Remark. The principal curvatures are the roots of $t^2 - 2Ht + K = 0$, so $\kappa_{1,2} = H \pm \sqrt{H^2 - K}$ and in particular $H^2 \geq K$ always, with equality exactly at umbilic points.

§4.4 Curvature in Local Coordinates

We can now compute K and H from the two fundamental forms. This is the formula that makes everything computable.

Proposition 4.12

In any allowable parametrisation,

$$K = \frac{LN - M^2}{EG - F^2}, \qquad H = \frac{LG - 2MF + NE}{2(EG - F^2)}.$$

Proof. Write the matrix of S_p in the basis $\{\sigma_u, \sigma_v\}$ as $A = (a_{ij})$, so that

$$-n_u = a_{11}\sigma_u + a_{21}\sigma_v, \qquad -n_v = a_{12}\sigma_u + a_{22}\sigma_v.$$

Taking the inner product of each with σ_u and with σ_v, and using the identities $-\langle n_u, \sigma_u\rangle = L$ and so on, we obtain the matrix identity

$$\begin{pmatrix} L & M \\ M & N \end{pmatrix} = \begin{pmatrix} E & F \\ F & G \end{pmatrix} A,$$

that is, $A = \mathrm{I}^{-1}\mathrm{II}$. Taking determinants gives

$$K = \det A = \frac{LN - M^2}{EG - F^2},$$

and computing $A = \frac{1}{EG-F^2}\begin{pmatrix} G & -F \\ -F & E \end{pmatrix}\begin{pmatrix} L & M \\ M & N \end{pmatrix}$ explicitly and taking half the trace gives the stated H. □

Example 4.13 (The Sphere)

For the unit sphere with outward normal, $n(p) = p$, so $Dn|_p = \mathrm{id}$ and $S_p = -\mathrm{id}$. Hence $\kappa_1 = \kappa_2 = -1$, $K = 1$ and $H = -1$. Every point is umbilic. With the inward normal we would get $\kappa_i = +1$; either way $K = 1$. For the sphere of radius a the same computation gives $K = 1/a^2$: big spheres are less curved.

Example 4.14 (The Cylinder)

For the cylinder $\sigma(u, v) = (a \cos u, a \sin u, v)$ we found $E = a^2$, $F = 0$, $G = 1$. Also $\sigma_{uv} = \sigma_{vv} = 0$ and $\sigma_{uu} = (-a \cos u, -a \sin u, 0)$, while $n = (\cos u, \sin u, 0)$ is the outward normal, so

$$L = -a, \quad M = N = 0.$$

Therefore $K = 0$ and $H = -1/(2a)$. The principal curvatures are $-1/a$ (around the cylinder) and 0 (along it). So the cylinder is *flat* – consistent with the fact that it is locally isometric to the plane, as we saw, and a first hint of the Theorema Egregium.

Example 4.15 (The Torus)

For the torus $\sigma(u, v) = ((a + r \cos u) \cos v, (a + r \cos u) \sin v, r \sin u)$ we have $E = r^2$, $F = 0$, $G = (a + r \cos u)^2$, and one computes

$$L = r, \quad M = 0, \quad N = (a + r \cos u) \cos u,$$

so

$$K = \frac{r(a + r \cos u) \cos u}{r^2(a + r \cos u)^2} = \frac{\cos u}{r(a + r \cos u)}.$$

Thus the outer half of the torus ($|u| < \pi/2$) has $K > 0$, the inner half has $K < 0$, and the top and bottom circles $u = \pm\pi/2$ have $K = 0$. This matches the intuition that the outside of a doughnut is dome-like and the inside is saddle-like.

Example 4.16 (Surfaces of Revolution)

Let $\sigma(u, v) = (f(u) \cos v, f(u) \sin v, g(u))$ with unit speed profile, so $(f')^2 + (g')^2 = 1$ and $\mathrm{I} = \,du^2 + f^2 \,dv^2$. A short computation gives

$$L = f'g'' - f''g', \quad M = 0, \quad N = fg',$$

and hence

$$K = \frac{(f'g'' - f''g')g'}{f}.$$

Differentiating $(f')^2 + (g')^2 = 1$ gives $f'f'' + g'g'' = 0$, from which $(f'g'' - f''g')g' = -f''$, and so the memorable formula

$$K = -\frac{f''}{f}.$$

§4.5 Elliptic, Hyperbolic and Parabolic Points

The sign of the Gaussian curvature controls the local shape of the surface, via the Taylor expansion we used to motivate the second fundamental form.

Definition 4.17

A smooth surface in $\mathbb{R}^3$ with $K \equiv 0$ is called **flat**.

Definition 4.18

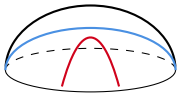
Let Σ be a smooth surface in $\mathbb{R}^3$, and let $p \in \Sigma$. We say p is

(i) **elliptic** if $K(p) > 0$;

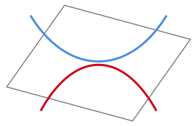
(ii) **hyperbolic** if $K(p) < 0$;

(iii) **parabolic** if $K(p) = 0$ but $S_p \neq 0$;

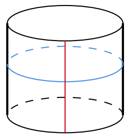
(iv) **planar** if $S_p = 0$.



elliptic: $K > 0$



hyperbolic: $K < 0$



parabolic: $K = 0$

Proposition 4.19

Let $p \in \Sigma$ be an elliptic point. Then there is a neighbourhood of p in Σ lying strictly on one side of the affine tangent plane $p + T_p\Sigma$ (apart from p itself). If p is hyperbolic, then every neighbourhood of p contains points on both sides.

Proof. Take an allowable parametrisation with $\sigma(0,0) = p$. From the Taylor expansion at the start of this section, the signed distance from $\sigma(h,\ell)$ to the affine tangent plane is

$$\langle \sigma(h,\ell) - p, n(p) \rangle = \frac{1}{2}\left(Lh^2 + 2Mh\ell + N\ell^2\right) + O\left((h^2+\ell^2)^{3/2}\right),$$

that is, $\frac{1}{2}\mathrm{II}_p(w,w) + o(\|w\|^2)$ where $w = h\sigma_u + \ell\sigma_v$.

If $K(p) > 0$ then $LN - M^2 > 0$, so the quadratic form II_p is definite, and hence $|\mathrm{II}_p(w,w)| \geq c\|w\|^2$ for some $c > 0$ by compactness of the unit circle. The error term is $o(\|w\|^2)$, so for $\|w\|$ small the sign of the whole expression is that of II_p, which is constant.

If $K(p) < 0$ then II_p is indefinite, so there are unit vectors $w_\pm$ with $\mathrm{II}_p(w_+,w_+) > 0 > \mathrm{II}_p(w_-,w_-)$, and the same estimate applied along the two directions produces points on both sides arbitrarily close to p. $\square$

Example 4.20 (The Monkey Saddle)

Let Σ be the graph of $z = x^3 - 3xy^2$. At the origin all first and second derivatives of the height function vanish, so $\mathrm{II}_0 = 0$ and the origin is a planar point. Writing $x = r\cos\theta$, $y = r\sin\theta$ gives $z = r^3\cos 3\theta$, so the surface crosses its tangent plane six times around the origin – a saddle with room for two legs and a tail.

§4.6 Curvature as Area Distortion

The determinant of a linear map is the factor by which it scales areas, and the Gauss map's derivative is a map between two-dimensional spaces. That gives the following interpretation of K, which is essentially Gauss's original definition.

Theorem 4.21

Let Σ be a smooth surface in $\mathbb{R}^3$ and $p \in \Sigma$ with $K(p) \neq 0$. Let (A_i) be a decreasing sequence of neighbourhoods of p shrinking to p, in the sense that for every $\varepsilon > 0$ we have $A_i \subseteq B(p, \varepsilon)$ for all large i. Then

$$|K(p)| = \lim_{i\to\infty} \frac{\text{area}_{S^2}(n(A_i))}{\text{area}_{\Sigma}(A_i)}.$$

Proof. Fix an allowable parametrisation σ near p. The area element of Σ is $\|\sigma_u \times \sigma_v\|$, and the composite $n \circ \sigma$ parametrises the image on the sphere, with area element $\|n_u \times n_v\|$, both with respect to $du\, dv$. But $n_u = -S(\sigma_u)$ and $n_v = -S(\sigma_v)$, so

$$n_u \times n_v = \det(S)\, \sigma_u \times \sigma_v = K\, \sigma_u \times \sigma_v.$$

Hence

$$\frac{\text{area}_{S^2}(n(A_i))}{\text{area}_{\Sigma}(A_i)} = \frac{\int_{\sigma^{-1}(A_i)} |K|\, \|\sigma_u \times \sigma_v\|}{\int_{\sigma^{-1}(A_i)} \|\sigma_u \times \sigma_v\|},$$

a weighted average of $|K|$ over a region shrinking to p. Since K is continuous and $K(p) \neq 0$ (so n is a local diffeomorphism near p and the areas above are genuinely areas of the image), this tends to $|K(p)|$. □

Informally, then, the Gaussian curvature is an infinitesimal measure of how much the Gauss map distorts area. On a sphere of radius a a patch of area A maps to a patch of area A/a^2; on a cylinder every patch maps into a single curve, of area zero, which is why $K = 0$.

§4.7 Compact Surfaces Have an Elliptic Point

Here is a first genuinely global statement, and a good illustration of a technique we will use again: put the surface inside the smallest sphere you can and look at where they touch.

Theorem 4.22

Every compact smooth surface $\Sigma \subseteq \mathbb{R}^3$ has an elliptic point. In particular, no compact surface in $\mathbb{R}^3$ has $K \leq 0$ everywhere.

Proof. The function $p \mapsto \|p\|^2$ is continuous on the compact set Σ, so it attains its maximum at some p_0, at distance $R = \|p_0\|$ from the origin. Then Σ is contained in the closed ball of radius R and touches the sphere S_R of radius R at p_0.

Let γ be any unit speed curve in Σ with $\gamma(0) = p_0$. The function $h(t) = \|\gamma(t)\|^2$ has a maximum at $t = 0$, so $\dot{h}(0) = 2\langle \gamma(0), \dot{\gamma}(0)\rangle = 0$ – which says p_0 is orthogonal to $T_{p_0}\Sigma$, so the unit normal there is $n(p_0) = \pm p_0/R$; choose the orientation so that $n(p_0) = -p_0/R$ (the inward normal). Also

$$0 \geq \tfrac{1}{2}\ddot{h}(0) = \langle \dot{\gamma}(0), \dot{\gamma}(0)\rangle + \langle \gamma(0), \ddot{\gamma}(0)\rangle = 1 + \langle p_0, \ddot{\gamma}(0)\rangle.$$

Now the normal curvature of γ at p_0 is $\kappa_n = \langle \ddot{\gamma}(0), n(p_0)\rangle = -\langle \ddot{\gamma}(0), p_0\rangle/R$, so the displayed inequality reads $R\kappa_n \geq 1$, i.e.

$$\kappa_n \geq \frac{1}{R}$$

for every unit tangent direction at p_0. In particular both principal curvatures at p_0 are at least $1/R > 0$, so

$$K(p_0) = \kappa_1\kappa_2 \geq \frac{1}{R^2} > 0,$$

and p_0 is elliptic. □

Remark. This is the germ of several deeper global theorems. For instance, a compact surface with K constant must have $K > 0$ by the above, and one can then show it is a sphere; and Hilbert's theorem states that there is no complete surface in $\mathbb{R}^3$ with constant negative curvature, so the hyperbolic plane of Section 10 can never be realised as a surface in $\mathbb{R}^3$. We will not prove either here.

§5 The Theorema Egregium

The Gaussian curvature was defined using the Gauss map, which needs a normal vector, which needs an ambient $\mathbb{R}^3$. So on the face of it K measures how the surface sits inside space. Gauss's remarkable theorem – *theorema egregium* really does mean 'remarkable theorem', and he named it himself – says that this appearance is deceptive: K can be computed from the first fundamental form alone, and so is unchanged by any isometry.

The strategy is exactly the one that worked for curves. There we differentiated the Frenet frame and expressed the answer back in the frame; here we differentiate the frame $\{\sigma_u, \sigma_v, n\}$ and express the answer back in that frame. The coefficients which appear are the Christoffel symbols, and the compatibility conditions coming from equality of mixed partials are the Gauss and Codazzi–Mainardi equations.

§5.1 Christoffel Symbols

Let $\sigma : V \to U \subseteq \Sigma$ be an allowable parametrisation. At each point $\{\sigma_u, \sigma_v, n\}$ is a basis of $\mathbb{R}^3$, so we may expand the second derivatives of σ in it.

Definition 5.1 (Christoffel Symbols)
The **Christoffel symbols** of Σ in the parametrisation σ are the functions Γ^k_{ij} defined by

$$\begin{aligned}
\sigma_{uu} &= \Gamma^1_{11}\sigma_u + \Gamma^2_{11}\sigma_v + Ln,\\
\sigma_{uv} &= \Gamma^1_{12}\sigma_u + \Gamma^2_{12}\sigma_v + Mn,\\
\sigma_{vv} &= \Gamma^1_{22}\sigma_u + \Gamma^2_{22}\sigma_v + Nn.
\end{aligned}$$

The normal components are the coefficients of the second fundamental form, by definition. Note that $\sigma_{uv} = \sigma_{vu}$, so the symbols are symmetric in their lower indices, $\Gamma^k_{12} = \Gamma^k_{21}$: there are six of them in all.

The whole point is that the Christoffel symbols are *intrinsic*.

Lemma 5.2

The Christoffel symbols are determined by E, F, G and their first partial derivatives.

Proof. Take the inner product of each of the three defining equations with σ_u and with σ_v; the normal terms drop out. Using

$$\langle \sigma_{uu}, \sigma_u \rangle = \tfrac{1}{2} E_u, \quad \langle \sigma_{uu}, \sigma_v \rangle = F_u - \tfrac{1}{2} E_v, \quad \langle \sigma_{uv}, \sigma_u \rangle = \tfrac{1}{2} E_v,$$

$$\langle \sigma_{uv}, \sigma_v \rangle = \tfrac{1}{2} G_u, \quad \langle \sigma_{vv}, \sigma_u \rangle = F_v - \tfrac{1}{2} G_u, \quad \langle \sigma_{vv}, \sigma_v \rangle = \tfrac{1}{2} G_v,$$

each of which follows by differentiating the definitions of E, F, G, we obtain three linear systems

$$\begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} \Gamma^1_{11} \\ \Gamma^2_{11} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} E_u \\ F_u - \frac{1}{2} E_v \end{pmatrix}, \quad \begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} \Gamma^1_{12} \\ \Gamma^2_{12} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} E_v \\ \frac{1}{2} G_u \end{pmatrix},$$

$$\begin{pmatrix} E & F \\ F & G \end{pmatrix} \begin{pmatrix} \Gamma^1_{22} \\ \Gamma^2_{22} \end{pmatrix} = \begin{pmatrix} F_v - \frac{1}{2} G_u \\ \frac{1}{2} G_v \end{pmatrix}.$$

Since $EG - F^2 > 0$ the matrix is invertible, and Cramer's rule solves each system in terms of E, F, G and their first derivatives. $\square$

Example 5.3 (Orthogonal Parametrisations)

When $F = 0$ the systems above are diagonal and we can read off everything at once:

$$\Gamma^1_{11} = \frac{E_u}{2E}, \quad \Gamma^2_{11} = -\frac{E_v}{2G}, \quad \Gamma^1_{12} = \frac{E_v}{2E}, \quad \Gamma^2_{12} = \frac{G_u}{2G}, \quad \Gamma^1_{22} = -\frac{G_u}{2E}, \quad \Gamma^2_{22} = \frac{G_v}{2G}.$$

§5.2 The Gauss and Codazzi–Mainardi Equations

We have written down $\sigma_{uu}, \sigma_{uv}, \sigma_{vv}$ and (from the shape operator) n_u, n_v in terms of the frame. The only constraint left to impose is that mixed partial derivatives commute, and this is where all the content is.

Theorem 5.4 (Gauss Equation)

In any allowable parametrisation,

$$EK = \left(\Gamma^2_{11}\right)_v - \left(\Gamma^2_{12}\right)_u + \Gamma^1_{11}\Gamma^2_{12} - \Gamma^1_{12}\Gamma^2_{11} + \Gamma^2_{11}\Gamma^2_{22} - \left(\Gamma^2_{12}\right)^2.$$

Proof. Write $n_u = a_{11}\sigma_u + a_{21}\sigma_v$ and $n_v = a_{12}\sigma_u + a_{22}\sigma_v$; from the previous section the matrix (a_{ij}) of $Dn|_p = -S_p$ is $-\mathrm{I}^{-1}\mathrm{II}$, so

$$a_{21} = \frac{FL - EM}{EG - F^2}, \qquad a_{22} = \frac{FM - EN}{EG - F^2}.$$

Now differentiate the first Christoffel identity with respect to v and the second with respect to u:

$$\sigma_{uuv} = \left(\Gamma^1_{11}\right)_v \sigma_u + \Gamma^1_{11}\sigma_{uv} + \left(\Gamma^2_{11}\right)_v \sigma_v + \Gamma^2_{11}\sigma_{vv} + L_v n + L n_v,$$

$$\sigma_{uvu} = \left(\Gamma^1_{12}\right)_u \sigma_u + \Gamma^1_{12}\sigma_{uu} + \left(\Gamma^2_{12}\right)_u \sigma_v + \Gamma^2_{12}\sigma_{uv} + M_u n + M n_u.$$

These are equal. Expanding $\sigma_{uu}, \sigma_{uv}, \sigma_{vv}, n_u, n_v$ in the frame and comparing the coefficients of σ_v (which is legitimate since the frame is a basis) gives

$$\left(\Gamma^2_{11}\right)_v + \Gamma^1_{11}\Gamma^2_{12} + \Gamma^2_{11}\Gamma^2_{22} + La_{22} = \left(\Gamma^2_{12}\right)_u + \Gamma^1_{12}\Gamma^2_{11} + \left(\Gamma^2_{12}\right)^2 + Ma_{21}.$$

Finally

$$Ma_{21} - La_{22} = \frac{M(FL - EM) - L(FM - EN)}{EG - F^2} = \frac{E(LN - M^2)}{EG - F^2} = EK,$$

and rearranging gives the claim. □

Theorem 5.5 (Theorema Egregium)
The Gaussian curvature of a smooth surface in $\mathbb{R}^3$ depends only on the first fundamental form. Consequently, if $f : \Sigma \to \Sigma'$ is an isometry then $K' \circ f = K$; that is, Gaussian curvature is an isometry invariant.

Proof. The right hand side of the Gauss equation involves only Christoffel symbols and their derivatives, hence only E, F, G and their derivatives up to second order. Since $E = \langle \sigma_u, \sigma_u \rangle > 0$ we may divide, and K is expressed intrinsically.

For the second statement, if f is an isometry and σ is an allowable parametrisation of Σ, then $f \circ \sigma$ is an allowable parametrisation of Σ' with the same first fundamental form, so the two curvature computations produce the same function. □

Comparing the coefficients of n instead of σ_v, and doing the same with $\sigma_{uvv} = \sigma_{vvu}$, gives the remaining compatibility conditions.

Theorem 5.6 (Codazzi–Mainardi Equations)
In any allowable parametrisation,

$$\begin{aligned} L_v - M_u &= L\Gamma^1_{12} + M\left(\Gamma^2_{12} - \Gamma^1_{11}\right) - N\Gamma^2_{11}, \\ M_v - N_u &= L\Gamma^1_{22} + M\left(\Gamma^2_{22} - \Gamma^1_{12}\right) - N\Gamma^2_{12}. \end{aligned}$$

Proof. Compare the coefficients of n in the two expansions of $\sigma_{uuv} = \sigma_{uvu}$ displayed above: this gives $\Gamma^1_{11}M + \Gamma^2_{11}N + L_v = \Gamma^1_{12}L + \Gamma^2_{12}M + M_u$, which rearranges to the first equation. Expanding $\sigma_{uvv} = \sigma_{vvu}$ in the same way and comparing n coefficients gives $\Gamma^1_{12}M + \Gamma^2_{12}N + M_v = \Gamma^1_{22}L + \Gamma^2_{22}M + N_u$, which is the second. □

Remark. The Gauss and Codazzi–Mainardi equations are precisely the integrability conditions for the system defining σ, and one can prove a converse: given smooth functions E, F, G (with $E, G, EG - F^2 > 0$) and L, M, N on a simply connected open $V \subseteq \mathbb{R}^2$ satisfying these equations, there is a smooth surface, unique up to a rigid motion of $\mathbb{R}^3$, with these as its fundamental forms. This is the *fundamental theorem of surface theory*, the exact analogue of the fundamental theorem of curve theory from Section 2. We will not prove it.

§5.3 Useful Special Cases

In practice one almost always arranges $F = 0$, and then the Gauss equation collapses to something usable.

Corollary 5.7 (Orthogonal Gauss Formula)
If σ is an orthogonal parametrisation ($F \equiv 0$) then

$$K = -\frac{1}{2\sqrt{EG}}\left[\left(\frac{E_v}{\sqrt{EG}}\right)_v + \left(\frac{G_u}{\sqrt{EG}}\right)_u\right].$$

Proof. Substitute the orthogonal Christoffel symbols computed above into the Gauss equation and simplify; it is a slightly tedious but entirely mechanical calculation. □

Corollary 5.8
If $\mathrm{I} = \mathrm{d}u^2 + G(u,v)\,\mathrm{d}v^2$, then with $h = \sqrt{G}$,

$$K = -\frac{h_{uu}}{h}.$$

Proof. Put $E = 1$ in the orthogonal Gauss formula: the first bracket vanishes and $G_u/\sqrt{G} = 2h_u$, so $K = -(2h_u)_u/(2h) = -h_{uu}/h$. □

Corollary 5.9 (Isothermal Case)
If σ is isothermal, so $E = G = \lambda^2$ and $F = 0$, then

$$K = -\frac{1}{\lambda^2}\nabla^2(\log\lambda),$$

where $\nabla^2 = \partial_u^2 + \partial_v^2$ is the flat Laplacian.

Proof. Here $\sqrt{EG} = \lambda^2$, so $E_v/\sqrt{EG} = 2\lambda_v/\lambda = 2(\log\lambda)_v$ and likewise $G_u/\sqrt{EG} = 2(\log\lambda)_u$. Substituting into the orthogonal Gauss formula gives the result. □

Example 5.10 (The Sphere, Intrinsically)
The unit sphere has $\mathrm{I} = \mathrm{d}u^2 + \cos^2 u\,\mathrm{d}v^2$ in latitude-longitude coordinates. So $h = \cos u$ and

$$K = -\frac{h_{uu}}{h} = -\frac{-\cos u}{\cos u} = 1,$$

with not a single normal vector in sight.

Example 5.11 (Surfaces of Revolution, Intrinsically)
A surface of revolution with unit speed profile has $\mathrm{I} = \mathrm{d}u^2 + f(u)^2\,\mathrm{d}v^2$, so immediately $K = -f''/f$, recovering the formula of the previous section by a purely intrinsic route.

Example 5.12 (Why the Sphere Cannot be Flattened)

A piece of the plane has $K \equiv 0$ and a piece of the unit sphere has $K \equiv 1$. By the Theorema Egregium no open subset of the sphere is isometric to an open subset of the plane. Every flat map of the world lies about distances – and now we know it is not the cartographers' fault.

Example 5.13 (The Cylinder Again)

The cylinder has $\mathrm{I} = a^2 \, \mathrm{d}u^2 + \mathrm{d}v^2$, with constant coefficients, so $K \equiv 0$: the cylinder is intrinsically flat, exactly as its local isometry with the plane demands. Note that the *mean* curvature does not have this property: the plane has $H \equiv 0$ and the cylinder has $H = -1/2a \neq 0$, so H is emphatically not intrinsic.

§6 Geodesics

§6.1 Length, Energy and Variations

We are now going to discuss a particular geometric phenomenon, *geodesics*, which are curves that are (in some sense) the shortest path between points.

Fix a smooth surface Σ and points $p, q \in \Sigma$, and let $\Omega(p, q)$ denote the set of smooth curves $\gamma : [a, b] \to \Sigma$ with $\gamma(a) = p$ and $\gamma(b) = q$. We would like to find the critical points of the length functional L on $\Omega(p, q)$. The trouble is that L has a square root in it, which makes differentiating it unpleasant, and it is also insensitive to reparametrisation, so its critical points are never isolated. Both problems are fixed by working with a different functional.

Definition 6.1 (Energy)

The **energy** of a curve $\gamma : [a, b] \to \mathbb{R}^3$ is given by

$$\mathcal{E}(\gamma) = \int_a^b \left\|\gamma'(t)\right\|^2 \, \mathrm{d}t.$$

We write $\mathcal{E}$ rather than E to avoid a clash with the first fundamental form. Energy is sensitive to reparametrisation, but the two functionals are related by Cauchy–Schwarz.

Lemma 6.2

For any $\gamma \in \Omega(p, q)$ we have $L(\gamma)^2 \leq (b - a)\mathcal{E}(\gamma)$, with equality if and only if γ has constant speed.

Proof. Apply Cauchy–Schwarz to $\|\gamma'\|$ and the constant function 1 on $[a, b]$:

$$L(\gamma)^2 = \left(\int_a^b \|\gamma'\| \cdot 1 \, \mathrm{d}t\right)^2 \leq (b - a) \int_a^b \left\|\gamma'\right\|^2 \, \mathrm{d}t = (b - a)\mathcal{E}(\gamma),$$

with equality exactly when $\|\gamma'\|$ is a constant multiple of 1. □

Since every curve can be reparametrised to have constant speed without changing its length, minimising L is the same problem as minimising $\mathcal{E}$ over constant speed curves, and the latter is the more tractable one.

Definition 6.3 (One-Parameter Variation)

Let $\gamma : [a, b] \to \Sigma$, where Σ is a smooth surface in $\mathbb{R}^3$. A **one-parameter variation** (with fixed endpoints) of γ is a smooth map $\Gamma : (-\varepsilon, \varepsilon) \times [a, b] \to \Sigma$ such that, writing $\gamma_s = \Gamma(s, \cdot)$, we have $\gamma_0 = \gamma$ and $\gamma_s(a), \gamma_s(b)$ are independent of s.

Definition 6.4 (Geodesic)

A smooth curve $\gamma : [a, b] \to \Sigma$ is a **geodesic** if, for every variation (γ_s) of γ with fixed endpoints, we have $\frac{\mathrm{d}}{\mathrm{d}s}\big|_{s=0} \mathcal{E}(\gamma_s) = 0$; that is, if γ is a critical point of the energy functional on $\Omega(\gamma(a), \gamma(b))$.

§6.2 The Geodesic Equations

So how do we actually find geodesics? It turns out to be *relatively* easy, using the geodesic equations.

Suppose we have some curve γ with image contained in the image of an allowable parametrisation σ. Then for sufficiently small s we can write $\gamma_s(t) = \sigma(u(s, t), v(s, t))$. Let the first fundamental form with respect to σ be

$$E \, \mathrm{d}u^2 + 2F \, \mathrm{d}u \, \mathrm{d}v + G \, \mathrm{d}v^2.$$

Then by definition,

$$\mathcal{E}(\gamma_s) = \int_a^b \left(E\dot{u}^2 + 2F\dot{u}\dot{v} + G\dot{v}^2\right) \, \mathrm{d}t,$$

where the dots are $\partial/\partial t$ and E, F, G are evaluated at $(u(s, t), v(s, t))$. Differentiating under the integral sign and integrating by parts – the boundary terms vanish because the endpoints are fixed – we get

$$\frac{\mathrm{d}}{\mathrm{d}s}\mathcal{E}(\gamma_s)\bigg|_{s=0} = \int_a^b \left(A\frac{\partial u}{\partial s} + B\frac{\partial v}{\partial s}\right) \, \mathrm{d}t\bigg|_{s=0}$$

where

$$A = E_u\dot{u}^2 + 2F_u\dot{u}\dot{v} + G_u\dot{v}^2 - 2\frac{\mathrm{d}}{\mathrm{d}t}(E\dot{u} + F\dot{v}),$$
$$B = E_v\dot{u}^2 + 2F_v\dot{u}\dot{v} + G_v\dot{v}^2 - 2\frac{\mathrm{d}}{\mathrm{d}t}(F\dot{u} + G\dot{v}).$$

Since $\partial u/\partial s$ and $\partial v/\partial s$ at $s = 0$ can be prescribed to be any pair of smooth functions vanishing at a and b, the integral vanishes for all variations exactly when $A = B = 0$. These give us the *geodesic equations*.

Theorem 6.5 (Geodesic Equations)

A smooth curve $\gamma(t) = \sigma(u(t), v(t))$ is a geodesic if and only if it satisfies the geodesic equations

$$\frac{\mathrm{d}}{\mathrm{d}t}(E\dot{u} + F\dot{v}) = \frac{E_u\dot{u}^2 + 2F_u\dot{u}\dot{v} + G_u\dot{v}^2}{2},$$

$$\frac{\mathrm{d}}{\mathrm{d}t}(F\dot{u} + G\dot{v}) = \frac{E_v\dot{u}^2 + 2F_v\dot{u}\dot{v} + G_v\dot{v}^2}{2}.$$

Unlike length, energy is sensitive to reparameterisation, but Cauchy–Schwarz gives us that $L(\gamma)^2 \leq \mathcal{E}(\gamma)(b-a)$, with equality if and only if γ is parametrised proportional to arc length.

Corollary 6.6

If γ has constant speed and locally minimises length, then it is a geodesic. Further, if γ minimises energy globally, then it minimises length globally and is parametrised with constant speed.

Expanding the derivatives on the left and using the formulae for the Christoffel symbols, the geodesic equations take an equivalent and often more convenient form.

Corollary 6.7

A curve $\gamma(t) = \sigma(u(t), v(t))$ is a geodesic if and only if

$$\begin{aligned} \ddot{u} + \Gamma^1_{11}\dot{u}^2 + 2\Gamma^1_{12}\dot{u}\dot{v} + \Gamma^1_{22}\dot{v}^2 &= 0, \\ \ddot{v} + \Gamma^2_{11}\dot{u}^2 + 2\Gamma^2_{12}\dot{u}\dot{v} + \Gamma^2_{22}\dot{v}^2 &= 0. \end{aligned}$$

In this form the geodesic equations are visibly a second order system of ODEs, with smooth coefficients depending only on the first fundamental form. Picard's theorem therefore gives us existence, uniqueness and smooth dependence on initial data all at once.

Proposition 6.8 (Local Existence and Uniqueness of Geodesics)

Let Σ be a smooth surface, $p \in \Sigma$ and $w \in T_p\Sigma$. Then there is $\varepsilon > 0$ and a unique geodesic $\gamma : (-\varepsilon, \varepsilon) \to \Sigma$ with $\gamma(0) = p$ and $\dot{\gamma}(0) = w$. Moreover γ depends smoothly on (p, w).

Proof. Choose an allowable parametrisation about p. In it the geodesic equations are a second order ODE system $\ddot{x} = h(x, \dot{x})$ with h smooth, which is equivalent to a first order system in $(x, \dot{x})$ with smooth right hand side. Picard's theorem gives a unique local solution with any prescribed initial position and velocity, and the standard theory gives smooth dependence on those initial conditions. □

Note also the important fact that geodesics automatically have constant speed. This will follow immediately from the characterisation in the next subsection.

§6.3 Geodesics Are Straightest

In $\mathbb{R}^2$, the geodesics are not just locally shortest, they are also locally *straightest*. We would intuitively expect this to hold on other surfaces too: the change in the tangent vector should be as small as possible subject to the constraint that we lie on the surface. That is, the acceleration should be entirely used up in staying on the surface.

Theorem 6.9

Let Σ be a smooth surface in $\mathbb{R}^3$. A smooth curve $\gamma : [a, b] \to \Sigma$ is a geodesic if and only if $\ddot{\gamma}(t)$ is everywhere normal to the surface Σ, that is, $\ddot{\gamma}(t) \perp T_{\gamma(t)}\Sigma$ for all t.

Proof. Let Γ be a variation of γ with fixed endpoints and put $W(t) = \frac{\partial \Gamma}{\partial s}\big|_{s=0}$, a vector field along γ tangent to Σ and vanishing at the endpoints. Differentiating under the integral sign,

$$\frac{\mathrm{d}}{\mathrm{d}s}\bigg|_{s=0} \mathcal{E}(\gamma_s) = \int_a^b 2\left\langle \frac{\partial^2 \Gamma}{\partial s \partial t}, \frac{\partial \Gamma}{\partial t}\right\rangle_{s=0} \mathrm{d}t = 2\int_a^b \left\langle \dot{W}, \dot{\gamma}\right\rangle \mathrm{d}t = -2\int_a^b \langle W, \ddot{\gamma}\rangle \,\mathrm{d}t,$$

integrating by parts and using $W(a) = W(b) = 0$.

If $\ddot{\gamma} \perp T_\gamma \Sigma$ everywhere then the integrand vanishes for every variation, so γ is a geodesic. Conversely, suppose γ is a geodesic but $\ddot{\gamma}(t_0)$ has a non-zero tangential part Z for some t_0. Choose a smooth bump function $\chi \geq 0$ supported in a small neighbourhood of t_0 with $\chi(t_0) = 1$, and let W be the tangential vector field along γ obtained by extending χZ (in a parametrisation, take $W = \chi(t)(\alpha \sigma_u + \beta \sigma_v)$ with α, β the constant coordinates of Z). Then W arises from a genuine variation – take $\Gamma(s, t) = \sigma(u(t) + s\alpha\chi(t), v(t) + s\beta\chi(t))$ – and $\int \langle W, \ddot{\gamma}\rangle > 0$ if the support is small enough, contradicting criticality. □

Corollary 6.10

Every geodesic has constant speed.

Proof. $\frac{\mathrm{d}}{\mathrm{d}t}\langle \dot{\gamma}, \dot{\gamma}\rangle = 2\langle \ddot{\gamma}, \dot{\gamma}\rangle = 0$, since $\ddot{\gamma}$ is normal and $\dot{\gamma}$ is tangent. □

This characterisation makes many examples immediate.

Example 6.11 (Geodesics in the Plane)

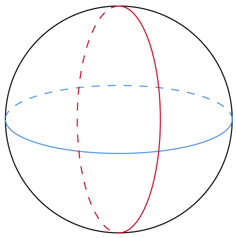
For a curve in a plane, $\ddot{\gamma}$ is normal to the plane if and only if $\ddot{\gamma} = 0$, so the geodesics are exactly the constant speed straight lines.

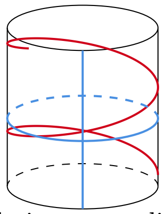
Example 6.12 (Geodesics on the Sphere)

Let γ be a unit speed geodesic on S^2. The normal at p is p itself, so $\ddot{\gamma} = \lambda\gamma$ for some function λ; taking the inner product with γ and using $\langle \gamma, \gamma\rangle = 1$, $\langle \dot{\gamma}, \gamma\rangle = 0$ and $\langle \ddot{\gamma}, \gamma\rangle = -1$ gives $\lambda = -1$. So $\ddot{\gamma} = -\gamma$, whose solutions are

$$\gamma(t) = \cos t\, \gamma(0) + \sin t\, \dot{\gamma}(0),$$

which traverses the great circle cut out by the plane through the origin spanned by $\gamma(0)$ and $\dot{\gamma}(0)$. So the geodesics on the sphere are exactly the constant speed parametrisations of great circles.





geodesics on a cylinder

Example 6.13 (Geodesics on the Cylinder)

Since the cylinder is locally isometric to the plane and geodesics are determined by the first fundamental form, the geodesics of the cylinder are the images of straight lines under the rolling-up map. These are the vertical rulings, the horizontal circles, and the helices. Alternatively, unroll: on the cylinder $\mathrm{I} = a^2\, \mathrm{d}u^2 + \mathrm{d}v^2$ has constant coefficients, so the geodesic equations read $\ddot{u} = \ddot{v} = 0$, giving u, v affine in t.

§6.4 Planes of Symmetry

There is a useful symmetry principle which lets us find geodesics with no computation at all.

Proposition 6.14

Let Σ be a smooth surface in $\mathbb{R}^3$ and let Π be a plane such that reflection in Π maps Σ to itself. Then every constant speed parametrisation of a connected component of $\Sigma \cap \Pi$ is a geodesic.

Proof. Let R denote reflection in Π and let γ be a unit speed parametrisation of a component of $\Sigma \cap \Pi$. Since R is an isometry of $\mathbb{R}^3$ fixing Σ setwise and fixing γ pointwise, R maps Σ to Σ and hence maps $T_{\gamma(t)}\Sigma$ to itself, and $R(\ddot{\gamma}) = (\ddot{R\gamma}) = \ddot{\gamma}$, so $\ddot{\gamma}$ lies in Π.

Now decompose $\ddot{\gamma}$ into its tangential and normal parts. The tangential part is orthogonal to $\dot{\gamma}$ (as γ has unit speed), and it lies in Π; but $T_{\gamma(t)}\Sigma \cap \Pi$ is the line spanned by $\dot{\gamma}(t)$, because Σ and Π meet transversally along γ. So the tangential part is both parallel and orthogonal to $\dot{\gamma}$, hence zero, and γ is a geodesic. $\square$

§6.5 Surfaces of Revolution

One common family of surfaces is the surfaces of revolution, obtained by rotating $\eta(u) = (f(u), 0, g(u))$ – where η is smooth and injective and $f(u) > 0$ – about the z-axis. We will assume throughout that η has unit speed, so that

$$\mathrm{I} = \mathrm{d}u^2 + f(u)^2\, \mathrm{d}v^2.$$

Definition 6.15

A circle obtained by rotating a point of η is called a **parallel**. A curve obtained by rotating η itself by a fixed angle about the z-axis is a **meridian**.

Every plane in $\mathbb{R}^3$ containing the z-axis is a plane of symmetry of Σ, and meets Σ exactly in a pair of meridians. So the previous proposition gives:

Corollary 6.16

All meridians of a surface of revolution, parametrised at constant speed, are geodesics.

Not all parallels are geodesics however – think of a circle of latitude on the sphere near the north pole, which is visibly not straight.

Lemma 6.17

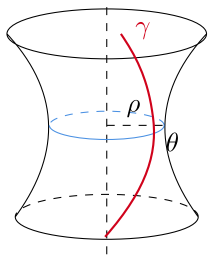
The parallel $u = u_0$, parametrised at constant speed, is a geodesic if and only if $f'(u_0) = 0$.

Proof. With $E = 1$, $F = 0$, $G = f^2$ the geodesic equations become

$$\ddot{u} = f f' \dot{v}^2, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\left(f^2 \dot{v}\right) = 0.$$

Along the parallel $u \equiv u_0$ with constant speed, $\dot{v}$ is a non-zero constant, so the second equation holds automatically and the first reduces to $0 = f(u_0) f'(u_0) \dot{v}^2$. As $f > 0$ and $\dot{v} \neq 0$, this holds precisely when $f'(u_0) = 0$: the parallel must sit at a critical radius. $\square$

Now consider a unit speed curve $\gamma(t)$ on Σ, making an angle θ with the parallel through each of its points, and let $\rho = f(u)$ be the radius of that parallel. There is a striking conservation law.



Theorem 6.18 (Clairaut's Relation)

If γ is a unit speed geodesic on a surface of revolution, then $\rho \cos\theta$ is constant along γ.

Proof. Write $\gamma(t) = \sigma(u(t), v(t))$, so $\dot{\gamma} = \sigma_u \dot{u} + \sigma_v \dot{v}$. The tangent to the parallel is σ_v, of length $\sqrt{G} = f = \rho$. Since $F = 0$ and $\|\dot{\gamma}\| = 1$,

$$\cos\theta = \frac{\langle \sigma_v, \dot{\gamma} \rangle}{\|\sigma_v\| \, \|\dot{\gamma}\|} = \frac{G\dot{v}}{\rho} = \rho \dot{v}.$$

So $\rho \cos\theta = \rho^2 \dot{v} = f^2 \dot{v}$, and the second geodesic equation says precisely that this is constant. $\square$

Example 6.19 (Trapped Geodesics)

Let γ be a geodesic on an ellipsoid of revolution which is not a meridian, so that $c = \rho\cos\theta > 0$. Since $|\cos\theta| \le 1$ we always have $\rho \ge c$, so the geodesic can never enter the polar caps where the parallels are narrower than c: it is trapped in a band between two parallels, oscillating between them forever. In particular, any geodesic passing through a pole must have $c = 0$ and hence be a meridian.

§6.6 The Covariant Derivative and Parallel Transport

The characterisation '$\ddot\gamma$ is normal' can be phrased more elegantly. The obstruction to $\dot\gamma$ being constant is entirely in the tangential direction, so let us give that tangential part a name.

Definition 6.20 (Vector Field Along a Curve)

Let $\gamma : I \to \Sigma$ be a smooth curve. A **vector field along** γ is a smooth map $W : I \to \mathbb{R}^3$ with $W(t) \in T_{\gamma(t)}\Sigma$ for every t.

Definition 6.21 (Covariant Derivative)

The **covariant derivative** of a vector field W along γ is

$$\frac{DW}{\mathrm{d}t} = \pi_{T_{\gamma(t)}\Sigma}\left(\frac{\mathrm{d}W}{\mathrm{d}t}\right),$$

the orthogonal projection of $\mathrm{d}W/\,\mathrm{d}t$ onto the tangent plane. We say W is **parallel** along γ if $DW/\,\mathrm{d}t \equiv 0$.

With this language, γ is a geodesic precisely when $D\dot\gamma/\,\mathrm{d}t = 0$: a geodesic is a curve whose velocity field is parallel along it. In the plane a vector field is parallel exactly when it is constant, so this really is the right generalisation of 'moving in a straight line at constant speed'.

Proposition 6.22

If V, W are parallel vector fields along γ then $\langle V(t), W(t)\rangle$ is constant. In particular parallel vector fields have constant length, and the angle between two of them is constant.

Proof. $\frac{\mathrm{d}}{\mathrm{d}t}\langle V, W\rangle = \langle \dot V, W\rangle + \langle V, \dot W\rangle$. Now $\dot V$ differs from $DV/\,\mathrm{d}t = 0$ by a normal vector, and W is tangent, so $\langle \dot V, W\rangle = 0$; similarly for the other term. □

The covariant derivative looks as though it depends on the embedding – we projected onto a tangent plane sitting in $\mathbb{R}^3$. It does not.

Lemma 6.23

The covariant derivative depends only on the first fundamental form. Explicitly, if

$\gamma(t) = \sigma(u(t), v(t))$ and $W = a\sigma_u + b\sigma_v$, then

$$\begin{aligned}\frac{DW}{\mathrm{d}t} &= \left(\dot{a} + \Gamma^1_{11} a\dot{u} + \Gamma^1_{12}(a\dot{v} + b\dot{u}) + \Gamma^1_{22} b\dot{v}\right)\sigma_u \\ &\quad + \left(\dot{b} + \Gamma^2_{11} a\dot{u} + \Gamma^2_{12}(a\dot{v} + b\dot{u}) + \Gamma^2_{22} b\dot{v}\right)\sigma_v.\end{aligned}$$

Proof. Differentiating $W = a\sigma_u + b\sigma_v$ along γ gives

$$\frac{\mathrm{d}W}{\mathrm{d}t} = \dot{a}\sigma_u + \dot{b}\sigma_v + a\left(\sigma_{uu}\dot{u} + \sigma_{uv}\dot{v}\right) + b\left(\sigma_{vu}\dot{u} + \sigma_{vv}\dot{v}\right).$$

Now expand $\sigma_{uu}, \sigma_{uv}, \sigma_{vv}$ using the Christoffel symbols and discard the n components, which is exactly what the projection does. Collecting terms gives the stated formula, and the Christoffel symbols depend only on E, F, G. □

Since $DW/\,\mathrm{d}t = 0$ is a linear first order ODE system in (a, b), Picard's theorem gives at once:

Proposition 6.24

Let $\gamma : I \to \Sigma$ be a smooth curve, $t_0 \in I$ and $w_0 \in T_{\gamma(t_0)}\Sigma$. Then there is a unique parallel vector field W along γ with $W(t_0) = w_0$, defined on all of I.

Definition 6.25 (Parallel Transport)

With notation as above, the map

$$\mathfrak{p} : T_{\gamma(t_0)}\Sigma \to T_{\gamma(t_1)}\Sigma, \qquad w_0 \mapsto W(t_1)$$

is called **parallel transport** along γ from $\gamma(t_0)$ to $\gamma(t_1)$.

Parallel transport is linear, since the defining ODE is linear, and it is a linear isometry by the proposition on inner products above. It is emphatically *path dependent*: transporting a vector around a closed loop on the sphere returns it rotated by an angle, and Gauss–Bonnet will tell us that the angle is the enclosed curvature.

§6.7 The Exponential Map

Geodesics give us a canonical way to move away from a point in a given direction, and hence a canonical way to put coordinates near a point.

Given $p \in \Sigma$ and $w \in T_p\Sigma$, let γ_w be the unique geodesic with $\gamma_w(0) = p$ and $\dot{\gamma}_w(0) = w$.

Definition 6.26 (Exponential Map)

The **exponential map** at p is

$$\exp_p(w) = \gamma_w(1),$$

defined for those $w \in T_p\Sigma$ for which γ_w exists up to time 1, and with $\exp_p(0) = p$.

Proposition 6.27

$\exp_p$ is defined and smooth on a neighbourhood of 0 in $T_p\Sigma$.

Proof. By the local existence theorem there is $\varepsilon > 0$ such that γ_w is defined on $(-2, 2)$ for every w in the closed disc $\{\|w\| \leq \varepsilon\}$: indeed, existence for a fixed time on a compact set of initial conditions follows from the local statement plus a compactness argument, and geodesics rescale, $\gamma_{\lambda w}(t) = \gamma_w(\lambda t)$, so shrinking ε buys more time. Smoothness is the smooth dependence of the solution of an ODE on its initial conditions. □

Proposition 6.28

$\exp_p$ is a diffeomorphism from a neighbourhood of $0 \in T_p\Sigma$ onto its image.

Proof. By the inverse function theorem it suffices to show $D\exp_p\big|_0$ is invertible. Fix $w \in T_p\Sigma$ and consider the curve $\alpha(t) = tw$ in $T_p\Sigma$. Then

$$\exp_p(\alpha(t)) = \exp_p(tw) = \gamma_{tw}(1) = \gamma_w(t),$$

so differentiating at $t = 0$ gives $D\exp_p\big|_0(w) = \dot\gamma_w(0) = w$. So $D\exp_p\big|_0$ is the identity. □

Definition 6.29 (Normal Neighbourhood)

The image of such a restriction of $\exp_p$ is a **normal neighbourhood** of p, and $\exp_p^{-1}$ gives **geodesic normal coordinates** on it.

Remark. The name comes from Lie theory. If $X \subseteq M(n)$ is a matrix group, with the inner product $\langle A, B\rangle = \mathrm{tr}(AB^\top)$, then for $A \in T_I X$ the curve $t \mapsto e^{tA}$ satisfies $\ddot\alpha = A^2\alpha$, which one checks is normal to X; so it is a geodesic, and $\exp_I(A) = e^A$ really is the matrix exponential.

§6.8 Geodesic Polar Coordinates

Composing the exponential map with polar coordinates on $T_p\Sigma$ gives an especially convenient parametrisation.

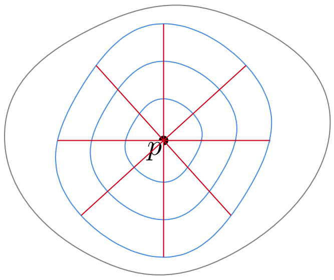
Definition 6.30 (Geodesic Polar Coordinates)

Choose an orthonormal basis of $T_p\Sigma$ and let $w(\theta)$ be the unit vector at angle θ. For ε small enough that $\exp_p$ is a diffeomorphism on the disc of radius ε, define

$$\sigma(r, \theta) = \exp_p(r\, w(\theta)) = \gamma_{w(\theta)}(r), \qquad 0 < r < \varepsilon.$$

These are **geodesic polar coordinates** centred at p. The curves $\theta = $ const are the **radial geodesics** and the curves $r = $ const are the **geodesic circles** about p.

Note that σ is a genuine allowable parametrisation away from $r = 0$ (the origin, as always with polar coordinates, is a removable defect).



Proposition 6.31 (The Gauss Lemma)
In geodesic polar coordinates the first fundamental form is

$$\mathrm{I} = \mathrm{d}r^2 + G(r,\theta)\,\mathrm{d}\theta^2,$$

that is, $E \equiv 1$ and $F \equiv 0$. Moreover

$$G(0,\theta) = 0, \qquad \left(\sqrt{G}\right)_r(0,\theta) = 1, \qquad K = -\frac{\left(\sqrt{G}\right)_{rr}}{\sqrt{G}}.$$

Proof. For fixed θ, $r \mapsto \sigma(r,\theta)$ is the unit speed geodesic $\gamma_{w(\theta)}$, so $E = \langle \sigma_r, \sigma_r\rangle = 1$.

For F, note σ_{rr} is the acceleration of a geodesic, hence normal to Σ, so $\langle \sigma_{rr}, \sigma_\theta\rangle = 0$, and

$$F_r = \langle \sigma_{rr}, \sigma_\theta\rangle + \langle \sigma_r, \sigma_{\theta r}\rangle = 0 + \tfrac{1}{2}\frac{\partial}{\partial\theta}\langle \sigma_r, \sigma_r\rangle = \tfrac{1}{2}E_\theta = 0.$$

So F is independent of r. Writing $\sigma_\theta = r\, D\exp_p\big|_{rw(\theta)}(w'(\theta))$, we see $\sigma_\theta \to 0$ as $r \to 0$, hence $F \to 0$; since F is constant in r this gives $F \equiv 0$. The same expression gives $G = r^2 \left\| D\exp_p\big|_{rw}(w')\right\|^2$, so $G(0,\theta) = 0$ and

$$\left(\sqrt{G}\right)_r(0,\theta) = \left\| D\exp_p\big|_0 (w'(\theta))\right\| = \left\|w'(\theta)\right\| = 1.$$

Finally the formula for K is the corollary of the Gauss equation for a metric of the form $\mathrm{d}r^2 + G\,\mathrm{d}\theta^2$. $\square$

The geometric content of $F \equiv 0$ is worth stating: *geodesic circles meet radial geodesics at right angles.* This is what makes the next result work.

Proposition 6.32 (Geodesics Locally Minimise Length)
Let q lie in a normal neighbourhood of p, and let γ be the radial geodesic from p to q. Then every piecewise smooth curve in Σ from p to q has length at least $L(\gamma)$, with equality only if it is a reparametrisation of γ.

Proof. Say $q = \sigma(r_0, \theta_0)$, so $L(\gamma) = r_0$. Let α be any piecewise smooth curve from p to q. Consider first the case that α stays inside the normal neighbourhood; write

$\alpha(t) = \sigma(r(t), \theta(t))$ (away from p). Then by the Gauss lemma

$$L(\alpha) = \int \sqrt{\dot{r}^2 + G\dot{\theta}^2}\, \mathrm{d}t \geq \int |\dot{r}|\, \mathrm{d}t \geq \left| \int \dot{r}\, \mathrm{d}t \right| = r_0,$$

with equality only if $\dot{\theta} \equiv 0$ and $\dot{r}$ never changes sign, i.e. only if α is a monotone reparametrisation of the radial geodesic.

If α leaves the normal neighbourhood, then it must cross the geodesic circle of radius r_1 for every $r_1 < \varepsilon$, and the same estimate applied to the initial segment shows $L(\alpha) \geq r_1$ for all such r_1, hence $L(\alpha) \geq \varepsilon > r_0$. □

Remark. 'Locally' is essential. On the sphere, the major arc of a great circle is a geodesic but is nowhere near the shortest path; and a geodesic on a cylinder can wind around many times. Geodesics are critical points of energy, and critical points need not be minima.

§6.9 Surfaces of Constant Curvature

Geodesic polar coordinates let us classify surfaces of constant curvature, locally.

Theorem 6.33 (Minding)

Let Σ be a smooth surface in $\mathbb{R}^3$ with constant Gaussian curvature K. Then Σ is locally isometric to

(i) the plane with $\mathrm{I} = \mathrm{d}u^2 + \mathrm{d}v^2$, if $K = 0$;

(ii) the sphere of radius $1/\sqrt{K}$, with $\mathrm{I} = \mathrm{d}u^2 + \frac{1}{K}\cos^2\left(\sqrt{K}u\right)\mathrm{d}v^2$, if $K > 0$;

(iii) the hyperbolic plane of curvature K, with $\mathrm{I} = \mathrm{d}u^2 + \frac{1}{|K|}\cosh^2\left(\sqrt{|K|}u\right)\mathrm{d}v^2$, if $K < 0$.

In particular any two surfaces with the same constant curvature are locally isometric.

Proof. Work in geodesic polar coordinates about a point, so $\mathrm{I} = \mathrm{d}r^2 + G\,\mathrm{d}\theta^2$ and, writing $h = \sqrt{G}$, the Gauss lemma gives

$$h_{rr} + Kh = 0, \qquad h(0,\theta) = 0, \qquad h_r(0,\theta) = 1.$$

This is a linear ODE in r for each fixed θ with prescribed initial data, so its solution is unique and independent of θ:

$$h(r) = \begin{cases} r & K = 0, \\ \frac{1}{\sqrt{K}}\sin\left(\sqrt{K}r\right) & K > 0, \\ \frac{1}{\sqrt{|K|}}\sinh\left(\sqrt{|K|}r\right) & K < 0. \end{cases}$$

So the first fundamental form in geodesic polar coordinates is completely determined by K, and any two surfaces of the same constant curvature have parametrisations with the same first fundamental form – hence are locally isometric. Identifying the resulting metrics with the standard models (after the substitution taking polar to the coordinates listed) gives the stated forms. □

Remark. Globally the situation is quite different. There are plenty of flat surfaces which are not isometric to the plane – the cylinder and the flat torus, for example – and Gauss–Bonnet will explain why the sphere is the only compact one with $K \equiv 1$. Meanwhile Hilbert's theorem says that no complete surface in $\mathbb{R}^3$ has constant negative curvature at all, so case (iii) can only ever be realised locally, on pieces such as the pseudosphere.

§7 Minimal Surfaces

Geodesics arose as the critical points of the length functional on curves. The obvious two-dimensional analogue is to look for critical points of the *area* functional on surfaces. The Euler–Lagrange equations are now a partial differential equation rather than an ordinary one, which makes the theory harder – but also much richer, because complex analysis turns out to solve it.

The physical model to keep in mind is a soap film spanning a wire loop: surface tension makes it minimise area subject to the boundary condition.

§7.1 The First Variation of Area

Let $\sigma : V \to \Sigma$ be an allowable parametrisation, let $D \subseteq V$ be a bounded domain with $\overline{D} \subseteq V$, and let $h : \overline{D} \to \mathbb{R}$ be smooth. We deform the surface in the normal direction, with the amount of deformation controlled by h.

Definition 7.1 (Normal Variation)
The **normal variation** of $\sigma(\overline{D})$ determined by h is the map

$$\sigma^t(u,v) = \sigma(u,v) + t\,h(u,v)\,n(\sigma(u,v)), \qquad t \in (-\varepsilon, \varepsilon).$$

For small t the vectors

$$\sigma^t_u = \sigma_u + th\,n_u + th_u\,n, \qquad \sigma^t_v = \sigma_v + th\,n_v + th_v\,n$$

remain linearly independent, so σ^t parametrises a genuine surface.

Proposition 7.2 (First Variation of Area)
With notation as above,

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \operatorname{area}\left(\sigma^t(D)\right) = -2\int_D hH\sqrt{EG - F^2}\,\mathrm{d}u\,\mathrm{d}v,$$

where H is the mean curvature of Σ.

Proof. Compute the first fundamental form of σ^t. Using $\langle n, \sigma_u\rangle = \langle n, \sigma_v\rangle = 0$ and the identities $\langle n_u, \sigma_u \rangle = -L$, $\langle n_u, \sigma_v \rangle = \langle n_v, \sigma_u \rangle = -M$, $\langle n_v, \sigma_v \rangle = -N$, we get

$$\begin{aligned} E^t &= E - 2thL + O(t^2), \\ F^t &= F - 2thM + O(t^2), \\ G^t &= G - 2thN + O(t^2), \end{aligned}$$

where the $O(t^2)$ terms are smooth in (u, v, t). Hence

$$\begin{aligned} E^t G^t - (F^t)^2 &= EG - F^2 - 2th\,(EN - 2FM + GL) + O(t^2) \\ &= (EG - F^2)\,(1 - 4thH) + O(t^2), \end{aligned}$$

using $H = (LG - 2MF + NE)/2(EG - F^2)$. Taking square roots and differentiating under the integral sign at $t = 0$,

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \int_D \sqrt{E^t G^t - (F^t)^2}\,\mathrm{d}u\,\mathrm{d}v = \int_D \frac{-4hH}{2} \sqrt{EG - F^2}\,\mathrm{d}u\,\mathrm{d}v,$$

which is the claim. □

Definition 7.3 (Minimal Surface)
A smooth surface in $\mathbb{R}^3$ is **minimal** if its mean curvature vanishes identically.

Corollary 7.4
A surface is minimal if and only if it is a critical point of area under all normal variations, i.e. the first variation vanishes for every choice of D and h.

Proof. If $H \equiv 0$ the first variation formula gives zero. Conversely, if $H(q) \neq 0$ for some q, choose by continuity a small domain D around $\sigma^{-1}(q)$ on which H does not vanish and put $h = H$; then

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \text{area} = -2 \int_D H^2 \sqrt{EG - F^2}\,\mathrm{d}u\,\mathrm{d}v < 0,$$

so the surface is not critical. □

Remark. As with geodesics, 'minimal' means *critical*, not minimising. A minimal surface need not minimise area even among nearby competitors, though small enough pieces of one always do.

The condition $H = 0$ says $\kappa_1 = -\kappa_2$, so at every point of a minimal surface the two principal curvatures are equal and opposite: minimal surfaces curve up as much as they curve down. In particular $K = \kappa_1 \kappa_2 = -\kappa_1^2 \leq 0$ everywhere, so every point of a minimal surface is hyperbolic or planar. Combined with the fact that every compact surface has an elliptic point, we get:

Corollary 7.5
There are no compact minimal surfaces in $\mathbb{R}^3$.

§7.2 Examples

Example 7.6 (The Plane)

A plane has $S_p = 0$ and so $H \equiv 0$. It is the trivial minimal surface, and the question that started the subject (asked by Lagrange in 1762) was whether there is any other.

Example 7.7 (The Catenoid)

Meusnier answered Lagrange in 1776. Consider the surface of revolution generated by the catenary $x = a\cosh(z/a)$, that is,

$$\sigma(u, v) = (a\cosh v \cos u, a\cosh v \sin u, av)\,.$$

One computes $E = G = a^2\cosh^2 v$ and $F = 0$, so this parametrisation is isothermal, and

$$L = -a, \quad M = 0, \quad N = a,$$

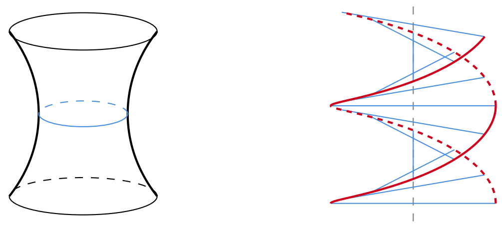
giving $H = (LG + NE)/2EG = 0$. So the catenoid is minimal. It is in fact the only minimal surface of revolution other than the plane – a fact one can prove by solving $H = 0$ for a profile curve, which reduces to $ff'' = 1 + (f')^2$ and hence to the catenary.

Example 7.8 (The Helicoid)

The helicoid is the ruled surface

$$\sigma(u, v) = (a\sinh v \cos u, a\sinh v \sin u, au)\,,$$

swept out by a horizontal line rotating at constant speed as it rises. Again $E = G = a^2\cosh^2 v$ and $F = 0$, so it too is isothermal with the same first fundamental form as the catenoid – and hence the two are locally isometric. It is minimal for the same reason. Physically, one can bend a catenoid soap film continuously into a helicoid without stretching it anywhere; the pictures of this deformation are worth looking up.



catenoid helicoid

Example 7.9 (Ruled Surfaces)

A **ruled surface** $\sigma(u, v) = \eta(u) + vw(u)$ is a union of straight lines. Every straight line on a surface is a geodesic and has vanishing normal curvature, so at every point of a ruled surface some normal curvature is zero, hence $\kappa_1\kappa_2 \leq 0$ and $K \leq 0$. The plane and the helicoid are the only ruled minimal surfaces in $\mathbb{R}^3$, a theorem of Catalan.

§7.3 Isothermal Coordinates and Harmonicity

The reason complex analysis enters is the following identity.

Proposition 7.10

Let σ be an isothermal parametrisation, so $E = G = \lambda^2$ and $F = 0$. Then

$$\nabla^2 \sigma := \sigma_{uu} + \sigma_{vv} = 2\lambda^2 H n.$$

Proof. Differentiating $\langle \sigma_u, \sigma_u \rangle = \langle \sigma_v, \sigma_v \rangle$ with respect to u gives $\langle \sigma_{uu}, \sigma_u \rangle = \langle \sigma_{uv}, \sigma_v \rangle$, and differentiating $\langle \sigma_u, \sigma_v \rangle = 0$ with respect to v gives $\langle \sigma_{uv}, \sigma_v \rangle = -\langle \sigma_u, \sigma_{vv} \rangle$. Adding,

$$\langle \sigma_{uu} + \sigma_{vv}, \sigma_u \rangle = 0,$$

and symmetrically $\langle \sigma_{uu} + \sigma_{vv}, \sigma_v \rangle = 0$. So $\nabla^2 \sigma$ is normal. Its normal component is $L + N$, and in the isothermal case

$$H = \frac{LG + NE}{2EG} = \frac{L + N}{2\lambda^2},$$

so $L + N = 2\lambda^2 H$ as claimed. □

Corollary 7.11

An isothermally parametrised surface is minimal if and only if the three coordinate functions of σ are harmonic.

Since every minimal surface admits isothermal coordinates – for minimal surfaces this is much easier than the general theorem – minimal surfaces are locally exactly the harmonic isothermal immersions. Because a non-constant harmonic function on a connected open set has no interior maximum, this gives another proof that no minimal surface is compact.

§7.4 The Weierstrass Representation

We can now write down *all* minimal surfaces, locally, in terms of holomorphic data.

Proposition 7.12

Let $\sigma(u, v) = (x, y, z)$ be a parametrisation of a surface in $\mathbb{R}^3$ and set

$$\theta_1 = x_u - i x_v, \qquad \theta_2 = y_u - i y_v, \qquad \theta_3 = z_u - i z_v.$$

Then

(i) σ is isothermal if and only if $\theta_1^2 + \theta_2^2 + \theta_3^2 = 0$;

(ii) if σ is isothermal, then the surface is minimal if and only if $\theta_1, \theta_2, \theta_3$ are holomorphic in $\xi = u + iv$.

Proof. For (i), expand:

$$\sum_j \theta_j^2 = \sum_j \left(x_u^{(j)}\right)^2 - \sum_j \left(x_v^{(j)}\right)^2 - 2i \sum_j x_u^{(j)} x_v^{(j)} = E - G - 2iF,$$

which vanishes exactly when $E = G$ and $F = 0$.

For (ii), the Cauchy–Riemann equations for $\theta_1 = x_u - ix_v$ are $x_{uu} = -x_{vv}$ and $x_{uv} = x_{vu}$. The second always holds and the first says x is harmonic, so θ_1 is holomorphic if and only if x is harmonic; likewise for θ_2, θ_3. Now apply the previous corollary. ☐

So the problem becomes: describe all triples of holomorphic functions with $\theta_1^2+\theta_2^2+\theta_3^2 = 0$. This is a conic, and conics can be rationally parametrised.

Lemma 7.13

Let $D \subseteq \mathbb{C}$ be a domain, g meromorphic on D and f holomorphic on D, such that f has a zero of order at least $2k$ wherever g has a pole of order k. Then

$$\theta_1 = \tfrac{1}{2} f\left(1 - g^2\right), \qquad \theta_2 = \tfrac{i}{2} f\left(1 + g^2\right), \qquad \theta_3 = fg$$

are holomorphic on D and satisfy $\theta_1^2 + \theta_2^2 + \theta_3^2 = 0$. Conversely, every triple of holomorphic functions satisfying this equation arises in this way, apart from the degenerate case $\theta_1 = i\theta_2$, $\theta_3 = 0$.

Proof. For the first part, the pole conditions make each θ_j holomorphic, and

$$\theta_1^2 + \theta_2^2 = \tfrac{f^2}{4}\left[(1-g^2)^2 - (1+g^2)^2\right] = -f^2 g^2 = -\theta_3^2.$$

Conversely, given holomorphic θ_j with $\sum \theta_j^2 = 0$, suppose $\theta_1 \neq i\theta_2$ and set

$$f = \theta_1 - i\theta_2, \qquad g = \frac{\theta_3}{f}.$$

Then $fg = \theta_3$, and from $\theta_1^2 + \theta_2^2 = -\theta_3^2$ we get $(\theta_1 - i\theta_2)(\theta_1 + i\theta_2) = -\theta_3^2$, so $\theta_1 + i\theta_2 = -fg^2$; solving the two linear equations for θ_1, θ_2 gives the stated formulae. Holomorphy of $\theta_1 + i\theta_2 = -fg^2$ is exactly the condition on the orders of zeros and poles. ☐

Proposition 7.14

Let $D \subseteq \mathbb{C}$ be simply connected, fix $\xi_0 \in D$, and let $\theta_1, \theta_2, \theta_3$ be holomorphic on D with $\sum \theta_j^2 = 0$. Define

$$x(u,v) = \operatorname{Re} \int_{\xi_0}^{u+iv} \theta_1 \, \mathrm{d}\xi, \quad y(u,v) = \operatorname{Re} \int_{\xi_0}^{u+iv} \theta_2 \, \mathrm{d}\xi, \quad z(u,v) = \operatorname{Re} \int_{\xi_0}^{u+iv} \theta_3 \, \mathrm{d}\xi.$$

Then $x_u - ix_v = \theta_1$, and similarly for y, z. Consequently, wherever $\sigma = (x,y,z)$ is an immersion, its image is a minimal surface.

Proof. Simple connectedness makes the integrals well defined and holomorphic. Write $\chi(\omega) = \int_{\xi_0}^{\omega} \theta_1 \, \mathrm{d}\xi = x + i\beta$; the Cauchy–Riemann equations for χ give $\beta_u = -x_v$, so

$$\theta_1(\omega) = \chi'(\omega) = x_u + i\beta_u = x_u - ix_v,$$

as claimed. The previous two results then say that σ is isothermal with harmonic components, hence minimal. □

Definition 7.15 (Weierstrass Data)

The pair (f, g) above is the **Weierstrass representation** of the resulting minimal surface.

By the existence of isothermal coordinates, *every* minimal surface has a local Weierstrass representation. Checking that a given (f, g) yields an embedding can be delicate; checking that it yields an immersion is easy, and amounts to asking that the zeros of f occur exactly at the poles of g and with exactly twice the order.

Example 7.16 (Enneper's Surface)

Take $D = \mathbb{C}$, $f(\xi) = 1$ and $g(\xi) = \xi$, so $\theta_1 = \frac{1}{2}(1 - \xi^2)$, $\theta_2 = \frac{i}{2}(1 + \xi^2)$, $\theta_3 = \xi$. Integrating from 0 and taking real parts gives

$$\sigma(u, v) = \frac{1}{2}\left(u - \frac{u^3}{3} + uv^2,\ -v - \frac{v^3}{3} - u^2 v,\ u^2 - v^2\right),$$

which is an immersion everywhere and an embedding near the origin. Its Gaussian curvature works out to $K = -16/(1 + u^2 + v^2)^4$.

Example 7.17 (Scherk's Surface)

Take $D = \{|\xi| < 1\}$, $f(\xi) = 4/(1 - \xi^4)$ and $g(\xi) = \xi$. The resulting surface satisfies the beautiful equation

$$\cos y = e^z \cos x,$$

is defined over the open square $(-\pi/2, \pi/2)^2$, and repeats periodically in a checkerboard pattern. It was the first new minimal surface found after the catenoid and helicoid, in 1835.

§7.5 Interpreting the Weierstrass Data

The function g has a lovely geometric meaning: it is the Gauss map, read through stereographic projection.

Proposition 7.18

Let σ be a minimal immersion with Weierstrass data (f, g), and let $\pi : S^2 \to \mathbb{C} \cup \{\infty\}$ be stereographic projection from the north pole. Then

$$g = \pi \circ n \circ \sigma,$$

and the conformal factor is $\lambda = \frac{1}{2}|f|\left(1 + |g|^2\right)$.

Proof. Since $\sigma_u - i\sigma_v = (\theta_1, \theta_2, \theta_3)$ and σ is isothermal,

$$\lambda^2 = E = \tfrac{1}{2}\langle \sigma_u - i\sigma_v, \sigma_u + i\sigma_v \rangle = \tfrac{1}{2}\left(|\theta_1|^2 + |\theta_2|^2 + |\theta_3|^2\right) = \left(\tfrac{|f|(1+|g|^2)}{2}\right)^2$$

after substituting the formulae for θ_j. A direct computation of the cross product gives

$$\sigma_u \times \sigma_v = \frac{|f|^2(1+|g|^2)}{4}\left(2\operatorname{Re} g,\ 2\operatorname{Im} g,\ |g|^2 - 1\right),$$

and normalising, the Gauss map is

$$n \circ \sigma = \frac{1}{1+|g|^2}\left(2\operatorname{Re} g,\ 2\operatorname{Im} g,\ |g|^2 - 1\right),$$

which is exactly $\pi^{-1}(g)$. □

Corollary 7.19

If $K \not\equiv 0$ on a connected minimal surface, then the zeros of K are isolated.

Proof. On the example sheet one computes $K = -\left(4|g'|/\left(|f|(1+|g|^2)^2\right)\right)^2$, so K vanishes exactly where g' does. Since g is meromorphic and non-constant, g' has isolated zeros by the principle of isolated zeros. □

Theorem 7.20

Let σ be a minimal immersion defined on all of $\mathbb{C}$. Then either its image lies in a plane, or its Gauss map omits at most two points of S^2.

Proof. If the degenerate case $\theta_1 = i\theta_2, \theta_3 = 0$ occurs then z is constant and the surface lies in a plane; so assume the Weierstrass data (f, g) exists. If g is constant then n is constant and again the image is planar. Otherwise $g : \mathbb{C} \to \mathbb{C} \cup \{\infty\}$ is a non-constant meromorphic function, so by Picard's theorem it omits at most two values of $\mathbb{C} \cup \{\infty\}$. Since $g = \pi \circ n \circ \sigma$ and π is a bijection, the Gauss map omits at most two points of S^2. □

Example 7.21

Enneper's surface has $g(\xi) = \xi$, which omits only ∞; its Gauss map misses exactly one point. The catenoid has $g(\xi) = \xi$ on $\mathbb{C} \setminus \{0\}$, whose Gauss map misses two points, the north and south poles. So the bound is achieved.

Remark. For complete non-planar minimal surfaces, Fujimoto proved in 1988 that the Gauss map can omit at most four points, sharpening earlier work of Xavier; and four is attained. This is a striking instance of the general principle that minimal surfaces inherit the rigidity of holomorphic functions.

§8 The Gauss–Bonnet Theorem

We come to the theorem the whole course has been building towards. Gauss–Bonnet relates an analytic quantity – the integral of the curvature – to a purely topological one, the Euler characteristic. Curvature is a delicate local invariant which changes if you dent the surface with a finger; the Euler characteristic is a crude count of holes which does not. The theorem says that if you push curvature down in one place, it has to pop up somewhere else.

We begin with the local statement, for which we need to measure how much a curve on a surface turns.

§8.1 Geodesic Curvature

Let Σ be an oriented surface and γ a unit speed curve on it. We have already seen that $\ddot{\gamma}$ splits into a normal part – the normal curvature $\kappa_n = \langle \ddot{\gamma}, n \rangle$ – and a tangential part, and that γ is a geodesic exactly when the tangential part vanishes. Since $\ddot{\gamma} \perp \dot{\gamma}$, the tangential part is a multiple of $n \times \dot{\gamma}$.

Definition 8.1 (Geodesic Curvature)
The **geodesic curvature** of a unit speed curve γ on an oriented surface Σ is

$$\kappa_g = \langle \ddot{\gamma}, n \times \dot{\gamma} \rangle.$$

So $\ddot{\gamma} = \kappa_g \, (n \times \dot{\gamma}) + \kappa_n \, n$, and therefore

$$\kappa^2 = \|\ddot{\gamma}\|^2 = \kappa_g^2 + \kappa_n^2 :$$

the curvature of a curve on a surface splits into the part forced on it by the surface bending, and the part due to the curve turning within the surface. Immediately from the definitions:

Proposition 8.2
A unit speed curve on Σ is a geodesic if and only if $\kappa_g \equiv 0$.

Example 8.3
For a curve in a plane, n is constant and κ_g is the signed curvature κ_s of Section 2. On the sphere, a circle of latitude at angle u from the equator has $\kappa_g = \tan u$: zero at the equator (a great circle) and blowing up at the poles.

The sign of κ_g depends on the orientation. More generally, given a unit vector field W along γ, the derivative $\dot{W}$ is orthogonal to W, so its tangential part is a multiple of $n \times W$; we write

$$\left[\frac{DW}{\mathrm{d}t}\right] = \left\langle \dot{W}, n \times W \right\rangle,$$

the **algebraic value** of the covariant derivative, so that $DW/\,\mathrm{d}t = [DW/\,\mathrm{d}t](n \times W)$. Then $\kappa_g = [D\dot{\gamma}/\,\mathrm{d}t]$. The following lemma is the technical heart of the local Gauss–Bonnet theorem: it says that the algebraic value measures the rate of turning relative to a reference field.

Lemma 8.4

Let V, W be smooth unit vector fields along a curve $\gamma : I \to \Sigma$, and let ψ be a smooth determination of the angle from V to W, that is, a smooth function with

$$W = \cos\psi\, V + \sin\psi\,(n \times V).$$

(Such a ψ exists and is unique up to adding a constant in $2\pi\mathbb{Z}$.) Then

$$\left[\frac{DW}{\mathrm{d}t}\right] - \left[\frac{DV}{\mathrm{d}t}\right] = \dot\psi.$$

Proof. Write $W = aV + b(n\times V)$ with $a^2+b^2 = 1$; existence of a smooth ψ follows by setting $\psi(t) = \psi_0 + \int_{t_0}^t (a\dot b - b\dot a)$ and checking as in the plane curve case. Abbreviate $iV = n \times V$, so that $n \times W = -\sin\psi\, V + \cos\psi\,(iV)$. Then

$$\dot W = -\dot\psi \sin\psi\, V + \dot\psi\cos\psi\,(iV) + \cos\psi\,\dot V + \sin\psi\,\frac{\mathrm{d}(iV)}{\mathrm{d}t},$$

and taking the inner product with $n \times W$, the first two terms contribute $\dot\psi(\sin^2\psi + \cos^2\psi) = \dot\psi$ while the last two contribute

$$\cos^2\psi\,\langle \dot V, iV\rangle - \sin^2\psi\,\langle V, \tfrac{\mathrm{d}}{\mathrm{d}t}(iV)\rangle = \left(\cos^2\psi + \sin^2\psi\right)\left[\frac{DV}{\mathrm{d}t}\right],$$

using $\langle V, iV\rangle = 0$. Adding gives the result. □

Proposition 8.5

Let σ be an orthogonal parametrisation compatible with the orientation, and let $\gamma(t) = \sigma(u(t), v(t))$ be a unit speed curve in its image. Then

$$\kappa_g = \frac{G_u \dot v - E_v \dot u}{2\sqrt{EG}} + \dot\psi,$$

where ψ is a smooth determination of the angle from σ_u to $\dot\gamma$.

Proof. Put $e_1 = \sigma_u/\sqrt{E}$ and $e_2 = \sigma_v/\sqrt{G}$, a positively oriented orthonormal frame. By the lemma it is enough to compute $[De_1/\,\mathrm{d}t] = \langle \dot e_1, e_2\rangle$. Now

$$\langle (e_1)_u, e_2\rangle = \frac{1}{\sqrt{EG}}\langle \sigma_{uu}, \sigma_v\rangle = \frac{1}{\sqrt{EG}}\left(F_u - \tfrac12 E_v\right) = -\frac{E_v}{2\sqrt{EG}},$$

using $F \equiv 0$, and similarly

$$\langle (e_1)_v, e_2\rangle = \frac{1}{\sqrt{EG}}\langle \sigma_{uv}, \sigma_v\rangle = \frac{G_u}{2\sqrt{EG}}.$$

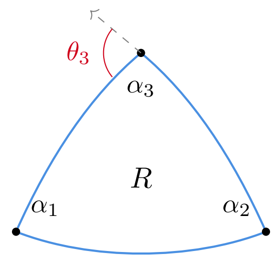
Hence $[De_1/\,\mathrm{d}t] = \dot u\langle (e_1)_u, e_2\rangle + \dot v\langle (e_1)_v, e_2\rangle$ is the first term, and the lemma supplies the $\dot\psi$. □

That first term is exactly the sort of thing Green's theorem eats, and the orthogonal Gauss formula is exactly what comes out. That is the whole proof of the local theorem.

§8.2 The Local Gauss–Bonnet Theorem

Definition 8.6 (Simple Region)

Let Σ be an oriented surface. A region $\overline{R} \subseteq \Sigma$ is **simple** if it is homeomorphic to a closed disc and its boundary is the image of a simple closed curve $\alpha : [0, L] \to \Sigma$, parametrised by arc length, which is smooth except at finitely many points called **vertices**. The **exterior angle** θ_i at a vertex is the angle from the incoming tangent to the outgoing tangent, taken in $(-\pi, \pi)$. We say α is **positively oriented** if $n \times \dot{\alpha}$ points into R.



We will use, without proof, the following fact, which is the surface version of Hopf's Umlaufsatz: if $\overline{R}$ is a simple region contained in the image of a parametrisation whose domain is a disc, and ψ is a piecewise smooth determination of the angle from σ_u to $\dot{\alpha}$ which jumps by θ_i at the ith vertex, then the total change is

$$\psi(L) - \psi(0) = 2\pi.$$

Theorem 8.7 (Local Gauss–Bonnet)

Let $\overline{R}$ be a simple region contained in the image of an orthogonal parametrisation compatible with the orientation, whose domain is a disc, and let $\alpha : [0, L] \to \Sigma$ be a positively oriented arc length parametrisation of ∂R with exterior angles $\theta_1, \dots, \theta_k$. Then

$$\int_0^L \kappa_g \, \mathrm{d}s + \int_R K \, \mathrm{d}A = 2\pi - \sum_{i=1}^{k} \theta_i.$$

Proof. Away from the vertices we have, by the proposition above,

$$\kappa_g = \frac{G_u \dot{v} - E_v \dot{u}}{2\sqrt{EG}} + \frac{\mathrm{d}\psi}{\mathrm{d}s}.$$

Integrating over the smooth arcs and summing, the last term contributes $\psi(L) - \psi(0)$ together with the jumps, that is, $2\pi - \sum_i \theta_i$ by the fact quoted above. For the first term, Green's theorem in the (u, v)-plane gives

$$\int_0^L \frac{G_u \dot{v} - E_v \dot{u}}{2\sqrt{EG}} \, \mathrm{d}s = \int_{\sigma^{-1}(R)} \left[\left(\frac{G_u}{2\sqrt{EG}} \right)_u + \left(\frac{E_v}{2\sqrt{EG}} \right)_v \right] \mathrm{d}u \, \mathrm{d}v,$$

and by the orthogonal Gauss formula the integrand is $-K\sqrt{EG}$. Since $F = 0$ we have $\sqrt{EG} = \sqrt{EG - F^2}$, so this is $-\int_R K \, \mathrm{d}A$. Rearranging gives the theorem. $\square$

Corollary 8.8 (Geodesic Triangles)

Let T be a geodesic triangle – a simple region whose three boundary arcs are geodesics – with interior angles $\alpha_1, \alpha_2, \alpha_3$. Then

$$\int_T K \, \mathrm{d}A = \alpha_1 + \alpha_2 + \alpha_3 - \pi.$$

Proof. The geodesic curvature vanishes on the boundary, and the exterior angles are $\theta_i = \pi - \alpha_i$. Substituting, $\int_T K \, \mathrm{d}A = 2\pi - \sum(\pi - \alpha_i) = \sum \alpha_i - \pi$. □

So the angle sum of a geodesic triangle exceeds π by exactly the total curvature it encloses. On the unit sphere this recovers the classical formula: the area of a spherical triangle is its angle excess. In the plane we recover Euclid.

§8.3 Triangulations and the Euler Characteristic

To globalise, we chop the surface into small pieces and add up.

Definition 8.9 (Triangulation)

A **triangulation** of a compact surface Σ (possibly with boundary) is a finite collection of topological triangles $T_1, \dots, T_F$ covering Σ, such that any two distinct triangles are either disjoint, meet in exactly one common vertex, or meet in exactly one common edge.

Definition 8.10 (Euler Characteristic)

The **Euler characteristic** of a triangulation is $\chi = V - E + F$, where V, E, F count vertices, edges and faces.

We will use the following facts from topology, which we do not prove.

Fact 8.11

Every compact smooth surface admits a triangulation whose edges are smooth curves, and every triangulation can be refined to a smooth one.

Fact 8.12 (Classification of Compact Surfaces)

Every compact connected orientable surface without boundary is diffeomorphic to a sphere with $g \geq 0$ handles attached, and this surface has Euler characteristic $2 - 2g$. In particular the Euler characteristic determines the surface up to diffeomorphism.

For the proof we will need triangulations by small triangles. Note first that subdividing a triangle into four by joining the midpoints of its edges adds 3 vertices, 6 edges and 3 faces, so it leaves χ unchanged. Repeating this finitely often makes all triangles as small as we like.

Proposition 8.13

Every point of a smooth surface has a neighbourhood admitting an orthogonal parametrisation.

Proof. Fix p and a unit speed geodesic γ through p with $\gamma(0) = p$. For each v, let γ^v be the unit speed geodesic with $\gamma^v(0) = \gamma(v)$ whose initial velocity is orthogonal to $\dot{\gamma}(v)$ and positively oriented. Define

$$\sigma(u, v) = \gamma^v(u).$$

By smooth dependence of solutions of the geodesic equations on initial conditions this is smooth near $(0, 0)$, and $\sigma_u(0, 0), \sigma_v(0, 0)$ are orthonormal, so by the inverse function theorem σ is a parametrisation near p. Finally $E = \|\sigma_u\|^2 = 1$ since the u-curves are unit speed geodesics, and

$$F_u = \langle \sigma_{uu}, \sigma_v \rangle + \langle \sigma_u, \sigma_{uv} \rangle = 0 + \tfrac{1}{2} E_v = 0,$$

so F is independent of u; and $F(0, v) = 0$ by construction. Hence $F \equiv 0$. These are called **geodesic normal** (or Fermi) coordinates. □

§8.4 The Global Gauss–Bonnet Theorem

Theorem 8.14 (Global Gauss–Bonnet)

Let Σ be a compact orientable smooth surface without boundary. Then for any triangulation of Σ with Euler characteristic χ,

$$\int_\Sigma K \, \mathrm{d}A = 2\pi\chi.$$

Proof. Cover Σ by finitely many open sets carrying orthogonal parametrisations, using the previous proposition and compactness, and let ε be a Lebesgue number for this cover. Take any smooth triangulation and subdivide repeatedly until every triangle has diameter less than ε; as noted, this does not change χ. Now every triangle T_i lies inside a single orthogonal patch, so local Gauss–Bonnet applies to it:

$$\int_{T_i} K \, \mathrm{d}A + \int_{\partial T_i} \kappa_g \, \mathrm{d}s = \alpha_{i,1} + \alpha_{i,2} + \alpha_{i,3} - \pi,$$

with ∂T_i positively oriented. Sum over i. On the left, the curvature integrals add up to $\int_\Sigma K \, \mathrm{d}A$. The geodesic curvature integrals cancel in pairs: every edge is shared by exactly two triangles and, because Σ is orientable, we may orient all the triangles compatibly, so each edge is traversed once in each direction and κ_g changes sign.

On the right, the angles at each vertex sum to 2π, so $\sum_{i,j} \alpha_{i,j} = 2\pi V$, and we subtract π once per face. Hence

$$\int_\Sigma K \, \mathrm{d}A = 2\pi V - \pi F.$$

Finally, each face has three edges and each edge lies in two faces, so $3F = 2E$, giving

$\pi F = 3\pi F - 2\pi F = 2\pi E - 2\pi F$ and therefore

$$\int_\Sigma K \,\mathrm{d}A = 2\pi V - 2\pi E + 2\pi F = 2\pi\chi. \qquad \square$$

Corollary 8.15

The Euler characteristic of a compact orientable smooth surface does not depend on the choice of triangulation.

Proof. The left hand side of Gauss–Bonnet does not mention the triangulation. $\square$

This is a genuinely startling deduction: a fact of pure combinatorial topology, proved by integrating curvature.

Theorem 8.16 (Gauss–Bonnet with Boundary)

Let R be a region with compact closure in an oriented surface Σ, whose boundary consists of n piecewise smooth simple closed curves $\alpha_1, \dots, \alpha_n$, positively oriented and parametrised by arc length, with exterior angles θ_j at the vertices. Then

$$\int_R K \,\mathrm{d}A + \sum_{i=1}^n \int_{\alpha_i} \kappa_g \,\mathrm{d}s = 2\pi\chi(R) - \sum_j \theta_j.$$

Proof. Identical to the previous proof, except that the geodesic curvature integrals along the boundary edges have no partner to cancel with, and the exterior angles at boundary vertices survive. $\square$

§8.5 Applications

Example 8.17 (Total Curvature of Standard Surfaces)

The sphere has $\chi = 2$, so $\int_{S^2} K \,\mathrm{d}A = 4\pi$ – which we can check directly, since $K \equiv 1$ and the area is 4π. The torus has $\chi = 0$, so $\int K \,\mathrm{d}A = 0$: the positive curvature on the outside exactly cancels the negative curvature on the inside, however you deform it. A surface of genus $g \geq 2$ has $\chi = 2 - 2g < 0$, so it must have negative curvature somewhere.

Corollary 8.18

A compact orientable surface with $K > 0$ everywhere is diffeomorphic to S^2.

Proof. By Gauss–Bonnet, $2\pi\chi = \int_\Sigma K \,\mathrm{d}A > 0$, so $\chi > 0$. But $\chi = 2 - 2g$ for some $g \geq 0$, so $\chi = 2$ and $g = 0$. By the classification of compact surfaces, Σ is diffeomorphic to S^2. $\square$

Corollary 8.19

A compact orientable surface with $K \leq 0$ everywhere has genus at least 1; and if

$K < 0$ everywhere it has genus at least 2.

Proof. $2\pi(2-2g) = \int K \, \mathrm{d}A \leq 0$ forces $g \geq 1$, and the strict inequality forces $2 - 2g < 0$, i.e. $g \geq 2$. □

Proposition 8.20

On a compact orientable surface with $K > 0$ everywhere, any two simple closed geodesics must intersect.

Proof. Suppose γ_1, γ_2 are disjoint simple closed geodesics. By the previous corollary Σ is a sphere, and two disjoint simple closed curves on a sphere bound a region R homeomorphic to an annulus, which has $\chi(R) = 0$. Applying Gauss–Bonnet with boundary to R, and noting that $\kappa_g = 0$ along both boundary curves and there are no vertices,

$$\int_R K \, \mathrm{d}A = 2\pi \cdot 0 = 0,$$

contradicting $K > 0$. □

Example 8.21 (The Catenoid)

The catenoid is homeomorphic to a cylinder and has $K < 0$ everywhere. A similar Gauss–Bonnet argument shows that such a surface carries at most one simple closed geodesic; and indeed the catenoid has exactly one, the circle at its waist.

Remark (Parallel Transport and Holonomy)**.** Local Gauss–Bonnet also computes the angle by which parallel transport around a closed curve rotates a tangent vector. If γ bounds a simple region R and V is parallel along γ, then the lemma above with $W = \dot{\gamma}$ shows that the total turning of V relative to $\dot{\gamma}$ is $\int \kappa_g$, and the computation in the proof shows that transporting once around γ rotates a vector by exactly $\int_R K \, \mathrm{d}A$. Carry a vector around a spherical triangle and it comes back rotated by the area of the triangle – this is what a Foucault pendulum is measuring.

§9 Total Curvature of Space Curves

We finish with two beautiful applications of the global theory to curves in $\mathbb{R}^3$. Both work by fattening the curve into a surface – a tube – and then applying what we know about the Gauss map.

Definition 9.1 (Total Curvature)

The **total curvature** of a closed unit speed curve $\alpha : [0, L] \to \mathbb{R}^3$ is $\int_0^L \kappa(s) \, \mathrm{d}s$.

For a plane curve, the total curvature is $\int |\kappa_s| \geq |\int \kappa_s| = 2\pi$ if the curve is simple, by Hopf's Umlaufsatz, with equality if and only if κ_s never changes sign, that is, if and only if the curve is convex. Fenchel's theorem says the same bound holds in space, and that only plane curves achieve it.

§9.1 Tubes

Definition 9.2 (Tube)

Let $\alpha : [0, L] \to \mathbb{R}^3$ be a closed unit speed curve with $\kappa > 0$ everywhere, with Frenet frame (T, N, B). For $r > 0$, the **tube** of radius r about α is the surface parametrised by

$$\sigma(s, v) = \alpha(s) + r\left(N(s)\cos v + B(s)\sin v\right).$$

For r small enough the tube is an embedded surface. Its geometry is entirely explicit.

Lemma 9.3

For r small enough that $r\kappa < 1$ everywhere, the tube T_r is a smooth compact surface with

$$EG - F^2 = r^2\left(1 - r\kappa\cos v\right)^2, \qquad n = -\left(N\cos v + B\sin v\right),$$

and Gaussian curvature

$$K = -\frac{\kappa\cos v}{r\left(1 - r\kappa\cos v\right)}.$$

Proof. Differentiating and using Frenet–Serret,

$$\sigma_s = \left(1 - r\kappa\cos v\right)T + r\tau\left(-N\sin v + B\cos v\right), \qquad \sigma_v = r\left(-N\sin v + B\cos v\right).$$

These are linearly independent when $r\kappa < 1$, and a short computation gives $E = (1 - r\kappa\cos v)^2 + r^2\tau^2$, $F = r^2\tau$, $G = r^2$, whence $EG - F^2 = r^2(1 - r\kappa\cos v)^2$. Also $\sigma_s \times \sigma_v = -r(1 - r\kappa\cos v)(N\cos v + B\sin v)$, giving the stated unit normal. Differentiating n and using Frenet–Serret again gives

$$n_s \times n_v = \kappa\cos v\left(N\cos v + B\sin v\right) = -\frac{\kappa\cos v}{r(1 - r\kappa\cos v)}\,\sigma_s \times \sigma_v,$$

and since $n_s \times n_v = K\,\sigma_s \times \sigma_v$, the formula for K follows. □

Note that $K \geq 0$ exactly when $\cos v \leq 0$, that is, on the 'outer' half of the tube – exactly as for the torus, which is the special case α a circle. We will also use, without proof, the following consequence of the stack of records theorem and Sard's theorem: for a compact oriented surface Σ in $\mathbb{R}^3$,

$$\int_\Sigma |K|\,\mathrm{d}A = \int_{S^2} \left|n^{-1}(y)\right|\,\mathrm{d}y,$$

which is to say that $\int |K|$ counts, with multiplicity, how often the Gauss map covers the sphere.

§9.2 Fenchel's Theorem

Theorem 9.4 (Fenchel)

Let α be a simple closed regular curve in $\mathbb{R}^3$ with nowhere vanishing curvature. Then its total curvature is at least 2π, with equality if and only if α is a convex plane curve.

Proof. Form the tube T_r for small r and let $R = \{p \in T_r : K(p) \geq 0\}$, which by the lemma is the set where $\cos v \leq 0$, i.e. $v \in [\pi/2, 3\pi/2]$. Then

$$\begin{aligned} \int_R K \, \mathrm{d}A &= \int_0^L \int_{\pi/2}^{3\pi/2} \frac{-\kappa \cos v}{r(1 - r\kappa \cos v)} \cdot r\,(1 - r\kappa \cos v)\; \mathrm{d}v \; \mathrm{d}s \\ &= -\int_0^L \kappa \; \mathrm{d}s \int_{\pi/2}^{3\pi/2} \cos v \; \mathrm{d}v, \end{aligned}$$

and since $\int_{\pi/2}^{3\pi/2} \cos v \; \mathrm{d}v = -2$, this is $2\int_0^L \kappa \; \mathrm{d}s$.

Now the Gauss map restricted to R is surjective onto S^2: given any unit vector y, the height function $s \mapsto \langle \alpha(s), y\rangle$ attains a maximum, and at that point the corresponding point of the tube in the direction y has outward normal y and non-negative curvature. Hence

$$2\int_0^L \kappa \; \mathrm{d}s = \int_R K \; \mathrm{d}A = \int_{S^2} \left| (n|_R)^{-1}(y) \right| \; \mathrm{d}y \geq \operatorname{area}(S^2) = 4\pi,$$

giving $\int \kappa \geq 2\pi$.

Suppose equality holds. Then n is injective on the set where $K > 0$, so distinct points of that set have distinct normals. It follows that at every such point p the whole tube lies on one side of the tangent plane at p – otherwise, using that n is a local diffeomorphism near p and that the zero set of K is nowhere dense, one produces two distinct points with positive curvature and the same normal. Taking limits, the same holds at the points where $K = 0$, which are exactly the points of the tube in the directions $\pm B$.

Fix s_0 and look at the circle $s = s_0$ of the tube. The two points on it in the directions $\pm B(s_0)$ have parallel tangent planes, and the tube is squeezed between them; hence α lies in the plane Π midway between. (If not, take s_1 maximising the distance from Π; then $T(s_1)$ is parallel to Π, and the circle $s = s_1$ then pokes out of the slab, a contradiction.) So α is a plane curve, and by the plane case discussed above, equality forces it to be convex. □

§9.3 The Fáry–Milnor Theorem

If a closed curve is knotted, it has to bend more. This was conjectured by Borsuk and proved independently by Fáry (1949) and Milnor (1950); Milnor was an undergraduate at the time.

Definition 9.5 (Unknotted)

A regular simple closed curve in $\mathbb{R}^3$ is **unknotted** if its image is the boundary of an embedded disc, and **knotted** otherwise.

Theorem 9.6 (Fáry–Milnor)

A knotted regular simple closed curve in $\mathbb{R}^3$ with nowhere vanishing curvature has total curvature at least 4π.

Proof. We prove the contrapositive: if the total curvature is less than 4π then the curve is unknotted.

Form the tube T_r as before. Computing $\int_{T_r} |K| \, \mathrm{d}A$ exactly as in Fenchel's proof, but now over all of $[0, 2\pi]$ in v and with $|\cos v|$ in place of $-\cos v$, gives

$$\int_{S^2} \left|n^{-1}(y)\right| \, \mathrm{d}y = \int_{T_r} |K| \, \mathrm{d}A = 4 \int_0^L \kappa \, \mathrm{d}s.$$

If $\int_0^L \kappa \, \mathrm{d}s < 4\pi$ then the right hand side is less than $16\pi = 4 \operatorname{area}(S^2)$, so the average number of preimages is less than 4. By Sard's theorem we may pick a regular value $y \in S^2$ of n with $\left|n^{-1}(y)\right| \leq 3$.

Now consider the height function $h(s) = \langle y, \alpha(s) \rangle$. We have $h'(s) = \langle y, T(s) \rangle$, and one checks from the formula for n that $h'(s) = 0$ precisely when the two points of the tube over s in the $\pm$ directions of the plane orthogonal to $T(s)$ hit y; so h has at most three critical points. But h is a smooth function on a circle, so it has at least one maximum and one minimum, and its critical points alternate between maxima and minima, hence are even in number. So h has exactly two critical points, one maximum and one minimum.

Therefore α consists of exactly two arcs joining the maximum to the minimum, on each of which h is strictly monotone. For each value c strictly between the minimum and the maximum, the plane $h = c$ meets α in exactly two points; joining them by a straight segment and letting c vary sweeps out an embedded disc with boundary α. Hence α is unknotted. $\square$

Remark. Milnor proved the stronger statement that the total curvature of a knotted curve is *strictly* greater than 4π, and that 4π is the best possible bound: a suitably flattened trefoil has total curvature as close to 4π as we please.

§10 Abstract Surfaces

So far, the geometry of our surfaces has been induced by the space that surface has been embedded in (which has so far been $\mathbb{R}^3$). Of course, it's possible to do geometry on surfaces that do not arise in this way. One way of specifying the geometry of a surface is with a *Riemannian metric*.

The point of doing this is not merely generality. We saw that constant curvature $K < 0$ is realised locally by pieces of surfaces in $\mathbb{R}^3$, but Hilbert's theorem says that no *complete* surface of constant negative curvature exists in $\mathbb{R}^3$. The hyperbolic plane, one of the three classical two-dimensional geometries, therefore only exists as an abstract surface – and once we allow abstract surfaces, it is easy to write down.

§10.1 Riemannian Metrics

Definition 10.1 (Riemannian Metric)

Let $V \subseteq \mathbb{R}^2$ be an open set. An (abstract) **Riemannian metric** on V is a smooth map from V to the set of positive definite symmetric bilinear forms, given by

$$p \mapsto \begin{pmatrix} E(p) & F(p) \\ F(p) & G(p) \end{pmatrix},$$

with $E, G, EG - F^2 > 0$.

If w is a tangent vector at $p \in V$, we can compute its infinitesimal length by

$$\|w\|^2 = w^\top \begin{pmatrix} E(p) & F(p) \\ F(p) & G(p) \end{pmatrix} w.$$

So, if $\gamma : [a, b] \to V$ is smooth,

$$L(\gamma) = \int_a^b \sqrt{E\dot{u}^2 + 2F\dot{u}\dot{v} + G\dot{v}^2}\, \mathrm{d}t,$$

where $\gamma(t) = (u(t), v(t))$. Areas are computed by $\int \sqrt{EG - F^2}\, \mathrm{d}u\, \mathrm{d}v$, exactly as before.

Definition 10.2 (Abstract Riemannian Surface)
An **abstract Riemannian surface** is an abstract smooth surface Σ together with a Riemannian metric in each chart, such that the metrics agree on overlaps: if φ is a transition map then

$$\begin{pmatrix} \tilde{E} & \tilde{F} \\ \tilde{F} & \tilde{G} \end{pmatrix} = (D\varphi)^\top \begin{pmatrix} E & F \\ F & G \end{pmatrix} (D\varphi).$$

Everything intrinsic that we have done goes through unchanged: the Christoffel symbols were defined purely in terms of E, F, G, hence so were the covariant derivative, parallel transport, the geodesic equations, and – by the Theorema Egregium – the Gaussian curvature, which we now simply *define* by the Gauss equation. Geodesic polar coordinates, the local and global Gauss–Bonnet theorems, and all their corollaries hold verbatim. What we lose is the second fundamental form, the mean curvature and the Gauss map, all of which needed an ambient space.

Example 10.3 (Surfaces in $\mathbb{R}^3$)
A smooth surface in $\mathbb{R}^3$ with its first fundamental form is an abstract Riemannian surface, and the transformation rule above is exactly how E, F, G transform.

Example 10.4 (The Flat Torus)
Take $\mathbb{R}^2$ with the Euclidean metric $\mathrm{d}u^2 + \mathrm{d}v^2$ and quotient by the lattice $\mathbb{Z}^2$. The result is a torus with a flat metric, $K \equiv 0$, which is consistent with Gauss–Bonnet since $\chi = 0$. This surface is *not* isometric to any torus of revolution in $\mathbb{R}^3$, since we showed such a torus has points of positive and of negative curvature. Indeed, no compact flat surface embeds in $\mathbb{R}^3$ at all, since compact surfaces in $\mathbb{R}^3$ have an elliptic point.

§10.2 The Length Metric

An abstract Riemannian surface carries a genuine metric space structure, obtained in the obvious way.

Definition 10.5 (Length Metric)

Let Σ be a connected abstract Riemannian surface. For $p, q \in \Sigma$ define

$$d(p,q) = \inf \{L(\gamma) : \gamma \text{ a piecewise smooth curve in } \Sigma \text{ from } p \text{ to } q\}.$$

Proposition 10.6

d is a metric on Σ, and it induces the original topology.

Proof. Symmetry and the triangle inequality are clear from concatenating and reversing curves, and $d(p,p) = 0$. The only issue is that $d(p,q) > 0$ for $p \neq q$. Take geodesic polar coordinates about p on a normal neighbourhood of radius ε small enough not to contain q. Any curve from p to q must cross every geodesic circle about p of radius less than ε, and by the argument used to prove that geodesics locally minimise length, such a curve has length at least ε. So $d(p,q) \geq \varepsilon > 0$. The same argument shows the metric balls of small radius about p are exactly the images under $\exp_p$ of the Euclidean balls, so the topologies agree. $\square$

Remark. A curve realising the infimum, if one exists, is necessarily a geodesic; but such a curve need not exist – delete a point from the plane and there is no shortest path between two points on opposite sides of the hole. The Hopf–Rinow theorem says that this is the only obstruction: if (Σ, d) is complete as a metric space then any two points are joined by a minimising geodesic.

§10.3 The Hyperbolic Plane

The simplest abstract Riemannian surface which is not a surface in $\mathbb{R}^3$ is also the most important.

Definition 10.7 (The Upper Half-Plane Model)

The **hyperbolic plane** is the upper half plane

$$\mathbb{H} = \{(u,v) \in \mathbb{R}^2 : v > 0\}$$

equipped with the Riemannian metric

$$\mathrm{I} = \frac{\mathrm{d}u^2 + \mathrm{d}v^2}{v^2}.$$

This is an isothermal metric with conformal factor $\lambda = 1/v$, so the corollary to the Gauss equation computes its curvature instantly.

Proposition 10.8

The hyperbolic plane has constant curvature $K \equiv -1$.

Proof. With $\lambda = 1/v$ we have $\log \lambda = -\log v$, so $\nabla^2 \log \lambda = -\partial_v^2 \log v = 1/v^2$. Hence

$$K = -\frac{1}{\lambda^2}\nabla^2(\log \lambda) = -v^2 \cdot \frac{1}{v^2} = -1. \qquad \square$$

Proposition 10.9

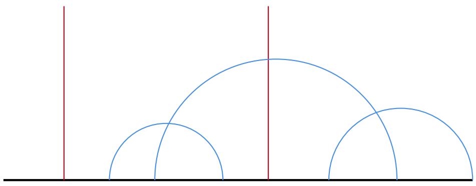
The geodesics of $\mathbb{H}$ are the vertical half-lines and the semicircles orthogonal to the real axis.

Proof. Here $E = G = 1/v^2$ and $F = 0$, and the geodesic equations read

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\dot u}{v^2}\right) = 0, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\dot v}{v^2}\right) = -\frac{\dot u^2 + \dot v^2}{v^3}.$$

The vertical line $u \equiv u_0$, $v = e^t$ satisfies both, so vertical lines are geodesics. For the general case, one can either integrate directly, or – more slickly – observe that the Möbius maps $z \mapsto (az+b)/(cz+d)$ with a, b, c, d real and $ad - bc > 0$ act on $\mathbb{H}$ by isometries (a computation with $\operatorname{Im}\left(\frac{az+b}{cz+d}\right) = \frac{\operatorname{Im} z}{|cz+d|^2}$), that this action is transitive on pairs (point, direction), and that such maps take lines and circles to lines and circles preserving angles. Since a vertical line is a geodesic and the images of vertical lines under this group are exactly the vertical lines and the semicircles meeting $\mathbb{R}$ at right angles, these are all the geodesics. $\square$

geodesics of $\mathbb{H}$



Theorem 10.10 (Area of a Hyperbolic Triangle)

A hyperbolic triangle with interior angles α, β, γ has area $\pi - \alpha - \beta - \gamma$.

Proof. Apply local Gauss–Bonnet to the triangle, whose sides are geodesics so $\kappa_g = 0$, with exterior angles $\pi - \alpha$ and so on:

$$\int_T K \,\mathrm{d}A = \alpha + \beta + \gamma - \pi.$$

But $K \equiv -1$, so the left hand side is $-\operatorname{area}(T)$, giving the result. $\square$

In particular the angle sum of a hyperbolic triangle is always less than π, and no hyperbolic triangle has area more than π. Compare the sphere, where $K \equiv 1$ and the area is the angle *excess*, and the plane, where $K \equiv 0$ and the angle sum is exactly π. All three statements are the same statement, with the sign of the curvature switched.

Remark. By Minding's theorem, any surface of constant curvature -1 is locally isometric to $\mathbb{H}$. The pseudosphere – the surface of revolution generated by the tractrix – is such a surface in $\mathbb{R}^3$, but it is incomplete: it has a singular circle where it cannot be continued. Hilbert's theorem says this is unavoidable, and that no complete surface of constant negative curvature exists in $\mathbb{R}^3$. So the hyperbolic plane is an object we can only reach by allowing abstract metrics – which is a fair summary of why one bothers with them.

Where Next

We have really only scratched the surface. The obvious next step is to allow more dimensions: the Christoffel symbols, the covariant derivative and parallel transport all generalise to n-dimensional Riemannian manifolds, with the Gaussian curvature replaced by the Riemann curvature tensor, and this is the subject of Part III Differential Geometry and of General Relativity. In another direction, the Gauss–Bonnet theorem is the first instance of an index theorem, expressing an analytic invariant in terms of a topological one; the Atiyah–Singer index theorem is the vast generalisation. And the complex-analytic side of minimal surface theory opens into the theory of Riemann surfaces. All of these begin exactly where we have stopped.